

STAT 210B: Theoretical Statistics

Lecture Notes

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These notes accompany the graduate course **STAT 210B: Theoretical Statistics** at UC Berkeley. Parts of the material overlap with standard references such as [3, 4] and related lecture notes [2]. Comments and corrections are welcome.

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1 Introduction

1.1 Why non-asymptotic?

Non-asymptotic bounds keep the sample size fixed and aim to track the effect of all influential parameters. This point of view is often more informative than taking limits, especially in modern high-dimensional settings where several parameters (dimension, noise level, model complexity) may grow simultaneously.

A simple example is mean estimation. Suppose we observe independent and identically distributed random variables $X_1, \dots, X_n$ with mean μ and (for simplicity) known variance σ^2 . A classical confidence interval is

$$\left[\frac{1}{n} \sum_{i=1}^n X_i - \frac{2\sigma}{\sqrt{n}}, \frac{1}{n} \sum_{i=1}^n X_i + \frac{2\sigma}{\sqrt{n}} \right].$$

By the Central Limit Theorem (CLT), this interval asymptotically contains μ with probability $1 - 2\Phi(-2) \approx 0.9545$, where Φ is the standard normal cumulative distribution function. Indeed, if $Z \sim \mathcal{N}(0, 1)$,

then

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma \sqrt{n}} \right| \geq t \right) = \mathbb{P}(|Z| \geq t) = 2\Phi(-t).$$

The point is not that the CLT is inaccurate; rather, it does not tell us how large n needs to be for a given distribution. For some distributions, the Gaussian approximation is excellent even for moderate n , while for others it can require much larger samples.

A second illustration comes from rare events. If $X_{n,1}, \dots, X_{n,n}$ are independent Bernoulli random variables with success probability p_n and $np_n \rightarrow \lambda$, then the sum behaves asymptotically as a Poisson random variable.

Theorem 1.1 (Poisson limit theorem). *Let $X_{n,i} \sim \text{Ber}(p_n)$ be independent. Assume that $np_n \rightarrow \lambda > 0$. Then*

$$\sum_{i=1}^n X_{n,i} \xrightarrow{d} \text{Poi}(\lambda) \quad \text{as } n \rightarrow \infty.$$

For instance, with $n = 50$ and $p = 1/50$, a Poisson approximation is typically more accurate than a Gaussian approximation. Non-asymptotic tools allow us to make such comparisons quantitatively at a finite sample size.

1.2 A motivating example: fixed-design Gaussian regression

A basic model where non-asymptotic bounds arise naturally is fixed-design Gaussian regression. We observe data $\{(Y_i, x_i)\}_{i=1}^n$ satisfying

$$Y_i = f(x_i) + \xi_i, \quad i = 1, \dots, n, \tag{1}$$

where $f : \mathcal{X} \rightarrow \mathbf{R}$ is unknown, the design points $x_i \in \mathcal{X}$ are fixed (nonrandom), and the noise variables ξ_i are independent $\mathcal{N}(0, \sigma^2)$.

Given an estimator $\hat{f}$, we measure error using the empirical loss

$$\|\hat{f} - f\|_n^2 = \frac{1}{n} \sum_{i=1}^n (\hat{f}(x_i) - f(x_i))^2, \quad \text{and the risk } \mathbb{E} \|\hat{f} - f\|_n^2.$$

A standard special case is linear regression: $x_i \in \mathbf{R}^d$ and $f(x_i) = \langle x_i, \theta^* \rangle$ for an unknown $\theta^* \in \mathbf{R}^d$. The least-squares estimator is

$$\hat{\theta} \in \arg \min_{\theta \in \mathbf{R}^d} \frac{1}{n} \sum_{i=1}^n (\langle x_i, \theta \rangle - Y_i)^2.$$

Assume that $\text{rank}(X) = d$, where $X \in \mathbf{R}^{n \times d}$ is the design matrix with rows $x_i^\top$. Then $\hat{\theta}$ is unique and one can show that

$$\mathbb{E} \|\hat{f} - f\|_n^2 = \frac{\sigma^2 d}{n}, \quad \text{where } \hat{f}(x_i) = \langle x_i, \hat{\theta} \rangle.$$

This is a typical non-asymptotic statement: it holds for every n and d (with $d \leq n$ and $\text{rank}(X) = d$), and it makes the dependence on n, d, σ^2 fully explicit. Here d, n , and σ^2 each play a specific role, and they can vary simultaneously in an arbitrary manner. In contrast, classical asymptotic analyses usually focus on specific limiting regimes, such as $n \rightarrow \infty$ with d fixed, or $n, d \rightarrow \infty$ with a prescribed relationship (for example, $d/n \rightarrow \gamma$). Even in this simple example, these regimes cover only a fraction of the possible interactions between d, σ , and n .

To put this viewpoint in context, we quote Pascal Massart [1]:

“Through our works, we have promoted a nonasymptotic approach in statistics which consists in taking the number of observations as it is and trying to evaluate the effect of all the influential parameters. At this very starting point, it seems to me that it is important to provide a first answer to the following question: why should we be interested by a nonasymptotic view for model selection at all? In my opinion, the motivation should neither be a strange interest for “small” sets of data nor a special taste for constants and inequalities rather than for limit theorems (although since mathematics is also a matter of taste, it is a possible way for getting involved in it. . .). On the contrary, the nonasymptotic point of view may turn to be especially relevant when the number of observations is large. It is indeed to fit large complex sets of data that one needs to deal with possibly huge collections of models at different scales. The nonasymptotic approach for model selection precisely allows the collection of models together with their dimensions to vary freely, letting the dimensions be possibly of the same order of magnitude as the number of observations.”

2 Concentration and sub-Gaussian tools

2.1 From Markov to Chernoff

Most concentration bounds begin with a refined application of Markov inequality. We start with two classical inequalities and then explain how exponential moments lead to much sharper bounds.

Proposition 2.1 (Markov inequality). *Let X be a non-negative random variable. Then for any $t > 0$,*

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}X}{t}.$$

A first consequence is Chebyshev inequality.

Proposition 2.2 (Chebyshev inequality). *Let X be a random variable with finite variance. Then for any $t > 0$,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

Proof. Apply Markov inequality to $(X - \mathbb{E}X)^2$. □

For light-tailed distributions, Chebyshev inequality is often far from optimal. For instance, Gaussian tails decay at an exponential rate in t^2 .

Proposition 2.3 (Gaussian tail bounds). *Let $Z \sim \mathcal{N}(0, 1)$. For any $t > 0$,*

$$\left(\frac{1}{t} - \frac{1}{t^3}\right) \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \leq \mathbb{P}(Z \geq t) \leq \frac{1}{t} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

Proof. We have

$$\mathbb{P}(Z \geq t) = \frac{1}{\sqrt{2\pi}} \int_t^\infty e^{-x^2/2} dx.$$

For the upper bound, substitute $x = t + y$:

$$\mathbb{P}(Z \geq t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \int_0^\infty e^{-ty} e^{-y^2/2} dy \leq \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \int_0^\infty e^{-ty} dy = \frac{1}{t} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

For the lower bound, define $f(x) = (\frac{1}{x} - \frac{1}{x^3})e^{-x^2/2}$ for $x > 0$. A direct computation gives $f'(x) = -(1 - 3x^{-4})e^{-x^2/2}$, and hence

$$\int_t^\infty (1 - 3x^{-4})e^{-x^2/2} dx = f(t) = \left(\frac{1}{t} - \frac{1}{t^3}\right)e^{-t^2/2}.$$

Since $(1 - 3x^{-4}) \leq 1$, we have

$$\int_t^\infty e^{-x^2/2} dx \geq \int_t^\infty (1 - 3x^{-4})e^{-x^2/2} dx = \left(\frac{1}{t} - \frac{1}{t^3}\right)e^{-t^2/2},$$

which yields the desired lower bound after multiplying by $1/\sqrt{2\pi}$. $\square$

The proof above is a good warm-up: for a Gaussian variable we can compute tails directly. For a general random variable, the analog of “completing the square” is to work with exponential moments.

For a real random variable Z , define its moment generating function (MGF) by

$$M_Z(\lambda) = \mathbb{E}e^{\lambda Z}, \quad \lambda \in \mathbf{R}.$$

The key feature is multiplicativity under independence: if $Z_1, \dots, Z_n$ are independent, then

$$M_{Z_1 + \dots + Z_n}(\lambda) = \prod_{i=1}^n M_{Z_i}(\lambda).$$

This simple identity is behind most concentration inequalities for sums.

For Gaussian random variables the MGF has an explicit closed form: if $Z \sim \mathcal{N}(0, \sigma^2)$, then

$$M_Z(\lambda) = \mathbb{E}e^{\lambda Z} = e^{\lambda^2 \sigma^2 / 2}, \quad \lambda \in \mathbf{R}.$$

2.2 Sub-Gaussian random variables

The Gaussian MGF suggests a natural class of random variables with comparable tail behavior.

Definition 2.4 (Sub-Gaussian random variable). A mean-zero random variable X is *sub-Gaussian with variance parameter σ^2* if

$$\mathbb{E}e^{\lambda X} \leq e^{\lambda^2 \sigma^2 / 2} \quad \text{for all } \lambda \in \mathbf{R}.$$

Sub-Gaussianity can be characterized in several equivalent ways. We record one standard collection of equivalences; see [3, Proposition 2.5.2] for a complete proof and sharp constants.

Proposition 2.5 (Equivalent definitions of sub-Gaussianity). *Let X be a random variable with $\mathbb{E}X = 0$. The following conditions are equivalent; moreover, the corresponding parameters are equivalent up to absolute constant factors.*

1. There exists $k_1 > 0$ such that for all $\lambda \in \mathbf{R}$,

$$\mathbb{E}e^{\lambda X} \leq e^{k_1^2 \lambda^2}.$$

2. There exists $k_2 > 0$ such that for all $t \geq 0$,

$$\mathbb{P}(|X| \geq t) \leq 2e^{-t^2/k_2^2}.$$

3. There exists $k_3 > 0$ such that for all $p \geq 1$,

$$\|X\|_{L_p} = (\mathbb{E}|X|^p)^{1/p} \leq k_3 \sqrt{p}.$$

4. There exists $k_5 > 0$ such that

$$\mathbb{E}e^{X^2/k_5^2} \leq 2.$$

Proof. We sketch two implications; the remaining ones are in [3, Proposition 2.5.2].

(1) $\Rightarrow$ (2) (Chernoff bound). For any $\lambda > 0$,

$$\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda X} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E}e^{\lambda X} \leq \exp(k_1^2 \lambda^2 - \lambda t).$$

Optimizing over λ yields $\mathbb{P}(X \geq t) \leq \exp(-t^2/(4k_1^2))$. Applying the same argument to $-X$ and using the union bound gives (2).

(2) $\Rightarrow$ (3). Using $\mathbb{E}|X|^p = \int_0^\infty \mathbb{P}(|X|^p \geq s) ds$ and the substitution $s = u^p$ gives

$$\mathbb{E}|X|^p = \int_0^\infty \mathbb{P}(|X| \geq u) p u^{p-1} du.$$

Assume without loss of generality that $k_2 = 1$. Then

$$\mathbb{E}|X|^p \leq 2p \int_0^\infty u^{p-1} e^{-u^2} du = p \int_0^\infty w^{p/2-1} e^{-w} dw = p \Gamma(p/2).$$

Standard bounds for the gamma function imply $\Gamma(p/2)^{1/p} \lesssim \sqrt{p}$, hence $\|X\|_{L_p} \lesssim \sqrt{p}$. $\square$

It is useful to keep a small supply of examples in mind; we will return to more examples throughout the course.

Example 2.6 (Gaussian random variables). If $Z \sim \mathcal{N}(0, \sigma^2)$, then Z is sub-Gaussian with variance parameter σ^2 , since $\mathbb{E}e^{\lambda Z} = e^{\lambda^2 \sigma^2/2}$ for all $\lambda \in \mathbf{R}$.

Lemma 2.7 (Hoeffding lemma (short proof, non-optimal constant)). *Let X be a random variable such that $\mathbb{E}X = 0$ and $X \in [a, b]$ almost surely. Then for all $\lambda \in \mathbf{R}$,*

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2(b-a)^2}{2}\right).$$

Proof. Let X' be an independent copy of X . By Jensen inequality (conditioning on X and using convexity of the exponential),

$$\mathbb{E}e^{\lambda X} = \mathbb{E}\left[e^{\lambda(X - \mathbb{E}[X'|X])}\right] \leq \mathbb{E}\left[e^{\lambda(X - X')}\right].$$

Set $Y = X - X'$. The random variable Y is symmetric: Y and $-Y$ have the same distribution, since swapping X and X' changes Y to $-Y$ without changing the joint law. Therefore,

$$\mathbb{E}e^{\lambda Y} = \mathbb{E}\left[\frac{e^{\lambda Y} + e^{-\lambda Y}}{2}\right] = \mathbb{E}[\cosh(\lambda Y)].$$

We use the elementary bound $\cosh(u) \leq e^{u^2/2}$ for all $u \in \mathbf{R}$. Indeed, comparing Taylor series,

$$\cosh(u) = \sum_{k=0}^{\infty} \frac{u^{2k}}{(2k)!} \leq \sum_{k=0}^{\infty} \frac{u^{2k}}{2^k k!} = e^{u^2/2},$$

where we used $(2k)! \geq 2^k k!$. Since $|Y| \leq b - a$ almost surely, we conclude that

$$\mathbb{E}e^{\lambda X} \leq \mathbb{E}e^{\lambda Y} = \mathbb{E} \cosh(\lambda Y) \leq \mathbb{E}e^{\lambda^2 Y^2/2} \leq e^{\lambda^2(b-a)^2/2}.$$

□

Remark 2.8. The constant in Theorem 2.7 is not optimal. A sharper argument gives $\mathbb{E}e^{\lambda X} \leq \exp(\lambda^2(b-a)^2/8)$; see, for example, [3, Proposition 2.5.2]. In most applications we only need the qualitative fact that bounded random variables are sub-Gaussian.

2.3 Sums of independent sub-Gaussians

One reason MGFs are convenient is that they interact well with independence. In particular, they lead to concentration bounds for sums that mirror the familiar identity $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i)$ for independent mean-zero random variables.

Proposition 2.9. *Let $X_1, \dots, X_n$ be independent random variables with means μ_i such that $X_i - \mu_i$ are sub-Gaussian with variance parameters σ_i^2 in the sense of Theorem 2.4. Then for any $t \geq 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n (X_i - \mu_i) \geq t\right) \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2}\right),$$

and

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mu_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2}\right).$$

Proof. By independence and the sub-Gaussian MGF bound,

$$\mathbb{E} \exp\left(\lambda \sum_{i=1}^n (X_i - \mu_i)\right) = \prod_{i=1}^n \mathbb{E} e^{\lambda(X_i - \mu_i)} \leq \exp\left(\frac{\lambda^2}{2} \sum_{i=1}^n \sigma_i^2\right).$$

Apply Markov inequality to $\exp(\lambda \sum_{i=1}^n (X_i - \mu_i))$ and optimize over $\lambda > 0$ to obtain the upper tail. The lower tail follows by applying the same argument to $-\sum_i (X_i - \mu_i)$ and then using the union bound. □

As a clean corollary, bounded random variables satisfy Hoeffding-type concentration.

Proposition 2.10 (Hoeffding inequality (with a non-optimal constant)). *Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}X_i = \mu_i$ and $X_i \in [a_i, b_i]$ almost surely. Then for any $t \geq 0$,*

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mu_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n (b_i - a_i)^2}\right).$$

Proof. By Theorem 2.7, each $X_i - \mu_i$ is sub-Gaussian with variance parameter $\sigma_i^2 = (b_i - a_i)^2$. Plug into Theorem 2.9. □

Example 2.11 (Averages of random signs). Let $\varepsilon_1, \dots, \varepsilon_n$ be independent random signs taking values ± 1 with probability $1/2$. Since $\varepsilon_i \in [-1, 1]$ and $\mathbb{E}\varepsilon_i = 0$, Theorem 2.10 gives, for $t \geq 0$,

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n \varepsilon_i\right| \geq t\right) \leq 2 \exp\left(-\frac{nt^2}{8}\right).$$

Equivalently, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \right| \leq \sqrt{\frac{8 \log(2/\delta)}{n}}.$$

For comparison, Chebyshev inequality only yields $\mathbb{P}(|\frac{1}{n} \sum_{i=1}^n \varepsilon_i| \geq t) \leq 1/(nt^2)$.

2.4 The sub-Gaussian norm

The MGF definition is convenient for sums, but it is sometimes useful to work with a norm that measures the “size” of tails. The standard choice is the Orlicz ψ_2 norm.

Definition 2.12 (Sub-Gaussian norm). For a random variable X (not necessarily centered), define its sub-Gaussian norm by

$$\|X\|_{\psi_2} = \inf \left\{ t > 0 : \mathbb{E} \exp\left(\frac{X^2}{t^2}\right) \leq 2 \right\}.$$

It is standard (and can be checked directly) that $X \mapsto \|X\|_{\psi_2}$ defines a norm on random variables modulo almost sure equality; see [3, Section 2.5].

Proposition 2.13 (Equivalent characterizations). *There exist absolute constants $c, C > 0$ such that for any random variable X :*

1. for all $t \geq 0$, $\mathbb{P}(|X| \geq t) \leq 2 \exp(-ct^2/\|X\|_{\psi_2}^2)$;
2. $\mathbb{E} \exp(X^2/\|X\|_{\psi_2}^2) \leq 2$;
3. for all $p \geq 1$, $\|X\|_{L_p} \leq C\|X\|_{\psi_2} \sqrt{p}$;
4. if $\mathbb{E}X = 0$, then for all $\lambda \in \mathbf{R}$, $\mathbb{E} \exp(\lambda X) \leq \exp(C\lambda^2\|X\|_{\psi_2}^2)$.

Remark 2.14. Items (1)–(3) do not require $\mathbb{E}X = 0$. Proofs and sharper constants can be found in [3, Proposition 2.5.2].

Centering does not significantly change the ψ_2 size. Indeed, if $\|X\|_{\psi_2} < \infty$, then $\|X - \mathbb{E}X\|_{\psi_2} \leq 2\|X\|_{\psi_2}$. Indeed, for any $t > \|X\|_{\psi_2}$ we have $\mathbb{E} \exp(X^2/t^2) \leq 2$, and Jensen gives $\exp((\mathbb{E}X)^2/t^2) \leq \mathbb{E} \exp(X^2/t^2) \leq 2$, hence $\|\mathbb{E}X\|_{\psi_2} = |\mathbb{E}X|/\sqrt{\log 2} \leq t$. Therefore $\|X - \mathbb{E}X\|_{\psi_2} \leq \|X\|_{\psi_2} + \|\mathbb{E}X\|_{\psi_2} \leq \|X\|_{\psi_2} + t$, and letting $t \rightarrow \|X\|_{\psi_2}$ yields the claim.

Example 2.15 (Bounded random variables). If $|X| \leq M$ almost surely, then $\|X\|_{\psi_2} \leq M/\sqrt{\log 2}$. Indeed, $\exp(X^2/t^2) \leq \exp(M^2/t^2)$ almost surely, so choosing $t = M/\sqrt{\log 2}$ makes $\mathbb{E} \exp(X^2/t^2) \leq 2$.

The next proposition is the ψ_2 -norm analogue of the variance additivity identity. It says that independent sub-Gaussian variables combine in an “ ℓ_2 ” fashion.

Proposition 2.16 (Sums of independent sub-Gaussians). *There exists an absolute constant $C > 0$ such that if $X_1, \dots, X_n$ are independent mean-zero random variables, then*

$$\left\| \sum_{i=1}^n X_i \right\|_{\psi_2}^2 \leq C \sum_{i=1}^n \|X_i\|_{\psi_2}^2.$$

Proof. By Theorem 2.13(4), for each i and all $\lambda \in \mathbf{R}$,

$$\mathbb{E} e^{\lambda X_i} \leq \exp(C_0 \lambda^2 \|X_i\|_{\psi_2}^2)$$

for an absolute constant C_0 . Using independence and multiplying MGFs,

$$\mathbb{E} \exp\left(\lambda \sum_{i=1}^n X_i\right) = \prod_{i=1}^n \mathbb{E} e^{\lambda X_i} \leq \exp\left(C_0 \lambda^2 \sum_{i=1}^n \|X_i\|_{\psi_2}^2\right).$$

Apply Theorem 2.5 ((1) to (4)) and use the definition of $\|\cdot\|_{\psi_2}$ to conclude the proof. $\square$

As a corollary, we have the following form of Hoeffding's inequality.

Corollary 2.17 (Hoeffding's inequality in terms of ψ_2 norms). *Let $X_1, \dots, X_n$ be independent random variables with $\max_i \|X_i\|_{\psi_2} < \infty$. Then for any $t \geq 0$,*

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mathbb{E} X_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{ct^2}{\sum_{i=1}^n \|X_i\|_{\psi_2}^2}\right),$$

where $c > 0$ is an absolute constant.

2.5 Khintchine-type bounds

The sub-Gaussian norm controls moments, and this leads to a useful inequality for linear combinations. In particular, for independent sub-Gaussian summands the natural geometry is Euclidean: the size of $\sum_i a_i X_i$ is governed by $\|a\|_2$.

Proposition 2.18 (Khintchine inequality for sub-Gaussian variables). *Let $X_1, \dots, X_n$ be independent mean-zero sub-Gaussian random variables such that $\text{Var}(X_i) = 1$ for all i . Let $a = (a_1, \dots, a_n) \in \mathbf{R}^n$ and $p \geq 2$. Then*

$$\left(\sum_{i=1}^n a_i^2\right)^{1/2} \leq \left\|\sum_{i=1}^n a_i X_i\right\|_{L_p} \leq C K \sqrt{p} \left(\sum_{i=1}^n a_i^2\right)^{1/2},$$

where $K = \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}$ and $C > 0$ is an absolute constant.

Proof. For the lower bound, $\|\cdot\|_{L_p} \geq \|\cdot\|_{L_2}$ for $p \geq 2$, hence

$$\begin{aligned} \left\|\sum_{i=1}^n a_i X_i\right\|_{L_p} &\geq \left(\mathbb{E} \left|\sum_{i=1}^n a_i X_i\right|^2\right)^{1/2} = \sqrt{\text{Var}\left(\sum_{i=1}^n a_i X_i\right)} \\ &= \left(\sum_{i=1}^n a_i^2 \text{Var}(X_i)\right)^{1/2} = \left(\sum_{i=1}^n a_i^2\right)^{1/2}. \end{aligned}$$

For the upper bound, Theorem 2.13(3) gives $\|Y\|_{L_p} \leq C_1 \sqrt{p} \|Y\|_{\psi_2}$ for an absolute constant C_1 (see also [3, Proposition 2.6.1]). Apply this to $Y = \sum_{i=1}^n a_i X_i$ and use Theorem 2.16:

$$\begin{aligned} \left\| \sum_{i=1}^n a_i X_i \right\|_{L_p} &\leq C_1 \sqrt{p} \left\| \sum_{i=1}^n a_i X_i \right\|_{\psi_2} \\ &\leq C_2 \sqrt{p} \left(\sum_{i=1}^n \|a_i X_i\|_{\psi_2}^2 \right)^{1/2} \\ &= C_2 \sqrt{p} \left(\sum_{i=1}^n a_i^2 \|X_i\|_{\psi_2}^2 \right)^{1/2} \leq C_2 K \sqrt{p} \left(\sum_{i=1}^n a_i^2 \right)^{1/2}. \end{aligned}$$

□

Example 2.19 (Random signs). Let $\varepsilon_1, \dots, \varepsilon_n$ be independent random signs. By Theorem 2.15, $\|\varepsilon_i\|_{\psi_2} \leq 1/\sqrt{\log 2}$. Thus Theorem 2.18 yields, for $p \geq 2$,

$$\left\| \sum_{i=1}^n a_i \varepsilon_i \right\|_{L_p} \leq C \sqrt{p} \left(\sum_{i=1}^n a_i^2 \right)^{1/2}.$$

In particular, taking $p = 2$ and expanding the square gives the exact identity $\mathbb{E}(\sum_i a_i \varepsilon_i)^2 = \sum_i a_i^2$.

3 Sub-exponential random variables

Sub-Gaussian tails are not the whole story. A common and very important class of random variables has tails closer to $\exp(-ct)$, which is still fast but weaker than $\exp(-ct^2)$. This class naturally appears when we look at quadratic functionals (squares, energies) and products.

Example 3.1 (A centered χ^2 variable is not sub-Gaussian). Let $Z \sim \mathcal{N}(0, 1)$ and define $X = Z^2 - 1$ so that $\mathbb{E}X = 0$. Then, for $\lambda < 1/2$,

$$\mathbb{E}e^{\lambda X} = \mathbb{E}e^{\lambda(Z^2-1)} = e^{-\lambda} \mathbb{E}e^{\lambda Z^2} = e^{-\lambda} \frac{1}{\sqrt{1-2\lambda}},$$

and $\mathbb{E}e^{\lambda X} = +\infty$ for $\lambda \geq 1/2$. Thus X cannot satisfy a sub-Gaussian MGF bound for all $\lambda \in \mathbf{R}$.

3.1 The sub-exponential norm

Definition 3.2 (Sub-exponential norm). For a random variable X , define its sub-exponential norm by

$$\|X\|_{\psi_1} = \inf \left\{ t > 0 : \mathbb{E} \exp\left(\frac{|X|}{t}\right) \leq 2 \right\}.$$

Proposition 3.3 (Equivalent characterizations). *There exist absolute constants $c, C > 0$ such that for any random variable X :*

1. for all $t \geq 0$, $\mathbb{P}(|X| \geq t) \leq 2 \exp(-ct/\|X\|_{\psi_1})$;
2. $\mathbb{E} \exp(|X|/\|X\|_{\psi_1}) \leq 2$;
3. for all $p \geq 1$, $\|X\|_{L_p} \leq Cp \|X\|_{\psi_1}$;

4. if $\mathbb{E}X = 0$, then for all λ with $|\lambda| \leq c/\|X\|_{\psi_1}$,

$$\mathbb{E}e^{\lambda X} \leq \exp(C\lambda^2\|X\|_{\psi_1}^2).$$

The sub-exponential class contains the sub-Gaussian class.

Proposition 3.4. *Any sub-Gaussian random variable is sub-exponential, and $\|X\|_{\psi_1} \lesssim \|X\|_{\psi_2}$.*

It is often convenient to remember that taking squares converts ψ_2 into ψ_1 .

Proposition 3.5. *For any random variable X ,*

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2.$$

Proof. By definition,

$$\|X^2\|_{\psi_1} = \inf \left\{ t > 0 : \mathbb{E} \exp\left(\frac{X^2}{t}\right) \leq 2 \right\}.$$

Writing $t = s^2$ shows that the condition $\mathbb{E} \exp(X^2/t) \leq 2$ is equivalent to $\mathbb{E} \exp(X^2/s^2) \leq 2$. $\square$

Proposition 3.6. *If X and Y are random variables (not necessarily independent), then*

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}.$$

Proof. Let $a = \|X\|_{\psi_2}$ and $b = \|Y\|_{\psi_2}$. By Young inequality, $|uv| \leq \frac{u^2}{2} + \frac{v^2}{2}$ for all real u, v . With $u = X/a$ and $v = Y/b$, we obtain

$$\frac{|XY|}{ab} \leq \frac{X^2}{2a^2} + \frac{Y^2}{2b^2}.$$

Therefore,

$$\mathbb{E} \exp\left(\frac{|XY|}{ab}\right) \leq \mathbb{E} \left[\exp\left(\frac{X^2}{2a^2}\right) \exp\left(\frac{Y^2}{2b^2}\right) \right].$$

By Cauchy–Schwarz and the definition of $\|\cdot\|_{\psi_2}$,

$$\mathbb{E} \exp\left(\frac{|XY|}{ab}\right) \leq \left(\mathbb{E} \exp\left(\frac{X^2}{a^2}\right) \right)^{1/2} \left(\mathbb{E} \exp\left(\frac{Y^2}{b^2}\right) \right)^{1/2} \leq \sqrt{2} \cdot \sqrt{2} = 2.$$

Thus $\|XY\|_{\psi_1} \leq ab$ by Theorem 3.2. $\square$

3.2 Examples

It is useful to keep a few canonical sub-exponential distributions in mind. The following examples can be verified directly from Theorem 3.2.

Example 3.7 (Exponential random variable). Let $Y \sim \text{Exp}(\lambda)$ with rate $\lambda > 0$, i.e. $\mathbb{P}(Y \geq y) = e^{-\lambda y}$ for $y \geq 0$. Then Y is sub-exponential and

$$\|Y\|_{\psi_1} = \frac{2}{\lambda}.$$

Indeed, since $Y \geq 0$,

$$\mathbb{E} \exp\left(\frac{|Y|}{t}\right) = \mathbb{E} \exp\left(\frac{Y}{t}\right) = \frac{\lambda}{\lambda - 1/t}, \quad t > \frac{1}{\lambda}.$$

The condition $\mathbb{E} \exp(Y/t) \leq 2$ is equivalent to $t \geq 2/\lambda$, which proves the claim.

Example 3.8 (Laplace distribution). Let X have the Laplace (double exponential) distribution with density $\frac{1}{2b}e^{-|x|/b}$, $b > 0$. Then $|X|$ is exponential with rate $1/b$, and therefore

$$\|X\|_{\psi_1} = \| |X| \|_{\psi_1} = 2b.$$

Example 3.9 (Squares and products). If $Z \sim \mathcal{N}(0, 1)$, then Z is sub-Gaussian and hence Z^2 is sub-exponential by Theorem 3.5. More generally, if X and Y are sub-Gaussian, then the product XY is sub-exponential by Theorem 3.6. These two operations (squaring and taking products) are a common way sub-exponential variables enter high-dimensional probability.

3.3 Bernstein inequality

We now turn to Bernstein-type inequalities. The ψ_1 form in Theorem 3.10 is convenient because it only assumes sub-exponential tails and it automatically interpolates between a *quadratic* regime (typical of Gaussian behavior) and a *linear* regime (typical of purely exponential tails). In many applications, however, it is also helpful to have a version that makes the role of the *variance* explicit. We record both viewpoints and then explain how to translate tail bounds into high-probability statements.

Theorem 3.10 (Bernstein's inequality (ψ_1 form)). *Let $X_1, \dots, X_n$ be independent mean-zero sub-exponential random variables. Then for all $t \geq 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-c \min\left(\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_{1 \leq i \leq n} \|X_i\|_{\psi_1}}\right)\right),$$

and therefore, by symmetry and the union bound,

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left(-c \min\left(\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_{1 \leq i \leq n} \|X_i\|_{\psi_1}}\right)\right),$$

where $c > 0$ is an absolute constant.

Proof. This is [3, Theorem 2.8.1]. The proof applies Chernoff's method to $\exp(\lambda \sum_{i=1}^n X_i)$, bounds the MGF using sub-exponentiality, and optimizes over λ . $\square$

A variance-aware bound for Bernoulli. For two-point distributions one can sometimes obtain MGF bounds that adapt sharply to the parameter. A useful example is the following inequality.

Lemma 3.11 (Kearns–Saul inequality). *Let $p \in (0, 1)$ and $\lambda \in \mathbf{R}$. Then*

$$p \exp(\lambda(1-p)) + (1-p) \exp(-\lambda p) \leq \exp\left(\lambda^2 \frac{1-2p}{4 \log\left(\frac{1-p}{p}\right)}\right).$$

(The right-hand side is interpreted by continuity at $p = \frac{1}{2}$.)

Example 3.12 (Centered Bernoulli). Let $Y \sim \text{Ber}(p)$ and set $X = Y - p$. Then $\mathbb{E}X = 0$ and

$$\text{Var}(X) = p(1-p).$$

By Theorem 3.11,

$$\mathbb{E}e^{\lambda X} = p e^{\lambda(1-p)} + (1-p) e^{-\lambda p} \leq \exp\left(\lambda^2 \frac{1-2p}{4 \log\left(\frac{1-p}{p}\right)}\right).$$

When p is close to 0 or 1, the variance $p(1-p)$ is small and the bound becomes correspondingly tighter; at $p = \frac{1}{2}$ it matches the familiar proxy variance $1/4$.

3.4 A variance form of sub-exponentiality

Besides the norm-based definition of sub-exponentiality, it is often convenient to work with a two-parameter MGF bound. This separates a *variance scale* and a *tail scale*.

Definition 3.13 ((σ^2, b) -sub-exponential). A mean-zero random variable X is sub-exponential with parameters (σ^2, b) if for all λ with $|\lambda| < 1/b$,

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}\right).$$

A simple sufficient condition for Theorem 3.13 is a moment growth assumption of Bernstein type.

Definition 3.14 (Bernstein moment condition). Let X be a random variable with mean $\mu = \mathbb{E}X$ and variance $\sigma^2 = \text{Var}(X)$. We say that X satisfies Bernstein's moment condition with parameter b if for every integer $k \geq 2$,

$$|\mathbb{E}[(X - \mu)^k]| \leq \frac{1}{2} k! \sigma^2 b^{k-2}.$$

Boundedness is a clean way to ensure Theorem 3.14, though it is far from necessary.

Proposition 3.15. Assume that $|X - \mu| \leq B$ almost surely. Then X satisfies Bernstein's moment condition with parameter $b = B/3$.

Proof. For $k \geq 2$,

$$|\mathbb{E}[(X - \mu)^k]| \leq \mathbb{E}[(X - \mu)^2 |X - \mu|^{k-2}] \leq \mathbb{E}[(X - \mu)^2] B^{k-2} = \sigma^2 B^{k-2}.$$

For $k \geq 3$, note that $k! \geq 2 \cdot 3^{k-2}$, hence

$$\sigma^2 B^{k-2} \leq \frac{1}{2} k! \sigma^2 (B/3)^{k-2}.$$

This yields the claim (and $k = 2$ is immediate). □

Lemma 3.16. Let X be mean-zero with variance σ^2 . If X satisfies Bernstein's moment condition with parameter b , then X is (σ^2, b) -sub-exponential in the sense of Theorem 3.13.

Proof. Fix $|\lambda| < 1/b$ and expand the MGF into a Taylor series:

$$\mathbb{E}e^{\lambda X} = 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{\lambda^k \mathbb{E}[X^k]}{k!}.$$

Using Theorem 3.14 (with $\mu = 0$) and taking absolute values,

$$\mathbb{E}e^{\lambda X} \leq 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{|\lambda|^k}{k!} \cdot \frac{1}{2} k! \sigma^2 b^{k-2} = 1 + \frac{\lambda^2 \sigma^2}{2} \left(1 + \sum_{k=3}^{\infty} (b|\lambda|)^{k-2}\right).$$

The geometric series sums to $\sum_{k=3}^{\infty} (b|\lambda|)^{k-2} = \frac{b|\lambda|}{1-b|\lambda|}$, so

$$\mathbb{E}e^{\lambda X} \leq 1 + \frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}.$$

Finally, $1 + u \leq e^u$ for all $u \in \mathbf{R}$, hence

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}\right),$$

which is exactly Theorem 3.13. □

We now state the corresponding variance form of Bernstein's inequality.

Theorem 3.17 (Bernstein's inequality (variance form)). *Let $X_1, \dots, X_n$ be independent random variables with means $\mu_i = \mathbb{E}X_i$ and variances $\sigma_i^2 = \text{Var}(X_i)$. Assume that each centered variable $X_i - \mu_i$ satisfies Bernstein's moment condition with parameter b . Then for all $t \geq 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n (X_i - \mu_i) \geq t\right) \leq \exp\left(-\frac{t^2}{2(\sum_{i=1}^n \sigma_i^2 + bt)}\right),$$

and hence, by symmetry and the union bound,

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mu_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2(\sum_{i=1}^n \sigma_i^2 + bt)}\right).$$

Proof. This follows from Chernoff's method combined with Theorem 3.16; see, e.g., [4, Proposition 2.14] for details. $\square$

3.5 From tail bounds to high-probability bounds

A tail bound becomes a high-probability statement by solving for the deviation level. For example, if we know that

$$\mathbb{P}(S \geq t) \leq e^{-\psi(t)} \quad (t \geq 0)$$

for some increasing function ψ , then setting $\delta = e^{-\psi(t)}$ yields the equivalent form: with probability at least $1 - \delta$, one has $S < \psi^{-1}(\log(1/\delta))$. We will use this translation repeatedly.

3.6 Comparing Hoeffding and Bernstein on a centered Bernoulli

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p) - p$ and set $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Since $-p \leq X_i \leq 1 - p$, Hoeffding's inequality gives, for all $t \geq 0$,

$$\mathbb{P}(|\bar{X}_n| \geq t) \leq 2 \exp(-2nt^2). \quad (2)$$

Equivalently, with probability at least $1 - \delta$,

$$|\bar{X}_n| \leq \sqrt{\frac{\log(2/\delta)}{2n}}. \quad (3)$$

This bound depends on n and δ , but not on p .

Bernstein's inequality, on the other hand, incorporates the variance $\text{Var}(X_i) = p(1 - p)$. Moreover, since $|X_i| \leq 1$, Theorem 3.15 yields Bernstein's moment condition with $b = 1/3$. Applying Theorem 3.17 gives

$$\mathbb{P}(|\bar{X}_n| \geq t) \leq 2 \exp\left(-\frac{nt^2}{2p(1-p) + 2t/3}\right). \quad (4)$$

Solving for t in terms of δ yields: with probability at least $1 - \delta$,

$$|\bar{X}_n| \leq \sqrt{\left(\frac{\log(2/\delta)}{3n}\right)^2 + \frac{2p(1-p)\log(2/\delta)}{n}} + \frac{\log(2/\delta)}{3n}. \quad (5)$$

Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, one can further simplify this to

$$|\bar{X}_n| \leq \sqrt{\frac{2p(1-p)\log(2/\delta)}{n}} + \frac{2\log(2/\delta)}{3n}.$$

The comparison is intuitive: when p is close to $1/2$, the variance is large and Hoeffding can be tighter; when p is close to 0 or 1, the variance is small and Bernstein becomes preferable.

3.7 Another application: relative deviations for a binomial

It is sometimes more natural to control deviations relative to the mean. Let $X \sim \text{Bin}(n, p)$, so $X = \sum_{i=1}^n Y_i$ with $Y_i \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p)$ and $\mathbb{E}X = np$. Applying Theorem 3.17 to $Y_i - p$ (where $|Y_i - p| \leq 1$ and $\text{Var}(Y_i - p) = p(1 - p)$), we obtain: for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$|X - \mathbb{E}X| \leq \sqrt{2np(1-p)\log(2/\delta)} + \frac{2}{3}\log(2/\delta).$$

To relate this to $\mathbb{E}X$, use $\sqrt{ab} \leq \frac{1}{2}(\gamma a + b/\gamma)$ for $a, b \geq 0$ and $\gamma > 0$. With $a = 2\mathbb{E}X$ and $b = (1-p)\log(2/\delta)$, we get

$$\sqrt{2np(1-p)\log(2/\delta)} \leq \gamma \mathbb{E}X + \frac{(1-p)\log(2/\delta)}{2\gamma}.$$

Hence, with probability at least $1 - \delta$,

$$|X - \mathbb{E}X| \leq \gamma \mathbb{E}X + \log(2/\delta) \left(\frac{1-p}{2\gamma} + \frac{2}{3} \right).$$

If the rightmost term is at most $\gamma \mathbb{E}X$, then $|X - \mathbb{E}X| \leq 2\gamma \mathbb{E}X$, i.e.

$$(1 - 2\gamma)\mathbb{E}X \leq X \leq (1 + 2\gamma)\mathbb{E}X.$$

This is a typical route from an additive Bernstein bound to a multiplicative Chernoff-type statement. There is, however, a classical bound tailored specifically to sums of Bernoulli random variables, usually called a *multiplicative Chernoff bound*, with slightly sharper constants.

Theorem 3.18 (Multiplicative Chernoff bounds). *Let $Y_1, \dots, Y_n$ be independent Bernoulli random variables with $\mathbb{P}(Y_i = 1) = p_i$. Set $X = \sum_{i=1}^n Y_i$ and $\mu = \mathbb{E}X = \sum_{i=1}^n p_i$. Then, it holds that*

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) \leq \exp\left(-\frac{\varepsilon^2}{2 + \varepsilon}\mu\right), \quad 0 \leq \varepsilon,$$

and

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) \leq \exp\left(-\frac{\varepsilon^2}{2}\mu\right), \quad 0 \leq \varepsilon \leq 1.$$

Proof. For the upper tail, fix $\lambda > 0$ and apply Chernoff's method:

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) = \mathbb{P}\left(e^{\lambda X} \geq e^{\lambda(1+\varepsilon)\mu}\right) \leq \exp(-\lambda(1 + \varepsilon)\mu) \mathbb{E}e^{\lambda X}.$$

Since $Y_i \in \{0, 1\}$,

$$\mathbb{E}e^{\lambda Y_i} = (1 - p_i) + p_i e^\lambda = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1)),$$

and therefore, by independence,

$$\mathbb{E}e^{\lambda X} = \prod_{i=1}^n \mathbb{E}e^{\lambda Y_i} \leq \exp\left(\sum_{i=1}^n p_i(e^\lambda - 1)\right) = \exp(\mu(e^\lambda - 1)).$$

Hence

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) \leq \exp\left(\mu(e^\lambda - 1 - \lambda(1 + \varepsilon))\right).$$

Choosing $\lambda = \log(1 + \varepsilon)$ gives

$$\mathbb{P}(X \geq (1 + \varepsilon)\mu) \leq \exp\left(-\mu((1 + \varepsilon)\log(1 + \varepsilon) - \varepsilon)\right).$$

It remains to use the standard inequality, valid for all $\varepsilon \geq 0$,

$$(1 + \varepsilon) \log(1 + \varepsilon) - \varepsilon \geq \frac{\varepsilon^2}{2 + \varepsilon}.$$

One quick way to check it is to consider $h(u) = (1 + u) \log(1 + u) - u - \frac{u^2}{2+u}$, note that $h(0) = 0$, and compute $h'(u) = \log(1 + u) - \frac{u(4+u)}{(2+u)^2} \geq 0$ using $\log(1 + u) \geq \frac{2u}{2+u}$ for $u \geq 0$. This proves the first bound.

For the lower tail, apply the same argument to $e^{-\lambda X}$ with $\lambda > 0$:

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) = \mathbb{P}\left(e^{-\lambda X} \geq e^{-\lambda(1-\varepsilon)\mu}\right) \leq \exp(\lambda(1 - \varepsilon)\mu) \mathbb{E}e^{-\lambda X}.$$

As before, $\mathbb{E}e^{-\lambda Y_i} = 1 + p_i(e^{-\lambda} - 1) \leq \exp(p_i(e^{-\lambda} - 1))$, hence $\mathbb{E}e^{-\lambda X} \leq \exp(\mu(e^{-\lambda} - 1))$ and therefore

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) \leq \exp\left(\mu(e^{-\lambda} - 1 + \lambda(1 - \varepsilon))\right).$$

Choosing $\lambda = -\log(1 - \varepsilon)$ (valid for $0 \leq \varepsilon < 1$) yields

$$\mathbb{P}(X \leq (1 - \varepsilon)\mu) \leq \exp\left(-\mu((1 - \varepsilon) \log(1 - \varepsilon) + \varepsilon)\right).$$

Finally, for $0 \leq \varepsilon \leq 1$ one has

$$(1 - \varepsilon) \log(1 - \varepsilon) + \varepsilon \geq \frac{\varepsilon^2}{2},$$

which follows by considering $g(u) = (1 - u) \log(1 - u) + u - \frac{u^2}{2}$ on $[0, 1]$, noting $g(0) = 0$ and $g'(u) = -\log(1 - u) - u \geq 0$ because $-\log(1 - u) \geq u$. This gives the second bound. $\square$

Remark 3.19. Compared to the Bernstein-based relative deviation bound above, Theorem 3.18 is specialized to Bernoulli sums and typically gives sharper constants in the multiplicative regime.

3.8 Application: Statistical learning theory

This section illustrates how the concentration tools developed so far translate into simple learning guarantees. We begin with the classical finite-class setting, and then record a few useful inequalities for maxima of random variables. We conclude with McDiarmid's inequality, which provides concentration for functions of independent samples under a bounded-differences condition.

3.9 A finite hypothesis class in the realizable case

We consider a basic binary classification model. Let $\mathcal{X}$ be an instance space and let $\mathcal{F}$ be a finite class of functions $f : \mathcal{X} \rightarrow \{0, 1\}$, with $M = |\mathcal{F}|$. Assume there is an unknown “true” classifier $f^* \in \mathcal{F}$. We observe the labeled sample

$$(X_1, f^*(X_1)), \dots, (X_n, f^*(X_n)),$$

where $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{P}$ for an unknown distribution $\mathcal{P}$ on $\mathcal{X}$.

Definition 3.20 (Risk). The *risk* of $f \in \mathcal{F}$ is the probability of misclassification on a fresh draw from $\mathcal{P}$:

$$R(f) = \mathbb{P}_{X \sim \mathcal{P}}(f(X) \neq f^*(X)).$$

Definition 3.21 (Empirical risk). The *empirical risk* of $f \in \mathcal{F}$ is the fraction of mistakes on the observed sample:

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(f(X_i) \neq f^*(X_i)).$$

For a fixed f , the indicators $\mathbf{1}(f(X_i) \neq f^*(X_i))$ are i.i.d. Bernoulli random variables with mean $R(f)$. Thus $R_n(f)$ is a rescaled binomial variable, and Bernstein's inequality yields a variance-sensitive deviation bound.

Proposition 3.22 (A Bernstein bound for the empirical risk). *Fix $f \in \mathcal{F}$ and let $\delta \in (0, 1)$. With probability at least $1 - \delta$,*

$$R(f) - R_n(f) \leq \sqrt{\frac{2R(f)(1 - R(f)) \log(1/\delta)}{n}} + \frac{2 \log(1/\delta)}{3n} \leq \sqrt{\frac{2R(f) \log(1/\delta)}{n}} + \frac{2 \log(1/\delta)}{3n}. \quad (6)$$

Proof. Let $U_i = \mathbf{1}(f(X_i) \neq f^*(X_i))$, so $\mathbb{E}U_i = R(f)$ and $\text{Var}(U_i) = R(f)(1 - R(f))$. Applying Theorem 3.17 to the centered variables $U_i - \mathbb{E}U_i$ (noting $|U_i - \mathbb{E}U_i| \leq 1$) gives, for all $t \geq 0$,

$$\mathbb{P}(R(f) - R_n(f) \geq t) = \mathbb{P}\left(\sum_{i=1}^n (R(f) - U_i) \geq nt\right) \leq \exp\left(-\frac{nt^2}{2R(f)(1 - R(f)) + \frac{2}{3}t}\right).$$

Solving the inequality $\frac{nt^2}{2R(f)(1 - R(f)) + \frac{2}{3}t} \geq \log(1/\delta)$ for t yields the first bound in (6). The second bound follows from $1 - R(f) \leq 1$. $\square$

A standard way to make (6) uniform over the class is to apply the union bound.

Corollary 3.23 (Uniform control over a finite class). *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$, simultaneously for all $f \in \mathcal{F}$,*

$$R(f) - R_n(f) \leq \sqrt{\frac{2R(f) \log(M/\delta)}{n}} + \frac{2 \log(M/\delta)}{3n}.$$

Proof. Apply Theorem 3.22 with δ/M and take the union bound over the M functions in $\mathcal{F}$. $\square$

Now define the empirical risk minimizer

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} R_n(f).$$

In the realizable setting, $f^* \in \mathcal{F}$ satisfies $R_n(f^*) = 0$, so we may choose $\hat{f}$ so that $R_n(\hat{f}) = 0$. Applying Theorem 3.23 to $\hat{f}$ yields, with probability at least $1 - \delta$,

$$R(\hat{f}) \leq \sqrt{\frac{2R(\hat{f}) \log(M/\delta)}{n}} + \frac{2 \log(M/\delta)}{3n}.$$

This inequality is self-bounding and implies $R(\hat{f}) \lesssim \frac{\log(M/\delta)}{n}$. A simple (loose but convenient) way to see this is the following: set $a = \log(M/\delta)/n$. If $R(\hat{f}) > 10a$, then

$$\sqrt{2R(\hat{f})a} < \sqrt{\frac{2}{10}} R(\hat{f}) \quad \text{and} \quad \frac{2a}{3} < \frac{1}{15} R(\hat{f}),$$

so the right-hand side is strictly smaller than $R(\hat{f})$, a contradiction. Hence $R(\hat{f}) \leq 10a$.

There is also a very direct argument that gives the same dependence (and often better constants).

Proposition 3.24 (A direct counting argument). *Fix $\delta \in (0, 1)$ and let $\hat{f} \in \mathcal{F}$ satisfy $R_n(\hat{f}) = 0$. Then, with probability at least $1 - \delta$,*

$$R(\hat{f}) \leq \frac{\log M + \log(1/\delta)}{n}.$$

Proof. Fix $\varepsilon \in (0, 1)$. By the union bound,

$$\mathbb{P}(\exists f \in \mathcal{F} : R(f) \geq \varepsilon, R_n(f) = 0) \leq \sum_{f \in \mathcal{F}} \mathbb{P}(R(f) \geq \varepsilon, R_n(f) = 0).$$

For a fixed f with $R(f) \geq \varepsilon$, each draw X_i is correctly classified with probability at most $1 - \varepsilon$, and the events are independent across i , so

$$\mathbb{P}(R_n(f) = 0) \leq (1 - \varepsilon)^n \leq e^{-\varepsilon n}.$$

Thus

$$\mathbb{P}(\exists f \in \mathcal{F} : R(f) \geq \varepsilon, R_n(f) = 0) \leq M e^{-\varepsilon n}.$$

Choose $\varepsilon = (\log M + \log(1/\delta))/n$, so that $M e^{-\varepsilon n} = \delta$. On the complementary event (of probability at least $1 - \delta$), every f with $R_n(f) = 0$ must satisfy $R(f) \leq \varepsilon$. In particular, $R(\hat{f}) \leq \varepsilon$. $\square$

3.10 Maximal inequalities

In many problems we need to control not only a single random quantity, but the maximum over a collection. The next bounds are elementary, but surprisingly useful.

Theorem 3.25 (Maximum of sub-Gaussian variables). *Let $X_1, \dots, X_n$ be mean-zero (not necessarily independent) random variables such that*

$$\mathbb{E} e^{\lambda X_i} \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right) \quad \text{for all } \lambda \in \mathbf{R} \text{ and all } i.$$

Then

$$\mathbb{E} \left[\max_{1 \leq i \leq n} X_i \right] \leq \sqrt{2\sigma^2 \log n}.$$

Proof. Fix $\lambda > 0$. Since $\log$ is concave, Jensen's inequality yields

$$\mathbb{E} \left[\max_i X_i \right] = \frac{1}{\lambda} \mathbb{E} \left[\log(e^{\lambda \max_i X_i}) \right] \leq \frac{1}{\lambda} \log \mathbb{E} \left[e^{\lambda \max_i X_i} \right] = \frac{1}{\lambda} \log \mathbb{E} \left[\max_i e^{\lambda X_i} \right].$$

Using $\max_i a_i \leq \sum_i a_i$ for $a_i \geq 0$ and linearity of expectation,

$$\mathbb{E} \left[\max_i e^{\lambda X_i} \right] \leq \mathbb{E} \left[\sum_{i=1}^n e^{\lambda X_i} \right] = \sum_{i=1}^n \mathbb{E} e^{\lambda X_i} \leq n \exp\left(\frac{\lambda^2 \sigma^2}{2}\right).$$

Therefore,

$$\mathbb{E} \left[\max_i X_i \right] \leq \frac{\log n}{\lambda} + \frac{\lambda \sigma^2}{2}.$$

Optimizing over $\lambda > 0$ gives the claim. $\square$

Theorem 3.26 (Maximum of sub-exponential variables). *Let $X_1, \dots, X_n$ be mean-zero (not necessarily independent) random variables such that for some $\sigma^2, b \geq 0$,*

$$\mathbb{E} e^{\lambda X_i} \leq \exp\left(\frac{\lambda^2 \sigma^2 / 2}{1 - b\lambda}\right) \quad \text{for all } 0 < \lambda < 1/b \text{ and all } i,$$

with the convention that $1/0 = \infty$. Then there exists an absolute constant $C > 0$ such that

$$\mathbb{E} \left[\max_{1 \leq i \leq n} X_i \right] \leq C(\sigma \sqrt{\log n} + b \log n).$$

In particular, for an absolute constant $C' > 0$,

$$\mathbb{E} \left[\max_{1 \leq i \leq n} X_i \right] \leq C' \left(\max_{1 \leq i \leq n} \|X_i\|_{\psi_1} \right) \log n.$$

Proof. As in the proof of Theorem 3.25, for any $0 < \lambda < 1/b$ we have

$$\mathbb{E} \left[\max_i X_i \right] \leq \frac{1}{\lambda} \log \left(\sum_{i=1}^n \mathbb{E} e^{\lambda X_i} \right) \leq \frac{\log n}{\lambda} + \frac{\lambda \sigma^2}{2(1 - b\lambda)}.$$

Choose

$$\lambda = \min \left\{ \frac{\sqrt{2 \log n}}{\sigma}, \frac{1}{2b} \right\},$$

with the conventions $1/0 = \infty$ and $\sqrt{2 \log n}/0 = \infty$. Then $b\lambda \leq 1/2$, so $(1 - b\lambda)^{-1} \leq 2$. Substituting this choice of λ into the previous display gives the stated bound up to an absolute constant.

For the final statement, recall that $\|X\|_{\psi_1}$ controls sub-exponential MGF bounds up to absolute constants: indeed, if $\|X\|_{\psi_1} \leq K$, then one may take $\sigma^2 \asymp K^2$ and $b \asymp K$ in the hypothesis above. $\square$

Proposition 3.27 (A crude L_p bound for maxima). *For any random variables $X_1, \dots, X_n$ and any $p \geq 1$,*

$$\mathbb{E} \left[\max_{1 \leq i \leq n} |X_i| \right] \leq n^{1/p} \max_{1 \leq i \leq n} \|X_i\|_{L_p}.$$

Proof. Since $x \mapsto x^{1/p}$ is concave on $\mathbf{R}_+$, Jensen's inequality gives

$$\mathbb{E} \left[\max_i |X_i| \right] = \mathbb{E} \left[\left(\max_i |X_i|^p \right)^{1/p} \right] \leq \left(\mathbb{E} \left[\max_i |X_i|^p \right] \right)^{1/p}.$$

Moreover, $\max_i |X_i|^p \leq \sum_{i=1}^n |X_i|^p$, so

$$\mathbb{E} \left[\max_i |X_i|^p \right] \leq \sum_{i=1}^n \mathbb{E} |X_i|^p \leq n \max_i \mathbb{E} |X_i|^p.$$

Taking p -th roots yields the claim. $\square$

The next proposition complements Theorem 3.27 by showing that maxima inherit sub-Gaussian or log price.

Proposition 3.28 (ψ -norms of a maximum). *Let $X_1, \dots, X_n$ be random variables and set*

$$M = \max_{1 \leq i \leq n} |X_i|.$$

Then

$$\|M\|_{\psi_2} \leq C \sqrt{\log(2n)} \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}, \quad \|M\|_{\psi_1} \leq C \log(2n) \max_{1 \leq i \leq n} \|X_i\|_{\psi_1},$$

where $C > 0$ is an absolute constant.

Proof. We prove the ψ_2 statement; the ψ_1 case is analogous. Let $K = \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}$ and let $p \geq 1$. By the standard moment growth of sub-Gaussian random variables, for every i ,

$$\|X_i\|_{L_p} \leq C_1 \sqrt{p} K$$

for an absolute constant $C_1 > 0$.

Next, observe that the same argument used in the proof of Theorem 3.27 gives the stronger inequality

$$\|M\|_{L_p} = \left(\mathbb{E} \left[\max_i |X_i|^p \right] \right)^{1/p} \leq \left(\sum_{i=1}^n \mathbb{E} |X_i|^p \right)^{1/p} \leq n^{1/p} \max_{1 \leq i \leq n} \|X_i\|_{L_p}.$$

Combining with the moment bound above yields

$$\|M\|_{L_p} \leq C_1 n^{1/p} \sqrt{p} K.$$

Now set $p_0 = \lceil \log(2n) \rceil$. If $1 \leq p \leq p_0$, then by monotonicity of L_p norms,

$$\|M\|_{L_p} \leq \|M\|_{L_{p_0}} \leq C_1 n^{1/p_0} \sqrt{p_0} K \leq C_2 \sqrt{\log(2n)} K$$

for an absolute constant $C_2 > 0$, since $n^{1/p_0} \leq e$ and $p_0 \asymp \log(2n)$. If $p \geq p_0$, then still $n^{1/p} \leq n^{1/p_0} \leq e$, and hence

$$\|M\|_{L_p} \leq C_3 \sqrt{p} K$$

for an absolute constant $C_3 > 0$.

Finally, using the standard equivalence between ψ_2 norms and moment growth,

$$\|M\|_{\psi_2} \asymp \sup_{p \geq 1} p^{-1/2} \|M\|_{L_p},$$

we conclude that $\|M\|_{\psi_2} \leq C \sqrt{\log(2n)} K$. The ψ_1 bound follows similarly, using $\|X_i\|_{L_p} \leq C' p \|X_i\|_{\psi_1}$ and the equivalence $\|M\|_{\psi_1} \asymp \sup_{p \geq 1} p^{-1} \|M\|_{L_p}$. $\square$

Additional material

Sometimes we would like to keep a “variance term” while still using a soft (non-deterministic) notion of boundedness. A convenient way to do this is to truncate the summands and pay separately for the event that at least one summand is unusually large.

Proposition 3.29 (Bernstein with a ψ_1 -random bound). *Let $Y_1, \dots, Y_m$ be i.i.d. random variables with $\mathbb{E}Y_1 = 0$, and set*

$$\sigma^2 = \mathbb{E}Y_1^2, \quad M = \max_{1 \leq i \leq m} |Y_i|, \quad K = \|M\|_{\psi_1}.$$

Then there exists an absolute constant $c > 0$ such that for every $u > 0$,

$$\mathbb{P} \left(\frac{1}{m} \sum_{i=1}^m Y_i \geq u \right) \leq 2 \exp \left(-c m \left(\frac{u^2}{\sigma^2} \wedge \frac{u}{K} \right) \right).$$

Proof. If $\sigma^2 = 0$, then $Y_1 = 0$ almost surely and the claim is immediate. If $\sigma^2 = +\infty$ or $K = +\infty$, then the right-hand side equals 2 and the claim is again trivial. Assume henceforth that $0 < \sigma^2 < \infty$ and $0 < K < \infty$.

Let $t = mu$ and write $S = \sum_{i=1}^m Y_i$, so that

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m Y_i \geq u\right) = \mathbb{P}(S \geq t).$$

Fix a truncation level

$$\rho = 4K,$$

and define

$$X_i = Y_i \mathbf{1}\{|Y_i| \leq \rho\}, \quad Z_i = Y_i - X_i = Y_i \mathbf{1}\{|Y_i| > \rho\}.$$

Since $Y_i = X_i + Z_i$ and $\mathbb{E}Y_i = 0$, we have

$$S = \sum_{i=1}^m (X_i - \mathbb{E}X_i) + \sum_{i=1}^m (Z_i - \mathbb{E}Z_i).$$

Therefore,

$$\mathbb{P}(S \geq t) \leq \mathbb{P}\left(\sum_{i=1}^m (X_i - \mathbb{E}X_i) \geq \frac{t}{2}\right) + \mathbb{P}\left(\sum_{i=1}^m (Z_i - \mathbb{E}Z_i) \geq \frac{t}{2}\right). \quad (7)$$

For the truncated sum, $|X_i| \leq \rho$ almost surely, hence $|X_i - \mathbb{E}X_i| \leq 2\rho$. Moreover,

$$\text{Var}(X_i) \leq \mathbb{E}[X_i^2] \leq \mathbb{E}[Y_i^2] = \sigma^2, \quad \text{so} \quad \sum_{i=1}^m \text{Var}(X_i) \leq m\sigma^2.$$

Bernstein's inequality for bounded, independent, mean-zero summands gives

$$\mathbb{P}\left(\sum_{i=1}^m (X_i - \mathbb{E}X_i) \geq \frac{t}{2}\right) \leq \exp\left(-c_1 \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{\rho}\right)\right)$$

for an absolute constant $c_1 > 0$. It remains to control the remainder sum. Define

$$q(s) = \mathbb{P}(M > s), \quad p(s) = \mathbb{P}(|Y_1| > s).$$

Since the Y_i are i.i.d. and $M = \max_{1 \leq i \leq m} |Y_i|$, we have

$$q(s) = 1 - (1 - p(s))^m, \quad s \geq 0.$$

By the definition of $K = \|M\|_{\psi_1}$, one has $\mathbb{E} \exp(M/K) \leq 2$, hence

$$q(s) = \mathbb{P}(M > s) \leq 2e^{-s/K}, \quad s \geq 0. \quad (8)$$

In particular, $q(\rho) \leq 2e^{-4} < \frac{1}{2}$, and since q is decreasing, $q(s) \leq \frac{1}{2}$ for all $s \geq \rho$. For such s ,

$$1 - q(s) = (1 - p(s))^m \leq e^{-mp(s)},$$

so

$$p(s) \leq -\frac{1}{m} \log(1 - q(s)) \leq \frac{q(s)}{\mathcal{P}(1 - q(s))} \leq \frac{2}{m} q(s).$$

Combining with (8) yields

$$p(s) \leq \frac{4}{m} e^{-s/K}, \quad s \geq \rho. \quad (9)$$

Fix $\lambda \in (0, \frac{1}{2K}]$. Since $Z_1 = Y_1 \mathbf{1}\{|Y_1| > \rho\}$,

$$\mathbb{E}e^{\lambda Z_1} = 1 + \mathbb{E}\left[(e^{\lambda Y_1} - 1)\mathbf{1}\{|Y_1| > \rho\}\right] \leq 1 + \mathbb{E}\left[(e^{\lambda Y_1} - 1)\mathbf{1}\{Y_1 > \rho\}\right],$$

because $e^{\lambda y} - 1 \leq 0$ when $y \leq 0$. Using the tail integration formula for the increasing function $s \mapsto e^{\lambda s} - 1$ on $[\rho, \infty)$,

$$\mathbb{E}\left[(e^{\lambda Y_1} - 1)\mathbf{1}\{Y_1 > \rho\}\right] = (e^{\lambda \rho} - 1)\mathbb{P}(Y_1 > \rho) + \int_{\rho}^{\infty} \lambda e^{\lambda s} \mathbb{P}(Y_1 > s) ds.$$

Since $\mathbb{P}(Y_1 > s) \leq p(s)$ and (9) holds for $s \geq \rho$,

$$\mathbb{E}\left[(e^{\lambda Y_1} - 1)\mathbf{1}\{Y_1 > \rho\}\right] \leq \frac{4}{m} \left[(e^{\lambda \rho} - 1)e^{-\rho/K} + \int_{\rho}^{\infty} \lambda e^{\lambda s} e^{-s/K} ds \right].$$

Using $e^x - 1 \leq xe^x$ and $\lambda \leq 1/(2K)$,

$$(e^{\lambda \rho} - 1)e^{-\rho/K} \leq \lambda \rho e^{-(1/K - \lambda)\rho} \leq \lambda \rho,$$

and

$$\int_{\rho}^{\infty} \lambda e^{\lambda s} e^{-s/K} ds = \frac{\lambda}{1/K - \lambda} e^{-(1/K - \lambda)\rho} \leq 2\lambda K.$$

Therefore,

$$\mathbb{E}e^{\lambda Z_1} \leq 1 + \frac{C}{m} \lambda(\rho + K) \leq 1 + \frac{C'}{m} \lambda K \leq \exp\left(\frac{C'}{m} \lambda K\right), \quad (10)$$

for absolute constants $C, C' > 0$, where we used $\rho = 4K$.

For the centering term, set $V = |Y_1|$. Then $|Z_1| = V \mathbf{1}\{V > \rho\}$ and the identity

$$\mathbb{E}[V \mathbf{1}\{V > \rho\}] = \rho \mathbb{P}(V > \rho) + \int_{\rho}^{\infty} \mathbb{P}(V > s) ds$$

gives

$$\mathbb{E}|Z_1| = \rho p(\rho) + \int_{\rho}^{\infty} p(s) ds \leq \frac{C''}{m} K$$

by (9) and $\rho = 4K$, for an absolute constant $C'' > 0$. Hence, for $\lambda \leq 1/(2K)$,

$$\mathbb{E}e^{\lambda(Z_1 - \mathbb{E}Z_1)} = e^{-\lambda \mathbb{E}Z_1} \mathbb{E}e^{\lambda Z_1} \leq \exp(\lambda \mathbb{E}|Z_1|) \mathbb{E}e^{\lambda Z_1} \leq \exp\left(\frac{C_0}{m} \lambda K\right)$$

for an absolute constant $C_0 > 0$, using (10). By independence,

$$\mathbb{E} \exp\left(\lambda \sum_{i=1}^m (Z_i - \mathbb{E}Z_i)\right) = \left(\mathbb{E}e^{\lambda(Z_1 - \mathbb{E}Z_1)}\right)^m \leq \exp(C_0 \lambda K).$$

Choose $\lambda = \frac{\log 2}{C_0 K}$ (which is at most $1/(2K)$ after increasing C_0 if needed). Then the last bound is ≤ 2 , and Markov's inequality yields

$$\mathbb{P}\left(\sum_{i=1}^m (Z_i - \mathbb{E}Z_i) \geq \frac{t}{2}\right) \leq e^{-\lambda t/2} \mathbb{E} \exp\left(\lambda \sum_{i=1}^m (Z_i - \mathbb{E}Z_i)\right) \leq 2 \exp\left(-c_2 \frac{t}{K}\right)$$

for an absolute constant $c_2 > 0$.

Insert the bounds into (7) and use $\rho = 4K$ to obtain

$$\mathbb{P}(S \geq t) \leq \exp\left(-c \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right) + 2 \exp\left(-c \frac{t}{K}\right) \leq 3 \exp\left(-c \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right)$$

for an absolute constant $c > 0$. Since $\mathbb{P}(S \geq t) \leq 1$, we may write the previous bound as

$$\mathbb{P}(S \geq t) \leq \min\left\{1, 3 \exp\left(-c \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right)\right\},$$

and this is dominated by

$$2 \exp\left(-c' \min\left(\frac{t^2}{m\sigma^2}, \frac{t}{K}\right)\right)$$

after possibly replacing c by a smaller absolute constant $c' > 0$. Finally, substituting $t = mu$ yields

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m Y_i \geq u\right) \leq 2 \exp\left(-c' m \left(\frac{u^2}{\sigma^2} \wedge \frac{u}{K}\right)\right),$$

as claimed. □

3.11 Bounded differences and McDiarmid's inequality

We now record a concentration inequality for functions of independent samples under a simple stability condition.

Definition 3.30 (Bounded differences). Let $\mathcal{X}$ be a set and let $g : \mathcal{X}^n \rightarrow \mathbf{R}$. We say that g satisfies the bounded differences property with constants $C_1, \dots, C_n$ if for every $i \in [n]$,

$$\sup_{\substack{x_1, \dots, x_n \in \mathcal{X} \\ x'_i \in \mathcal{X}}} \left| g(x_1, \dots, x_n) - g(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n) \right| \leq C_i.$$

Theorem 3.31 (McDiarmid's inequality). Let $X_1, \dots, X_n$ be independent random variables taking values in $\mathcal{X}$, and let $g : \mathcal{X}^n \rightarrow \mathbf{R}$ satisfy the bounded differences property with constants $C_1, \dots, C_n$. Then for every $t \geq 0$,

$$\mathbb{P}(g(X_1, \dots, X_n) - \mathbb{E}g(X_1, \dots, X_n) \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n C_i^2}\right).$$

Proof. Let $g = g(X_1, \dots, X_n)$ and define the Doob martingale differences

$$Y_i = \mathbb{E}[g \mid X_1, \dots, X_i] - \mathbb{E}[g \mid X_1, \dots, X_{i-1}], \quad i = 1, \dots, n,$$

so that $g - \mathbb{E}g = \sum_{i=1}^n Y_i$ and $\mathbb{E}[Y_i \mid X_1, \dots, X_{i-1}] = 0$.

For each i , define

$$h_i(x_1, \dots, x_i) = \mathbb{E}[g \mid X_1 = x_1, \dots, X_i = x_i].$$

By the bounded differences property,

$$\sup_{x_i, x'_i \in \mathcal{X}} \left| h_i(X_1, \dots, X_{i-1}, x_i) - h_i(X_1, \dots, X_{i-1}, x'_i) \right| \leq C_i,$$

and therefore Y_i , conditional on $X_1, \dots, X_{i-1}$, is supported on an interval of length at most C_i . Hoeffding's lemma (in its standard conditional form) implies that for every $\lambda \in \mathbf{R}$,

$$\mathbb{E}[e^{\lambda Y_i} \mid X_1, \dots, X_{i-1}] \leq \exp\left(\frac{\lambda^2 C_i^2}{8}\right).$$

Iterating conditional expectations then gives

$$\mathbb{E}e^{\lambda(g - \mathbb{E}g)} = \mathbb{E}e^{\lambda \sum_{i=1}^n Y_i} \leq \exp\left(\frac{\lambda^2}{8} \sum_{i=1}^n C_i^2\right).$$

Finally, Chernoff's method yields, for any $\lambda > 0$,

$$\mathbb{P}(g - \mathbb{E}g \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n C_i^2\right),$$

and optimizing over λ gives the stated bound. $\square$

Remark 3.32. As a sanity check, if $g(X_1, \dots, X_n) = \sum_{i=1}^n X_i$ with $X_i \in [0, 1]$, then $C_i = 1$ and Theorem 3.31 recovers Hoeffding's inequality.

3.12 Application of bounded differences: kernel density estimation

We briefly illustrate how bounded differences can be used to control the *random fluctuations* of an estimator around its mean. Throughout this section we focus on the one-dimensional setting for clarity.

Let $X_1, \dots, X_n$ be i.i.d. samples from an unknown density f on $\mathbf{R}$. Given a kernel $K : \mathbf{R} \rightarrow [0, \infty)$ with $\int_{\mathbf{R}} K(u) du = 1$ and a bandwidth $h > 0$, define the kernel density estimator

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{j=1}^n K\left(\frac{x - X_j}{h}\right).$$

A natural way to measure accuracy is the L^1 error

$$\|\hat{f}_n - f\|_1 = \int_{\mathbf{R}} |\hat{f}_n(x) - f(x)| dx.$$

Since f is unknown, this quantity cannot be evaluated directly from the data; instead, we study its expectation $\mathbb{E}\|\hat{f}_n - f\|_1$ and ask how strongly $\|\hat{f}_n - f\|_1$ concentrates around this mean.

For a deterministic input $(x_1, \dots, x_n) \in \mathbf{R}^n$, define

$$g(x_1, \dots, x_n) = \|\hat{f}_n - f\|_1,$$

where $\hat{f}_n$ is computed from the sample $(x_1, \dots, x_n)$. To apply Theorem 3.31, we estimate how much g can change if a single coordinate is modified.

Fix $i \in [n]$ and let $x'_i \in \mathbf{R}$ be another value (keeping the other coordinates fixed). Write $\hat{f}'_n$ for the

estimator constructed from $(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)$. Then, by the triangle inequality,

$$\begin{aligned}
& \left| g(x_1, \dots, x_n) - g(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n) \right| \\
& \leq \int_{\mathbf{R}} |\widehat{f}_n(x) - \widehat{f}'_n(x)| \, dx \\
& = \frac{1}{nh} \int_{\mathbf{R}} \left| K\left(\frac{x - x_i}{h}\right) - K\left(\frac{x - x'_i}{h}\right) \right| \, dx \\
& \leq \frac{1}{nh} \int_{\mathbf{R}} K\left(\frac{x - x_i}{h}\right) \, dx + \frac{1}{nh} \int_{\mathbf{R}} K\left(\frac{x - x'_i}{h}\right) \, dx \\
& = \frac{2}{n},
\end{aligned}$$

where in the last step we used the change of variables $u = (x - x_i)/h$ and $\int_{\mathbf{R}} K(u) \, du = 1$ (and similarly for x'_i).

Therefore g satisfies the bounded differences property with constants $C_i = 2/n$. Applying Theorem 3.31 to g and to $-g$ and using the union bound, we obtain: for every $t \geq 0$,

$$\mathbb{P}\left(\left| \|\widehat{f}_n - f\|_1 - \mathbb{E}\|\widehat{f}_n - f\|_1 \right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (2/n)^2}\right) = 2 \exp\left(-\frac{nt^2}{2}\right).$$

In words: while $\mathbb{E}\|\widehat{f}_n - f\|_1$ captures the bias–variance tradeoff through the choice of bandwidth h , the random fluctuations of $\|\widehat{f}_n - f\|_1$ around its mean are, in this sense, of order $n^{-1/2}$.

4 Concentration of norms and random projections

We now return to a theme that appears repeatedly in high-dimensional statistics: for many “nice” random vectors $X \in \mathbf{R}^d$, the Euclidean norm $\|X\|_2$ is tightly concentrated. A basic example is a standard Gaussian vector, where $\|X\|_2^2$ is a sum of independent $\chi^2(1)$ variables.

4.1 Gaussian norm concentration via Bernstein

Let $X \sim \mathcal{N}(0, I_d)$, *i.e.* $X_1, \dots, X_d \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, and consider

$$\|X\|_2^2 = \sum_{i=1}^d X_i^2.$$

Since X_i is sub-Gaussian, X_i^2 is sub-exponential and $\|X_i^2\|_{\psi_1} = \|X_i\|_{\psi_2}^2 \lesssim 1$. By the triangle inequality for $\|\cdot\|_{\psi_1}$, this implies $\|X_i^2 - 1\|_{\psi_1} \lesssim 1$ as well. Applying Bernstein’s inequality (Theorem 3.10) to the independent mean-zero variables $X_i^2 - 1$ yields

$$\mathbb{P}\left(\left| \|X\|_2^2 - d \right| \geq t\right) \leq 2 \exp\left(-c \min\left(\frac{t^2}{d}, t\right)\right), \quad t \geq 0,$$

where $c > 0$ is an absolute constant. Equivalently, normalizing by d and restricting to $t \in (0, 1)$,

$$\mathbb{P}\left(\left| \frac{\|X\|_2^2}{d} - 1 \right| \geq t\right) \leq 2 \exp(-ct^2d). \tag{11}$$

This is a first concrete instance of a general phenomenon: in high dimension, sums of many weakly dependent terms often exhibit concentration at a scale set by the dimension.

4.2 Johnson–Lindenstrauss via a Gaussian projection

The concentration bound (11) immediately leads to a simple random construction of a dimension-reduction map that approximately preserves distances.

Fix $u_1, \dots, u_n \in \mathbf{R}^d$ and let $m \ll d$. Let $\Gamma \in \mathbf{R}^{m \times d}$ have i.i.d. $\mathcal{N}(0, 1)$ entries and define the random linear map

$$\Pi(v) = \frac{1}{\sqrt{m}} \Gamma v, \quad v \in \mathbf{R}^d.$$

For a fixed $v \neq 0$, the vector Γv is Gaussian with covariance $\|v\|_2^2 I_m$, hence

$$\Pi(v) \sim \mathcal{N}\left(0, \frac{\|v\|_2^2}{m} I_m\right) \quad \text{and} \quad \frac{\|\Pi(v)\|_2^2}{\|v\|_2^2} = \frac{1}{m} \left\| \Gamma \left(\frac{v}{\|v\|_2} \right) \right\|_2^2.$$

Applying (11) in dimension m yields that for any $t \in (0, 1)$,

$$\mathbb{P}\left(\left| \frac{\|\Pi(v)\|_2^2}{\|v\|_2^2} - 1 \right| \geq t\right) \leq 2 \exp(-ct^2 m).$$

Now apply the union bound over the $\binom{n}{2}$ difference vectors $u_i - u_j$. (If $u_i = u_j$, there is nothing to prove; the inequalities below hold trivially.) Thus

$$\mathbb{P}\left(\exists 1 \leq i < j \leq n \text{ with } u_i \neq u_j : \left| \frac{\|\Pi(u_i - u_j)\|_2^2}{\|u_i - u_j\|_2^2} - 1 \right| \geq t\right) \leq 2 \binom{n}{2} \exp(-ct^2 m).$$

In particular, if

$$m \geq \frac{1}{ct^2} \log\left(\frac{2\binom{n}{2}}{\delta}\right),$$

then with probability at least $1 - \delta$ we have simultaneously for all $1 \leq i < j \leq n$,

$$(1 - t)\|u_i - u_j\|_2^2 \leq \|\Pi(u_i) - \Pi(u_j)\|_2^2 \leq (1 + t)\|u_i - u_j\|_2^2,$$

where we used the linearity $\Pi(u_i) - \Pi(u_j) = \Pi(u_i - u_j)$. This is a standard Gaussian proof of the Johnson–Lindenstrauss lemma; the striking feature is that the required projection dimension m depends only logarithmically on n and does not depend on the ambient dimension d .

4.3 Concentration of $\|X\|_2$ for independent sub-Gaussian coordinates

The previous subsection focused on $\|X\|_2^2$. Often we would like a direct concentration statement for $\|X\|_2$ itself.

Proposition 4.1 (Concentration of $\|X\|_2$ for independent coordinates). *Let $X = (X_1, \dots, X_d) \in \mathbf{R}^d$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$ for all i . Set $K = \max_{1 \leq i \leq d} \|X_i\|_{\psi_2}$. Then*

$$\left| \|X\|_2 - \sqrt{d} \right|_{\psi_2} \leq CK^2,$$

where $C > 0$ is an absolute constant.

Proof. First note that $K \geq 1$. Indeed, since $\mathbb{E}X_i^2 = 1$, Jensen's inequality gives $\mathbb{E} \exp(X_i^2/t^2) \geq \exp(1/t^2)$. Therefore $\mathbb{E} \exp(X_i^2/t^2) \leq 2$ forces $t^2 \geq 1/\log 2$, so $\|X_i\|_{\psi_2} \geq 1/\sqrt{\log 2}$ for every i .

By Theorem 3.5, $\|X_i^2\|_{\psi_1} = \|X_i\|_{\psi_2}^2 \leq K^2$. Also, $\|1\|_{\psi_1} = 1/\log 2$. Using the triangle inequality for $\|\cdot\|_{\psi_1}$,

$$\|X_i^2 - 1\|_{\psi_1} \leq \|X_i^2\|_{\psi_1} + \|1\|_{\psi_1} \leq K^2 + \frac{1}{\log 2} \leq C_0 K^2$$

for an absolute constant $C_0 > 0$, where we used $K \geq 1$ in the last step.

Applying Bernstein's inequality (Theorem 3.10) to the independent mean-zero variables $X_i^2 - 1$ yields: for all $u \geq 0$,

$$\mathbb{P}\left(\left|\frac{\|X\|_2^2}{d} - 1\right| \geq u\right) = \mathbb{P}\left(\left|\sum_{i=1}^d (X_i^2 - 1)\right| \geq ud\right) \leq 2 \exp\left(-cd \min\left(\frac{u^2}{K^4}, \frac{u}{K^2}\right)\right), \quad (12)$$

where $c > 0$ is an absolute constant.

Next, for any $a \geq 0$ and $u \geq 0$,

$$|a - 1| \geq u \implies |a^2 - 1| = |a - 1||a + 1| \geq u \max\{1, u\} = \max\{u, u^2\}.$$

Applying this with $a = \|X\|_2/\sqrt{d}$ yields, for every $u \geq 0$,

$$\mathbb{P}\left(\left|\frac{\|X\|_2}{\sqrt{d}} - 1\right| \geq u\right) \leq \mathbb{P}\left(\left|\frac{\|X\|_2^2}{d} - 1\right| \geq \max\{u, u^2\}\right).$$

Combining this with (12), a simple case check shows that

$$d \min\left(\frac{\max\{u, u^2\}^2}{K^4}, \frac{\max\{u, u^2\}}{K^2}\right) \geq \frac{u^2 d}{K^4}, \quad u \geq 0,$$

and hence

$$\mathbb{P}\left(\left|\frac{\|X\|_2}{\sqrt{d}} - 1\right| \geq u\right) \leq 2 \exp\left(-c \frac{u^2 d}{K^4}\right), \quad u \geq 0.$$

Finally, substituting $u = t/\sqrt{d}$ gives, for all $t \geq 0$,

$$\mathbb{P}\left(\left|\|X\|_2 - \sqrt{d}\right| \geq t\right) \leq 2 \exp\left(-c \frac{t^2}{K^4}\right).$$

By the standard tail characterization of ψ_2 norms (Theorem 2.13), this implies $\|\|X\|_2 - \sqrt{d}\|_{\psi_2} \leq CK^2$ for an absolute constant $C > 0$. $\square$

4.4 Beyond independence: isotropic and sub-Gaussian random vectors

Many random vectors in high-dimensional statistics do not have independent coordinates. A convenient baseline is the following normalization.

Definition 4.2 (Isotropic random vector). A random vector $X \in \mathbf{R}^d$ is *isotropic* if $\mathbb{E}[XX^\top] = I_d$.

If Y has mean μ and invertible covariance matrix Σ , then $X = \Sigma^{-1/2}(Y - \mu)$ is isotropic. Isotropy, however, does not imply independence. For instance, if X is uniform on the sphere $\sqrt{d}S^{d-1}$, then X is isotropic, but its coordinates are strongly dependent.

To discuss concentration without independence, it is useful to formalize what it means for a random vector to have sub-Gaussian one-dimensional marginals.

Definition 4.3 (ψ_2 size of a random vector). For a random vector $X \in \mathbf{R}^d$, define

$$\|X\|_{\psi_2} = \sup_{v \in S^{d-1}} \|\langle v, X \rangle\|_{\psi_2}.$$

We call X *sub-Gaussian* if $\|X\|_{\psi_2} < \infty$.

A more refined, variance-sensitive formulation uses a covariance proxy.

Definition 4.4 (Sub-Gaussian with covariance proxy). Let $X \in \mathbf{R}^d$ be mean-zero and let $\Sigma \succeq 0$. We say that X is sub-Gaussian with covariance proxy Σ if for all $\lambda \in \mathbf{R}$ and all $v \in S^{d-1}$,

$$\mathbb{E} \exp(\lambda \langle v, X \rangle) \leq \exp\left(\frac{\lambda^2}{2} v^\top \Sigma v\right).$$

In particular, Theorem 4.4 implies $\text{Var}(\langle v, X \rangle) \leq v^\top \Sigma v$ for every v , and hence

$$\mathbb{E} \|X\|_2^2 = \text{Tr}(\text{Cov}(X)) \leq \text{Tr}(\Sigma).$$

Proposition 4.5 (Norm tail bound under a covariance proxy). *Let X be mean-zero and sub-Gaussian with covariance proxy Σ in the sense of Theorem 4.4. Then for every $\delta \in (0, 1)$,*

$$\mathbb{P}\left(\|X\|_2 \geq \sqrt{\text{Tr}(\Sigma)} + \sqrt{2 \lambda_{\max}(\Sigma) \log(1/\delta)}\right) \leq \delta,$$

where $\lambda_{\max}(\Sigma)$ denotes the largest eigenvalue.

Remark 4.6. We will return to the proof of this result later. It can be viewed as a dimension-free concentration statement for the 1-Lipschitz function $x \mapsto \|x\|_2$, with the scale governed by the operator norm $\lambda_{\max}(\Sigma)$ and the location controlled by $\text{Tr}(\Sigma)$.

5 A norm bound for sub-Gaussian random vectors

In Theorem 4.5 we stated a tail bound for $\|X\|_2$ under a covariance-proxy sub-Gaussian assumption. We now give a proof. The argument is pleasant for two reasons:

- it does *not* require independence of the coordinates of X ;
- it showcases a useful “change-of-measure” technique, expressed through a variational identity for log-moment generating functions.

5.1 Kullback–Leibler divergence

Let ρ and π be probability densities on a measurable set $\Theta \subseteq \mathbf{R}^d$. The *Kullback–Leibler divergence* (or *relative entropy*) is

$$\text{KL}(\rho\|\pi) = \int_{\Theta} \rho(\theta) \log\left(\frac{\rho(\theta)}{\pi(\theta)}\right) d\theta,$$

when ρ is absolutely continuous with respect to π ; otherwise $\text{KL}(\rho\|\pi) = +\infty$. As usual, the integrand is interpreted as 0 on the set where $\rho(\theta) = 0$.

Lemma 5.1 (Non-negativity of KL). *For any densities ρ, π as above, $\text{KL}(\rho\|\pi) \geq 0$, and $\text{KL}(\rho\|\pi) = 0$ if and only if $\rho = \pi$ almost everywhere.*

Proof. By Jensen's inequality (since $\log$ is concave),

$$-\text{KL}(\rho\|\pi) = \int_{\Theta} \rho(\theta) \log\left(\frac{\pi(\theta)}{\rho(\theta)}\right) d\theta \leq \log\left(\int_{\Theta} \rho(\theta) \frac{\pi(\theta)}{\rho(\theta)} d\theta\right) = \log\left(\int_{\Theta} \pi(\theta) d\theta\right) = 0.$$

Thus $\text{KL}(\rho\|\pi) \geq 0$. Moreover, strict concavity of $\log$ implies equality in Jensen's inequality only when $\pi(\theta)/\rho(\theta)$ is almost surely constant under ρ , which is equivalent to $\rho = \pi$ a.e. $\square$

5.2 Donsker–Varadhan variational formula

The next lemma is a convenient way to “move” expectations between measures.

Lemma 5.2 (Donsker–Varadhan variational formula). *Let π be a probability density on $\Theta \subseteq \mathbf{R}^d$ and let $h : \Theta \rightarrow \mathbf{R}$ be measurable such that $\mathbb{E}_{\theta \sim \pi} e^{h(\theta)} < \infty$. Then*

$$\log \mathbb{E}_{\theta \sim \pi} e^{h(\theta)} = \sup_{\rho} \left\{ \mathbb{E}_{\theta \sim \rho} h(\theta) - \text{KL}(\rho\|\pi) \right\},$$

where the supremum is over all probability densities ρ such that $\text{KL}(\rho\|\pi) < \infty$.

Proof. Define the tilted density

$$\pi'(\theta) = \frac{\pi(\theta) e^{h(\theta)}}{\mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)}}.$$

Then π' is a probability density on Θ . For any density ρ with $\text{KL}(\rho\|\pi) < \infty$,

$$\begin{aligned} 0 \leq \text{KL}(\rho\|\pi') &= \mathbb{E}_{\theta \sim \rho} \log\left(\frac{\rho(\theta)}{\pi'(\theta)}\right) \\ &= \mathbb{E}_{\theta \sim \rho} \log\left(\frac{\rho(\theta)}{\pi(\theta)} \cdot \frac{\mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)}}{e^{h(\theta)}}\right) \\ &= \text{KL}(\rho\|\pi) + \log \mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)} - \mathbb{E}_{\theta \sim \rho} h(\theta). \end{aligned}$$

Rearranging gives

$$\log \mathbb{E}_{\vartheta \sim \pi} e^{h(\vartheta)} \geq \mathbb{E}_{\theta \sim \rho} h(\theta) - \text{KL}(\rho\|\pi).$$

Since this holds for every admissible ρ , it holds for the supremum. Finally, equality is achieved by taking $\rho = \pi'$, because then $\text{KL}(\pi'\|\pi') = 0$ by Theorem 5.1. $\square$

5.3 A uniform change-of-measure inequality

We now combine Theorem 5.2 with Markov's inequality to obtain a bound that holds simultaneously over all choices of ρ .

Lemma 5.3. *Let π be a probability density on $\Theta \subseteq \mathbf{R}^d$. Let X be a random variable (or random vector), and let $f(X, \theta)$ be a measurable function. Assume that the expectations below are finite. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ (over the draw of X), the following holds: for every probability density ρ such that $\text{KL}(\rho\|\pi) < \infty$,*

$$\mathbb{E}_{\theta \sim \rho} f(X, \theta) \leq \mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{f(X, \theta)}] + \text{KL}(\rho\|\pi) + \log(1/\delta). \quad (13)$$

Proof. Define

$$h_X(\theta) = f(X, \theta) - \log \mathbb{E}[e^{f(X, \theta)}],$$

where the expectation is with respect to the randomness of X . Now set

$$Y_X = \sup_{\rho} \left\{ \mathbb{E}_{\theta \sim \rho} h_X(\theta) - \text{KL}(\rho \| \pi) \right\}.$$

By Theorem 5.2, we also have

$$Y_X = \log \mathbb{E}_{\theta \sim \pi} e^{h_X(\theta)}.$$

A direct computation shows $\mathbb{E}[e^{Y_X}] = 1$:

$$\begin{aligned} \mathbb{E}[e^{Y_X}] &= \mathbb{E} \left[\mathbb{E}_{\theta \sim \pi} e^{h_X(\theta)} \right] \\ &= \mathbb{E}_{\theta \sim \pi} \mathbb{E} \left[\exp(f(X, \theta) - \log \mathbb{E} e^{f(X, \theta)}) \right] \\ &= \mathbb{E}_{\theta \sim \pi} \left[\frac{\mathbb{E} e^{f(X, \theta)}}{\mathbb{E} e^{f(X, \theta)}} \right] = 1. \end{aligned}$$

By Markov's inequality, for every $t \geq 0$,

$$\mathbb{P}(Y_X \geq t) = \mathbb{P}(e^{Y_X} \geq e^t) \leq e^{-t} \mathbb{E}[e^{Y_X}] = e^{-t}.$$

Taking $t = \log(1/\delta)$ yields $\mathbb{P}(Y_X \geq \log(1/\delta)) \leq \delta$. Equivalently, with probability at least $1 - \delta$,

$$Y_X \leq \log(1/\delta).$$

Unpacking the definition of Y_X , this means that with probability at least $1 - \delta$, for every admissible ρ ,

$$\mathbb{E}_{\theta \sim \rho} h_X(\theta) - \text{KL}(\rho \| \pi) \leq \log(1/\delta).$$

Finally, substitute the definition of h_X and rearrange to obtain (13). □

5.4 A few Gaussian computations

We record several elementary facts that we will use in the proof of Theorem 4.5.

Lemma 5.4 (Quadratic forms of a spherical Gaussian). *Let $Y \sim \mathcal{N}(\mu, \sigma^2 I_d)$ and let $A \in \mathbf{R}^{d \times d}$ be any matrix. Then*

$$\mathbb{E}[Y^\top A Y] = \sigma^2 \text{Tr}(A) + \mu^\top A \mu.$$

Proof. Write $Y = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, I_d)$. Expanding and using $\mathbb{E}Z = 0$ gives

$$\mathbb{E}[Y^\top A Y] = \mu^\top A \mu + \sigma^2 \mathbb{E}[Z^\top A Z].$$

Finally, by the trace trick,

$$\mathbb{E}[Z^\top A Z] = \mathbb{E}[\text{Tr}(Z^\top A Z)] = \mathbb{E}[\text{Tr}(A Z Z^\top)] = \text{Tr}(A \mathbb{E}[Z Z^\top]) = \text{Tr}(A).$$

□

Lemma 5.5 (KL divergence between shifted Gaussians). *Let $\beta > 0$ and $\|v\|_2 = 1$. Let ρ and π be the densities of $\mathcal{N}(v, I_d/\beta)$ and $\mathcal{N}(0, I_d/\beta)$, respectively. Then*

$$\text{KL}(\rho \| \pi) = \frac{\beta}{2}.$$

Proof. For $x \in \mathbf{R}^d$, both densities share the same normalizing constant $c(\beta)$ and satisfy

$$\rho(x) = c(\beta) \exp\left(-\frac{\beta}{2}\|x - v\|_2^2\right), \quad \pi(x) = c(\beta) \exp\left(-\frac{\beta}{2}\|x\|_2^2\right).$$

Hence

$$\log \frac{\rho(x)}{\pi(x)} = -\frac{\beta}{2}(\|x - v\|_2^2 - \|x\|_2^2) = \frac{\beta}{2}(2\langle x, v \rangle - \|v\|_2^2).$$

If $X \sim \rho$ then $\mathbb{E}X = v$, and therefore

$$\text{KL}(\rho\|\pi) = \mathbb{E} \log \frac{\rho(X)}{\pi(X)} = \frac{\beta}{2}(2\langle \mathbb{E}X, v \rangle - \|v\|_2^2) = \frac{\beta}{2}(2\|v\|_2^2 - \|v\|_2^2) = \frac{\beta}{2}.$$

□

Lemma 5.6 (Two suprema). *Let $x \in \mathbf{R}^d$ and let $\Sigma \succeq 0$ be symmetric. Then*

$$\sup_{v \in S^{d-1}} \langle x, v \rangle = \|x\|_2, \quad \sup_{v \in S^{d-1}} v^\top \Sigma v = \lambda_{\max}(\Sigma).$$

Proof. The first identity is Cauchy–Schwarz, with equality at $v = x/\|x\|_2$ (when $x \neq 0$). For the second, diagonalize $\Sigma = U\Lambda U^\top$ with U orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$. Writing $u = U^\top v$ gives $\|u\|_2 = 1$ and

$$v^\top \Sigma v = u^\top \Lambda u = \sum_{i=1}^d \lambda_i u_i^2 \leq \lambda_{\max}(\Sigma) \sum_{i=1}^d u_i^2 = \lambda_{\max}(\Sigma),$$

with equality when v is a top eigenvector. □

5.5 Proof of Theorem 4.5

Proof of Theorem 4.5. Recall the assumption of Theorem 4.4: for every $v \in S^{d-1}$ and every $\lambda \in \mathbf{R}$,

$$\mathbb{E} \exp(\lambda \langle v, X \rangle) \leq \exp\left(\frac{\lambda^2}{2} v^\top \Sigma v\right).$$

Fix parameters $\alpha > 0$ and $\beta > 0$ and a direction $v \in S^{d-1}$ (to be chosen later). Consider the function $f(X, \theta) = \alpha \langle X, \theta \rangle$, defined on $\theta \in \mathbf{R}^d$.

Let π and ρ be the densities of the Gaussians $\mathcal{N}(0, I_d/\beta)$ and $\mathcal{N}(v, I_d/\beta)$, respectively. Applying Theorem 5.3 gives that, with probability at least $1 - \delta$,

$$\mathbb{E}_{\theta \sim \rho} f(X, \theta) \leq \mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{f(X, \theta)}] + \text{KL}(\rho\|\pi) + \log(1/\delta). \quad (14)$$

We evaluate the terms on the right-hand side. First, since $\mathbb{E}_{\theta \sim \rho} \theta = v$,

$$\mathbb{E}_{\theta \sim \rho} f(X, \theta) = \alpha \mathbb{E}_{\theta \sim \rho} \langle X, \theta \rangle = \alpha \langle X, v \rangle. \quad (15)$$

Second, using the sub-Gaussian property with direction $\theta/\|\theta\|_2$ and parameter $\lambda = \alpha\|\theta\|_2$ (the bound is trivial when $\theta = 0$),

$$\mathbb{E} e^{\alpha \langle X, \theta \rangle} = \mathbb{E} e^{\|\theta\|_2 \alpha \langle X, \theta/\|\theta\|_2 \rangle} \leq \exp\left(\frac{\alpha^2}{2} \theta^\top \Sigma \theta\right),$$

hence

$$\begin{aligned}
\mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{f(X, \theta)}] &= \mathbb{E}_{\theta \sim \rho} \log \mathbb{E}[e^{\alpha \langle X, \theta \rangle}] \leq \mathbb{E}_{\theta \sim \rho} \left[\frac{\alpha^2}{2} \theta^\top \Sigma \theta \right] \\
&= \frac{\alpha^2}{2} \mathbb{E}_{\theta \sim \rho} [\theta^\top \Sigma \theta] \\
&= \frac{\alpha^2}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + v^\top \Sigma v \right) \leq \frac{\alpha^2}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + \lambda_{\max}(\Sigma) \right),
\end{aligned} \tag{16}$$

where we used Theorems 5.4 and 5.6. Third, by Theorem 5.5,

$$\text{KL}(\rho \| \pi) = \frac{\beta}{2}. \tag{17}$$

Plugging (15)–(17) into (14), we obtain: with probability at least $1 - \delta$,

$$\alpha \langle X, v \rangle \leq \frac{\alpha^2}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + \lambda_{\max}(\Sigma) \right) + \frac{\beta}{2} + \log(1/\delta). \tag{18}$$

Divide by α and note that the right-hand side no longer depends on v after the bound $v^\top \Sigma v \leq \lambda_{\max}(\Sigma)$. Taking the supremum over $v \in S^{d-1}$ and using Theorem 5.6, we get that with probability at least $1 - \delta$,

$$\|X\|_2 \leq \frac{\alpha}{2} \left(\frac{\text{Tr}(\Sigma)}{\beta} + \lambda_{\max}(\Sigma) \right) + \frac{\beta}{2\alpha} + \frac{\log(1/\delta)}{\alpha}. \tag{19}$$

Now set $\gamma = \alpha/\beta$ (so $\beta/\alpha = 1/\gamma$) to rewrite

$$\|X\|_2 \leq \frac{\gamma}{2} \text{Tr}(\Sigma) + \frac{1}{2\gamma} + \frac{\alpha}{2} \lambda_{\max}(\Sigma) + \frac{\log(1/\delta)}{\alpha}.$$

We may optimize γ and α separately. Indeed, for $a, b > 0$, the function $x \mapsto ax + b/x$ on $(0, \infty)$ is minimized at $x^* = \sqrt{b/a}$, and its minimum value is $2\sqrt{ab}$. Applying this twice gives

$$\inf_{\gamma > 0} \left\{ \frac{\gamma}{2} \text{Tr}(\Sigma) + \frac{1}{2\gamma} \right\} = \sqrt{\text{Tr}(\Sigma)}, \quad \inf_{\alpha > 0} \left\{ \frac{\alpha}{2} \lambda_{\max}(\Sigma) + \frac{\log(1/\delta)}{\alpha} \right\} = \sqrt{2 \lambda_{\max}(\Sigma) \log(1/\delta)}.$$

Substituting these choices into (19) yields

$$\|X\|_2 \leq \sqrt{\text{Tr}(\Sigma)} + \sqrt{2 \lambda_{\max}(\Sigma) \log(1/\delta)},$$

which is exactly the claim. $\square$

5.6 Gaussian concentration for Lipschitz functions

Finally, we record a classical concentration inequality for Lipschitz functions of a Gaussian vector.

Definition 5.7 (Lipschitz function). A function $f : \mathbf{R}^d \rightarrow \mathbf{R}$ is L -Lipschitz (with respect to $\|\cdot\|_2$) if

$$|f(x) - f(y)| \leq L\|x - y\|_2 \quad \text{for all } x, y \in \mathbf{R}^d.$$

Theorem 5.8 (Gaussian concentration inequality). Let $Z \sim \mathcal{N}(0, I_d)$ and let $f : \mathbf{R}^d \rightarrow \mathbf{R}$ be L -Lipschitz. Then for every $t \geq 0$,

$$\mathbb{P}(f(Z) - \mathbb{E}f(Z) \geq t) \leq \exp\left(-\frac{t^2}{2L^2}\right),$$

and consequently

$$\mathbb{P}(|f(Z) - \mathbb{E}f(Z)| \geq t) \leq 2 \exp\left(-\frac{t^2}{2L^2}\right).$$

We will often apply this inequality to the Euclidean norm. Two simple auxiliary facts are helpful.

Lemma 5.9 (Existence of $\Sigma^{1/2}$). *If $\Sigma \succeq 0$ is symmetric, then there exists a unique symmetric matrix $\Sigma^{1/2} \succeq 0$ such that $(\Sigma^{1/2})^2 = \Sigma$.*

Proof. Diagonalize $\Sigma = U\Lambda U^\top$ with U orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$, $\lambda_i \geq 0$. Then $\Sigma^{1/2} = U\Lambda^{1/2}U^\top$ with $\Lambda^{1/2} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_d})$ is symmetric, positive semidefinite, and satisfies $(\Sigma^{1/2})^2 = \Sigma$. Uniqueness holds among symmetric positive semidefinite square roots. $\square$

Lemma 5.10 (The norm is 1-Lipschitz). *For all $x, y \in \mathbf{R}^d$,*

$$|\|x\|_2 - \|y\|_2| \leq \|x - y\|_2.$$

Proof. By the triangle inequality, $\|x\|_2 = \|x - y + y\|_2 \leq \|x - y\|_2 + \|y\|_2$, which gives $\|x\|_2 - \|y\|_2 \leq \|x - y\|_2$. Swapping the roles of x and y yields the reverse inequality. $\square$

Corollary 5.11 (Concentration of $\|X\|_2$ for a Gaussian with covariance Σ). *Let $X \sim \mathcal{N}(0, \Sigma)$, where $\Sigma \succeq 0$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$|\|X\|_2 - \mathbb{E}\|X\|_2| \leq \sqrt{2 \lambda_{\max}(\Sigma) \log(2/\delta)}.$$

Proof. Let $Z \sim \mathcal{N}(0, I_d)$. Then X has the same distribution as $\Sigma^{1/2}Z$ (by Theorem 5.9). Define $f(z) = \|\Sigma^{1/2}z\|_2$. By Theorem 5.10,

$$|f(x) - f(y)| = |\|\Sigma^{1/2}x\|_2 - \|\Sigma^{1/2}y\|_2| \leq \|\Sigma^{1/2}(x - y)\|_2 \leq \sqrt{\lambda_{\max}(\Sigma)} \|x - y\|_2,$$

so f is $\sqrt{\lambda_{\max}(\Sigma)}$ -Lipschitz. Apply Theorem 5.8 and set $t = \sqrt{2 \lambda_{\max}(\Sigma) \log(2/\delta)}$. $\square$

5.7 Simple proof of the Gaussian concentration inequality with a non-optimal constant

We now give a self-contained proof of a Gaussian concentration bound using a useful interpolation trick. The argument is short and robust, but it produces a slightly weaker numerical constant than the sharp one in Theorem 5.8.

Lemma 5.12 (A convex comparison lemma). *Let $\varphi : \mathbf{R} \rightarrow \mathbf{R}$ be convex and let $f : \mathbf{R}^d \rightarrow \mathbf{R}$ be differentiable. If $Z, Y \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$ are independent, then*

$$\mathbb{E}[\varphi(f(Z) - \mathbb{E}f(Z))] \leq \mathbb{E}\left[\varphi\left(\frac{\pi}{2} \langle \nabla f(Z), Y \rangle\right)\right].$$

Proof. Let Z' be an independent copy of Z . Since $\mathbb{E}f(Z') = \mathbb{E}f(Z)$ and φ is convex, Jensen's inequality gives

$$\mathbb{E}[\varphi(f(Z) - \mathbb{E}f(Z))] = \mathbb{E}[\varphi(f(Z) - \mathbb{E}f(Z'))] \leq \mathbb{E}[\varphi(f(Z) - f(Z'))].$$

Fix $\theta \in [0, \pi/2]$ and define the interpolation

$$W(\theta) = Z \sin \theta + Z' \cos \theta, \quad W'(\theta) = Z \cos \theta - Z' \sin \theta.$$

Then $W(0) = Z'$ and $W(\pi/2) = Z$, hence by the fundamental theorem of calculus,

$$f(Z) - f(Z') = \int_0^{\pi/2} \frac{d}{d\theta} f(W(\theta)) d\theta = \int_0^{\pi/2} \langle \nabla f(W(\theta)), W'(\theta) \rangle d\theta,$$

where we used the chain rule.

Let $\Theta \sim \text{Unif}[0, \pi/2]$ be independent of (Z, Z') . Then

$$f(Z) - f(Z') = \frac{\pi}{2} \mathbb{E}_{\Theta} [\langle \nabla f(W(\Theta)), W'(\Theta) \rangle].$$

Applying Jensen's inequality once more (to φ) yields

$$\varphi(f(Z) - f(Z')) \leq \mathbb{E}_{\Theta} \left[\varphi \left(\frac{\pi}{2} \langle \nabla f(W(\Theta)), W'(\Theta) \rangle \right) \right].$$

Taking expectations over (Z, Z', Θ) gives

$$\mathbb{E}[\varphi(f(Z) - f(Z'))] \leq \mathbb{E} \left[\varphi \left(\frac{\pi}{2} \langle \nabla f(W(\Theta)), W'(\Theta) \rangle \right) \right].$$

Finally, for each fixed θ , the map $(Z, Z') \mapsto (W(\theta), W'(\theta))$ is an orthogonal transformation on $\mathbf{R}^{2d}$. Therefore $(W(\theta), W'(\theta))$ has the same distribution as (Z, Y) with $Z, Y \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$ independent. It follows that the last expectation equals $\mathbb{E}[\varphi(\frac{\pi}{2} \langle \nabla f(Z), Y \rangle)]$, which proves the claim. $\square$

We now derive a tail bound. Assume for the moment that f is differentiable.

Proof of a Gaussian concentration bound. Let $f : \mathbf{R}^d \rightarrow \mathbf{R}$ be L -Lipschitz. Then $\|\nabla f(x)\|_2 \leq L$ for all x : indeed, for any unit vector $u \in S^{d-1}$ and any $t \in \mathbf{R}$,

$$|f(x + tu) - f(x)| \leq L|t|,$$

and letting $t \rightarrow 0$ shows $|\langle \nabla f(x), u \rangle| \leq L$; taking the supremum over u yields $\|\nabla f(x)\|_2 \leq L$.

Apply Theorem 5.12 with $\varphi(s) = e^{\lambda s}$ for $\lambda \in \mathbf{R}$:

$$\mathbb{E} \exp(\lambda(f(Z) - \mathbb{E}f(Z))) \leq \mathbb{E} \exp\left(\frac{\pi\lambda}{2} \langle \nabla f(Z), Y \rangle\right).$$

Conditioning on Z , the random variable $\langle \nabla f(Z), Y \rangle$ is Gaussian with mean 0 and variance $\|\nabla f(Z)\|_2^2$. Hence

$$\mathbb{E} \left[\exp\left(\frac{\pi\lambda}{2} \langle \nabla f(Z), Y \rangle\right) \mid Z \right] = \exp\left(\frac{1}{2} \left(\frac{\pi\lambda}{2}\right)^2 \|\nabla f(Z)\|_2^2\right) \leq \exp\left(\frac{\pi^2 \lambda^2 L^2}{8}\right).$$

Taking expectation over Z gives the MGF bound

$$\mathbb{E} \exp(\lambda(f(Z) - \mathbb{E}f(Z))) \leq \exp\left(\frac{\pi^2 \lambda^2 L^2}{8}\right), \quad \lambda \in \mathbf{R}.$$

By Chernoff's method, for any $t \geq 0$ and any $\lambda > 0$,

$$\mathbb{P}(f(Z) - \mathbb{E}f(Z) \geq t) \leq \exp\left(-\lambda t + \frac{\pi^2 \lambda^2 L^2}{8}\right).$$

Optimizing over λ gives $\lambda^* = 4t/(\pi^2 L^2)$ and therefore

$$\mathbb{P}(f(Z) - \mathbb{E}f(Z) \geq t) \leq \exp\left(-\frac{2t^2}{\pi^2 L^2}\right).$$

Applying the same bound to $-f$ and using the union bound yields

$$\mathbb{P}(|f(Z) - \mathbb{E}f(Z)| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\pi^2 L^2}\right), \quad t \geq 0.$$

Finally, if f is merely Lipschitz (not differentiable), one can apply the argument to a smooth approximation of f and pass to the limit; we omit these standard approximation details. $\square$

Remark 5.13. The proof above yields the same Gaussian t^2/L^2 scaling as Theorem 5.8, but with a worse numerical constant in the exponent. Sharper constants can be obtained, for example, via Gaussian isoperimetry or the Gaussian log-Sobolev inequality.

6 Matrices and random matrix concentration

We now turn to random matrices. The first goal is to understand how the operator norm behaves for matrices with independent entries. The second goal is to revisit sample covariance matrices and show how the same variational method used earlier for vectors extends naturally to this setting.

Throughout, if $A \in \mathbf{R}^{m \times n}$, we write its singular value decomposition in the form

$$A = \sum_{j=1}^r s_j(A) u_j v_j^\top,$$

where $r = \text{rank}(A)$, the vectors $u_1, \dots, u_r \in \mathbf{R}^m$ and $v_1, \dots, v_r \in \mathbf{R}^n$ are orthonormal, and

$$s_1(A) \geq s_2(A) \geq \dots \geq s_r(A) > 0$$

are the singular values of A . The two most common norms for us will be the operator norm

$$\|A\|_{\text{op}} = \sup_{v \in S^{n-1}} \|Av\|_2 = \sup_{u \in S^{m-1}, v \in S^{n-1}} u^\top Av$$

and the Frobenius norm

$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2 \right)^{1/2}.$$

These are related to singular values by

$$\|A\|_{\text{op}} = s_1(A), \quad \|A\|_F^2 = \sum_{j=1}^r s_j(A)^2.$$

When A is square and symmetric, $\|A\|_{\text{op}}$ is the largest absolute value of an eigenvalue of A .

6.1 Euclidean nets, covering numbers, and a discretization lemma

A recurring theme in random matrix theory is that operator norms can be reduced to maxima over finite sets. The right finite sets are obtained from coverings of the Euclidean sphere.

Definition 6.1 (ε -net and covering number). Let (K, d) be a metric space and let $\varepsilon > 0$. A set $\mathcal{N} \subseteq K$ is called an ε -net of K if for every $x \in K$ there exists $y \in \mathcal{N}$ such that $d(x, y) \leq \varepsilon$. The *covering number* of K is

$$\mathcal{N}(K, d, \varepsilon) = \min\{|\mathcal{N}| : \mathcal{N} \subseteq K \text{ is an } \varepsilon\text{-net}\}.$$

Definition 6.2 (ε -separated set and packing number). Let (K, d) be a metric space and let $\varepsilon > 0$. A set $\mathcal{P} \subseteq K$ is called ε -separated if

$$d(x, y) > \varepsilon \quad \text{for all distinct } x, y \in \mathcal{P}.$$

The *packing number* of K is

$$\mathcal{P}(K, d, \varepsilon) = \max\{|\mathcal{P}| : \mathcal{P} \subseteq K \text{ is } \varepsilon\text{-separated}\}.$$

The two notions are equivalent up to harmless constants.

Lemma 6.3 (Packing and covering numbers). *For every metric space (K, d) and every $\varepsilon > 0$,*

$$\mathcal{P}(K, d, 2\varepsilon) \leq \mathcal{N}(K, d, \varepsilon) \leq \mathcal{P}(K, d, \varepsilon).$$

Proof. We begin with the right-hand inequality. Let $\mathcal{P} \subseteq K$ be a maximal ε -separated set. We claim that $\mathcal{P}$ is automatically an ε -net. Indeed, if some $x \in K$ were farther than ε from every point of $\mathcal{P}$, then $\mathcal{P} \cup \{x\}$ would still be ε -separated, contradicting maximality. Thus every maximal ε -separated set is an ε -net, and therefore

$$\mathcal{N}(K, d, \varepsilon) \leq \mathcal{P}(K, d, \varepsilon).$$

For the left-hand inequality, let $\mathcal{N}$ be an ε -net of K , and let $\mathcal{P} \subseteq K$ be 2ε -separated. Every point of $\mathcal{P}$ must lie in some ball of radius ε centered at a point of $\mathcal{N}$. Moreover, no such ball can contain two distinct points of $\mathcal{P}$, because otherwise those two points would be at distance at most 2ε , contradicting the separation property. Hence $|\mathcal{P}| \leq |\mathcal{N}|$. Taking the maximum over all 2ε -separated sets and the minimum over all ε -nets gives

$$\mathcal{P}(K, d, 2\varepsilon) \leq \mathcal{N}(K, d, \varepsilon).$$

□

We will mostly use these notions for Euclidean balls and spheres.

Lemma 6.4 (Covering number of the Euclidean ball). *Let*

$$B_2^d = \{x \in \mathbf{R}^d : \|x\|_2 \leq 1\}.$$

Then for every $\varepsilon > 0$,

$$\left(\frac{1}{\varepsilon}\right)^d \leq \mathcal{N}(B_2^d, \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon}\right)^d.$$

Proof. For the lower bound, let $\mathcal{N}$ be any ε -net of B_2^d . Then

$$B_2^d \subseteq \bigcup_{x \in \mathcal{N}} (x + \varepsilon B_2^d).$$

Taking volumes and using subadditivity,

$$\text{Vol}(B_2^d) \leq \sum_{x \in \mathcal{N}} \text{Vol}(x + \varepsilon B_2^d) = |\mathcal{N}| \varepsilon^d \text{Vol}(B_2^d),$$

because translations do not change volume. Therefore $|\mathcal{N}| \geq \varepsilon^{-d}$, and taking the minimum over all ε -nets proves the lower bound.

For the upper bound, let $\mathcal{P} \subseteq B_2^d$ be a maximal ε -separated set. By Theorem 6.3, it is enough to bound $|\mathcal{P}|$. The Euclidean balls

$$x + \frac{\varepsilon}{2} B_2^d, \quad x \in \mathcal{P},$$

are pairwise disjoint. Also, because $\|x\|_2 \leq 1$ for every $x \in B_2^d$,

$$x + \frac{\varepsilon}{2} B_2^d \subseteq \left(1 + \frac{\varepsilon}{2}\right) B_2^d.$$

Hence

$$|\mathcal{P}| \text{Vol}\left(\frac{\varepsilon}{2} B_2^d\right) \leq \text{Vol}\left(\left(1 + \frac{\varepsilon}{2}\right) B_2^d\right).$$

Using the scaling of volume gives

$$|\mathcal{P}| \leq \left(\frac{1 + \varepsilon/2}{\varepsilon/2} \right)^d = \left(1 + \frac{2}{\varepsilon} \right)^d.$$

Now apply Theorem 6.3. □

Remark 6.5. Since $S^{d-1} \subseteq B_2^d$, the same upper bound also applies to the sphere:

$$\mathcal{N}(S^{d-1}, \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon} \right)^d.$$

The next observation is the basic bridge between nets and operator norms.

Proposition 6.6 (Approximating the operator norm by a net). *Let $A \in \mathbf{R}^{m \times n}$ and let $\mathcal{M} \subseteq S^{m-1}$ and $\mathcal{N} \subseteq S^{n-1}$ be ε -nets with $\varepsilon \in (0, \frac{1}{2})$. Then*

$$\sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top A v| \leq \|A\|_{\text{op}} \leq \frac{1}{1 - 2\varepsilon} \sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top A v|.$$

Proof. The lower bound is immediate from the definition of $\|A\|_{\text{op}}$.

For the upper bound, fix $v \in S^{n-1}$. Choose $v_0 \in \mathcal{N}$ such that $\|v - v_0\|_2 \leq \varepsilon$. Then

$$\|Av\|_2 \leq \|A(v - v_0)\|_2 + \|Av_0\|_2 \leq \varepsilon \|A\|_{\text{op}} + \|Av_0\|_2.$$

Taking the supremum over $v \in S^{n-1}$ yields

$$\|A\|_{\text{op}} \leq \varepsilon \|A\|_{\text{op}} + \sup_{w \in \mathcal{N}} \|Aw\|_2,$$

hence

$$(1 - \varepsilon) \|A\|_{\text{op}} \leq \sup_{w \in \mathcal{N}} \|Aw\|_2.$$

Now fix $w \in \mathcal{N}$. For every $u \in S^{m-1}$ choose $u_0 \in \mathcal{M}$ with $\|u - u_0\|_2 \leq \varepsilon$. Then

$$|u^\top Aw| \leq |u_0^\top Aw| + |(u - u_0)^\top Aw| \leq \sup_{z \in \mathcal{M}} |z^\top Aw| + \varepsilon \|Aw\|_2.$$

Taking the supremum over $u \in S^{m-1}$ gives

$$\|Aw\|_2 \leq \sup_{z \in \mathcal{M}} |z^\top Aw| + \varepsilon \|Aw\|_2,$$

and therefore

$$(1 - \varepsilon) \|Aw\|_2 \leq \sup_{z \in \mathcal{M}} |z^\top Aw|.$$

Now take the supremum over $w \in \mathcal{N}$ to obtain

$$(1 - \varepsilon) \sup_{w \in \mathcal{N}} \|Aw\|_2 \leq \sup_{z \in \mathcal{M}, w \in \mathcal{N}} |z^\top Aw|.$$

Combining the last two estimates yields

$$(1 - \varepsilon)^2 \|A\|_{\text{op}} \leq \sup_{z \in \mathcal{M}, w \in \mathcal{N}} |z^\top Aw|.$$

Finally, since $\varepsilon \in (0, \frac{1}{2})$,

$$(1 - \varepsilon)^2 \geq 1 - 2\varepsilon,$$

which gives the claimed inequality. □

6.2 Upper bounds for matrices with independent entries

We now apply the net argument to random matrices with independent sub-Gaussian entries.

Theorem 6.7 (Operator norm of a matrix with independent entries). *Let $X = (X_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$ be an $m \times n$ random matrix whose entries are independent, mean-zero random variables. Assume that*

$$\max_{1 \leq i \leq m, 1 \leq j \leq n} \|X_{ij}\|_{\psi_2} \leq K$$

for some $K > 0$. Then there exists an absolute constant $C > 0$ such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\|X\|_{\text{op}} \leq CK \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right).$$

Proof. Let $\mathcal{M} \subseteq S^{m-1}$ and $\mathcal{N} \subseteq S^{n-1}$ be $\frac{1}{4}$ -nets. By Theorem 6.4 and the remark after it, we may choose them so that

$$|\mathcal{M}| \leq 9^m, \quad |\mathcal{N}| \leq 9^n.$$

Then Theorem 6.6 gives

$$\|X\|_{\text{op}} \leq 2 \sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v|.$$

Fix a pair $(u, v) \in \mathcal{M} \times \mathcal{N}$. Since

$$u^\top X v = \sum_{i=1}^m \sum_{j=1}^n u_i v_j X_{ij},$$

this is a sum of independent mean-zero sub-Gaussian random variables. By Theorem 2.16,

$$\begin{aligned} \|u^\top X v\|_{\psi_2}^2 &\leq C_1 \sum_{i=1}^m \sum_{j=1}^n \|u_i v_j X_{ij}\|_{\psi_2}^2 \\ &\leq C_1 K^2 \sum_{i=1}^m \sum_{j=1}^n u_i^2 v_j^2 \\ &= C_1 K^2 \left(\sum_{i=1}^m u_i^2 \right) \left(\sum_{j=1}^n v_j^2 \right) = C_1 K^2, \end{aligned}$$

because $u \in S^{m-1}$ and $v \in S^{n-1}$. Thus there exists an absolute constant $c_1 > 0$ such that

$$\mathbb{P}(|u^\top X v| \geq s) \leq 2 \exp\left(-c_1 \frac{s^2}{K^2}\right), \quad s \geq 0.$$

Now apply the union bound over all pairs $(u, v) \in \mathcal{M} \times \mathcal{N}$:

$$\mathbb{P}\left(\sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v| \geq s\right) \leq 2 \cdot 9^{m+n} \exp\left(-c_1 \frac{s^2}{K^2}\right).$$

Choose

$$s = C_2 K \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right)$$

with C_2 large enough. Since

$$(\sqrt{m} + \sqrt{n})^2 \geq m + n,$$

the right-hand side is at most δ . Therefore, with probability at least $1 - \delta$,

$$\sup_{u \in \mathcal{M}, v \in \mathcal{N}} |u^\top X v| \leq C_2 K \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right).$$

Using the net-to-operator estimate finishes the proof. $\square$

The bound is already very informative in a number of familiar situations.

Example 6.8 (Rectangular Gaussian matrices). Let G be an $m \times n$ matrix with independent $\mathcal{N}(0, 1)$ entries. Since each entry has bounded ψ_2 norm by an absolute constant, Theorem 6.7 gives

$$\|G\|_{\text{op}} \leq C \left(\sqrt{m} + \sqrt{n} + \sqrt{\log(1/\delta)} \right)$$

with probability at least $1 - \delta$. This has the correct order in m and n .

Example 6.9 (Symmetric matrices and Wigner-type ensembles). Suppose W is an $n \times n$ symmetric random matrix such that the upper-triangular entries

$$\{W_{ij} : 1 \leq i < j \leq n\}$$

are independent mean-zero sub-Gaussian random variables, the diagonal entries are also independent mean-zero sub-Gaussian random variables, and all these variables have ψ_2 norm at most K .

To reduce to the independent-entry setting, define an upper-triangular matrix Y by

$$Y_{ij} = W_{ij} \quad (i < j), \quad Y_{ii} = \frac{W_{ii}}{2}, \quad Y_{ij} = 0 \quad (i > j).$$

Then $W = Y + Y^\top$, so

$$\|W\|_{\text{op}} \leq \|Y\|_{\text{op}} + \|Y^\top\|_{\text{op}} = 2\|Y\|_{\text{op}}.$$

Since the entries of Y are independent and $\max_{i,j} \|Y_{ij}\|_{\psi_2} \leq CK$, Theorem 6.7 yields

$$\|W\|_{\text{op}} \leq CK \left(\sqrt{n} + \sqrt{\log(1/\delta)} \right)$$

with probability at least $1 - \delta$.

In particular, for a standard Wigner matrix with Gaussian upper-triangular entries, the operator norm is of order $\sqrt{n}$. After dividing by $\sqrt{n}$, one obtains a random matrix whose norm remains of constant order.

6.3 Operator norm of the sample covariance matrix

We now return to sample covariance matrices.

Let X be a mean-zero random vector in $\mathbf{R}^d$ with covariance matrix

$$\Sigma = \mathbb{E}[X X^\top].$$

Assume that there exists a constant $L \geq 1$ such that for every $v \in \mathbf{R}^d$,

$$\|\langle X, v \rangle\|_{\psi_2} \leq L (\mathbb{E} \langle X, v \rangle^2)^{1/2} = L (v^\top \Sigma v)^{1/2}. \quad (20)$$

Let $X_1, \dots, X_n$ be independent copies of X , and define the sample covariance matrix

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top.$$

We also recall the *effective rank*

$$r(\Sigma) = \frac{\text{Tr}(\Sigma)}{\|\Sigma\|_{\text{op}}},$$

with the convention that the problem is trivial when $\Sigma = 0$.

The key point is that bilinear forms of $X_i X_i^\top - \Sigma$ are not only sub-exponential; they satisfy a sub-gamma type moment bound (for small enough λ), which is exactly what we need for a Bernstein-style argument.

Lemma 6.10 (Sub-gamma behavior of centered bilinear forms). *Assume (20). For $\theta, \nu \in \mathbf{R}^d$, define*

$$Z(\theta, \nu) = \langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu.$$

Then there exist positive constants c_L, C_L , depending only on L , such that

$$\|Z(\theta, \nu)\|_{\psi_1} \leq C_L (\theta^\top \Sigma \theta + \nu^\top \Sigma \nu),$$

and consequently

$$\log \mathbb{E} e^{\lambda Z(\theta, \nu)} \leq C_L \lambda^2 (\theta^\top \Sigma \theta + \nu^\top \Sigma \nu)^2$$

for all

$$|\lambda| \leq \frac{c_L}{\theta^\top \Sigma \theta + \nu^\top \Sigma \nu}.$$

Proof. By (20),

$$\|\langle X, \theta \rangle\|_{\psi_2} \leq L(\theta^\top \Sigma \theta)^{1/2}, \quad \|\langle X, \nu \rangle\|_{\psi_2} \leq L(\nu^\top \Sigma \nu)^{1/2}.$$

Therefore, by Theorem 3.6,

$$\|\langle X, \theta \rangle \langle X, \nu \rangle\|_{\psi_1} \leq \|\langle X, \theta \rangle\|_{\psi_2} \|\langle X, \nu \rangle\|_{\psi_2} \leq L^2 (\theta^\top \Sigma \theta)^{1/2} (\nu^\top \Sigma \nu)^{1/2}.$$

For the deterministic centering term, we use $\|c\|_{\psi_1} = |c|/\log 2$ for constants c , and Cauchy–Schwarz:

$$|\theta^\top \Sigma \nu| \leq (\theta^\top \Sigma \theta)^{1/2} (\nu^\top \Sigma \nu)^{1/2} \leq \frac{1}{2} (\theta^\top \Sigma \theta + \nu^\top \Sigma \nu).$$

Combining these bounds with the triangle inequality for $\|\cdot\|_{\psi_1}$ gives

$$\|Z(\theta, \nu)\|_{\psi_1} \leq \|\langle X, \theta \rangle \langle X, \nu \rangle\|_{\psi_1} + \|\theta^\top \Sigma \nu\|_{\psi_1} \leq C_L (\theta^\top \Sigma \theta + \nu^\top \Sigma \nu)$$

for a constant C_L depending only on L .

Finally, $\mathbb{E} Z(\theta, \nu) = 0$ by definition, so the sub-exponential MGF control from Theorem 3.3(4) implies that for $|\lambda| \leq c/\|Z(\theta, \nu)\|_{\psi_1}$,

$$\log \mathbb{E} e^{\lambda Z(\theta, \nu)} \leq C \lambda^2 \|Z(\theta, \nu)\|_{\psi_1}^2.$$

Substituting the bound on $\|Z(\theta, \nu)\|_{\psi_1}$ and absorbing absolute constants into c_L, C_L yields the claim. $\square$

We are now ready for the main theorem.

Theorem 6.11 (Sample covariance in operator norm). *Assume (20), and let $\widehat{\Sigma}$ be the sample covariance matrix based on $X_1, \dots, X_n$. Then there exists a constant $C_L > 0$, depending only on L , such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \leq C_L \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma) + \log(1/\delta)}{n}} + \frac{r(\Sigma) + \log(1/\delta)}{n} \right).$$

Proof. If $\Sigma = 0$, then $X = 0$ almost surely, hence $\widehat{\Sigma} = 0$ and there is nothing to prove. Assume $\Sigma \neq 0$. Fix parameters $\beta > 0$ and $R > 0$ to be chosen later, and let

$$\pi = \mathcal{N}(0, \beta^{-1} I_d).$$

For each $u \in S^{d-1}$, define ρ_u as the law of $u + G$ conditioned on

$$\|\Sigma^{1/2} G\|_2 \leq R, \quad G \sim \mathcal{N}(0, \beta^{-1} I_d),$$

where $\Sigma^{1/2} \succeq 0$ is the symmetric square root of Σ . Equivalently, ρ_u has density

$$\frac{1}{p} \left(\frac{\beta}{2\pi} \right)^{d/2} \exp\left(-\frac{\beta}{2} \|\theta - u\|_2^2\right) \mathbf{1}\{\|\Sigma^{1/2}(\theta - u)\|_2 \leq R\}, \quad \theta \in \mathbf{R}^d,$$

where

$$p = \mathbb{P}(\|\Sigma^{1/2} G\|_2 \leq R).$$

Since the conditioning event is symmetric in G , we have $\mathbb{E}[G \mid \|\Sigma^{1/2} G\|_2 \leq R] = 0$, and therefore

$$\mathbb{E}_{\theta \sim \rho_u} \theta = u. \tag{21}$$

Now fix $u, v \in S^{d-1}$ and let

$$\rho_{u,v} = \rho_u \otimes \rho_v, \quad \pi_{2d} = \pi \otimes \pi.$$

Apply Theorem 5.3 on the parameter space $\mathbf{R}^d \times \mathbf{R}^d$ with

$$f((X_1, \dots, X_n), \theta, \nu) = \lambda \sum_{i=1}^n \left(\langle X_i, \theta \rangle \langle X_i, \nu \rangle - \theta^\top \Sigma \nu \right),$$

where $\lambda > 0$ will be chosen later. Using independence of $X_1, \dots, X_n$ to factorize the exponential moment, and using (21), we obtain that with probability at least $1 - \delta$, simultaneously for all $u, v \in S^{d-1}$,

$$\begin{aligned} \lambda u^\top (\widehat{\Sigma} - \Sigma) v &\leq \mathbb{E}_{(\theta, \nu) \sim \rho_{u,v}} \log \mathbb{E} \exp\left(\lambda (\langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu)\right) \\ &\quad + \frac{\text{KL}(\rho_{u,v} \parallel \pi_{2d}) + \log(1/\delta)}{n}, \end{aligned} \tag{22}$$

where X denotes an independent copy of X_1 .

If (θ, ν) lies in the support of $\rho_{u,v}$, then

$$\|\Sigma^{1/2}(\theta - u)\|_2 \leq R, \quad \|\Sigma^{1/2}(\nu - v)\|_2 \leq R.$$

Using $(a + b)^2 \leq 2a^2 + 2b^2$ in the Σ -inner product,

$$\theta^\top \Sigma \theta \leq 2(\theta - u)^\top \Sigma (\theta - u) + 2u^\top \Sigma u \leq 2R^2 + 2\|\Sigma\|_{\text{op}},$$

and similarly $\nu^\top \Sigma \nu \leq 2R^2 + 2\|\Sigma\|_{\text{op}}$. Choose

$$R^2 = 2\|\Sigma\|_{\text{op}}.$$

Then on the support of $\rho_{u,v}$,

$$\theta^\top \Sigma \theta + \nu^\top \Sigma \nu \leq 12\|\Sigma\|_{\text{op}}.$$

By Theorem 6.10, there exist constants $c_L, C_L > 0$ such that, for every

$$0 < \lambda \leq \frac{c_L}{\|\Sigma\|_{\text{op}}},$$

we have uniformly on the support of $\rho_{u,v}$,

$$\log \mathbb{E} \exp \left(\lambda (\langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu) \right) \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2.$$

Therefore,

$$\mathbb{E}_{(\theta, \nu) \sim \rho_{u,v}} \log \mathbb{E} \exp \left(\lambda (\langle X, \theta \rangle \langle X, \nu \rangle - \theta^\top \Sigma \nu) \right) \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2. \quad (23)$$

We claim that

$$\text{KL}(\rho_u \|\pi) = \log(1/p) + \frac{\beta}{2}.$$

Indeed, for θ in the support of ρ_u ,

$$\log \frac{\rho_u(\theta)}{\pi(\theta)} = \log(1/p) + \frac{\beta}{2} (\|\theta\|_2^2 - \|\theta - u\|_2^2) = \log(1/p) + \frac{\beta}{2} (2 \langle \theta, u \rangle - \|u\|_2^2).$$

Taking expectation under ρ_u , using (21) and $\|u\|_2 = 1$, yields

$$\text{KL}(\rho_u \|\pi) = \log(1/p) + \frac{\beta}{2} (2 \langle \mathbb{E}_{\rho_u} \theta, u \rangle - 1) = \log(1/p) + \frac{\beta}{2}.$$

Since $\rho_{u,v}$ and π_{2d} are product measures,

$$\text{KL}(\rho_{u,v} \|\pi_{2d}) = \text{KL}(\rho_u \|\pi) + \text{KL}(\rho_v \|\pi) = 2 \log(1/p) + \beta.$$

It remains to lower bound p . If $G \sim \mathcal{N}(0, \beta^{-1} I_d)$, then $\mathbb{E}[GG^\top] = \beta^{-1} I_d$, and therefore

$$\mathbb{E} \|\Sigma^{1/2} G\|_2^2 = \mathbb{E} \text{Tr}(\Sigma^{1/2} G G^\top \Sigma^{1/2}) = \text{Tr}(\Sigma \mathbb{E}[GG^\top]) = \frac{1}{\beta} \text{Tr}(\Sigma).$$

Choose

$$\beta = r(\Sigma) = \frac{\text{Tr}(\Sigma)}{\|\Sigma\|_{\text{op}}}.$$

Then $\mathbb{E} \|\Sigma^{1/2} G\|_2^2 = \|\Sigma\|_{\text{op}} = R^2/2$, and Markov's inequality yields

$$\mathbb{P}(\|\Sigma^{1/2} G\|_2 > R) \leq \frac{1}{2}, \quad \text{so} \quad p = \mathbb{P}(\|\Sigma^{1/2} G\|_2 \leq R) \geq \frac{1}{2}.$$

Since $\Sigma \neq 0$ implies $r(\Sigma) \geq 1$, we can bound

$$\text{KL}(\rho_{u,v} \|\pi_{2d}) = \beta + 2 \log(1/p) \leq r(\Sigma) + 2 \log 2 \leq C r(\Sigma),$$

and hence, for all $\delta \in (0, 1)$,

$$\text{KL}(\rho_{u,v} \|\pi_{2d}) + \log(1/\delta) \leq C(r(\Sigma) + \log(1/\delta)). \quad (24)$$

Insert (23) and (24) into (22). With probability at least $1 - \delta$, simultaneously for all $u, v \in S^{d-1}$,

$$\lambda u^\top (\widehat{\Sigma} - \Sigma) v \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2 + C_L \frac{r(\Sigma) + \log(1/\delta)}{n}, \quad 0 < \lambda \leq \frac{c_L}{\|\Sigma\|_{\text{op}}}.$$

The same inequality with $(-u, v)$ in place of (u, v) yields the corresponding lower bound, so we may write

$$\lambda |u^\top (\widehat{\Sigma} - \Sigma) v| \leq C_L \lambda^2 \|\Sigma\|_{\text{op}}^2 + C_L \frac{r(\Sigma) + \log(1/\delta)}{n}.$$

Divide by λ and take the supremum over $u, v \in S^{d-1}$. Using $\|A\|_{\text{op}} = \sup_{u, v \in S^{d-1}} |u^\top A v|$, we obtain

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \leq C_L \left(\lambda \|\Sigma\|_{\text{op}}^2 + \frac{r(\Sigma) + \log(1/\delta)}{\lambda n} \right), \quad 0 < \lambda \leq \frac{c_L}{\|\Sigma\|_{\text{op}}}.$$

Set $a = r(\Sigma) + \log(1/\delta)$ and choose

$$\lambda = \frac{1}{\|\Sigma\|_{\text{op}}} \min \left\{ c_L, \sqrt{\frac{a}{n}} \right\}.$$

If $\sqrt{a/n} \leq c_L$, then both terms in parentheses are at most $\|\Sigma\|_{\text{op}} \sqrt{a/n}$. If $\sqrt{a/n} > c_L$, then $a/n \geq c_L^2$ and the same bound is controlled by a constant multiple of $\|\Sigma\|_{\text{op}} a/n$. In either case,

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \leq C_L \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{a}{n}} + \frac{a}{n} \right),$$

and recalling $a = r(\Sigma) + \log(1/\delta)$ finishes the proof. $\square$

Remark 6.12. When $n \gtrsim r(\Sigma) + \log(1/\delta)$, the linear term is dominated by the square-root term, and the theorem simplifies to

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \lesssim \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma) + \log(1/\delta)}{n}}.$$

So the earlier bound appears as a clean large-sample corollary of the more precise statement above.

Example 6.13 (The isotropic case). If $\Sigma = I_d$, then $r(\Sigma) = d$, and Theorem 6.11 becomes: for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left\| \frac{1}{n} \sum_{i=1}^n X_i X_i^\top - I_d \right\|_{\text{op}} \leq C_L \left(\sqrt{\frac{d + \log(1/\delta)}{n}} + \frac{d + \log(1/\delta)}{n} \right).$$

This is the familiar operator-norm concentration of the empirical covariance around the identity.

6.4 Matrix Bernstein Inequality

The scalar Bernstein inequality from the previous section gives sharp control for sums of independent real random variables. There is a matrix analogue with essentially the same shape, but the route to it is genuinely different. The reason is noncommutativity. For real numbers,

$$e^{\theta(x_1 + \dots + x_n)} = \prod_{i=1}^n e^{\theta x_i},$$

so independence turns the moment generating function of a sum into a product of one-dimensional MGFs. For matrices this identity fails unless the summands commute. Thus the scalar Laplace-transform method does not pass to matrices verbatim: one needs new input from matrix analysis.

Throughout this section, matrices are real and symmetric unless explicitly stated otherwise. For a symmetric matrix $A \in \mathbf{R}^{d \times d}$, we write $A \succeq 0$ if $x^\top A x \geq 0$ for all $x \in \mathbf{R}^d$, and $A \preceq B$ if $B - A \succeq 0$. We also write

$$\|A\|_{\text{op}} = \max\{\lambda_{\max}(A), \lambda_{\max}(-A)\}.$$

One of the standard modern forms of matrix Bernstein is comparatively recent. The noncommutative concentration philosophy goes back to work of Ahlswede–Winter, while the now-classical short proof through matrix Laplace transforms was clarified and popularized by Tropp in the early 2010s. The statement is as follows.

Theorem 6.14 (Matrix Bernstein inequality). *Let $X_1, \dots, X_n$ be independent, mean-zero, symmetric $d \times d$ random matrices, and assume that*

$$\|X_i\|_{\text{op}} \leq K \quad \text{almost surely for all } i.$$

Define the variance proxy

$$V = \sum_{i=1}^n \mathbb{E}[X_i^2], \quad v = \|V\|_{\text{op}}.$$

Then for every $t \geq 0$,

$$\mathbb{P}\left(\lambda_{\max}\left(\sum_{i=1}^n X_i\right) \geq t\right) \leq d \exp\left(-\frac{t^2/2}{v + Kt/3}\right).$$

Consequently,

$$\mathbb{P}\left(\left\|\sum_{i=1}^n X_i\right\|_{\text{op}} \geq t\right) \leq 2d \exp\left(-\frac{t^2/2}{v + Kt/3}\right).$$

Equivalently, if

$$x = \log(2d/\delta),$$

then with probability at least $1 - \delta$,

$$\left\|\sum_{i=1}^n X_i\right\|_{\text{op}} \leq \frac{Kx}{3} + \sqrt{\frac{K^2 x^2}{9} + 2vx} \leq \sqrt{2vx} + \frac{2Kx}{3}.$$

Compared with the scalar Bernstein inequality, three new features appear. First, the role of the variance is played by the matrix quantity $V = \sum_i \mathbb{E}[X_i^2]$, and in the final bound only its operator norm matters. Second, boundedness still enters through the same scale parameter K . Third, we pay an additional dimension factor d (or $2d$ for the two-sided operator-norm form). If all matrices commute, then they are simultaneously diagonalizable and the result reduces essentially to applying scalar Bernstein in a common eigenbasis together with a union bound over the d diagonal coordinates.

6.5 Useful facts for the proof of Theorem 6.14

The proof uses several basic facts from matrix analysis. The first group concerns the spectral functional calculus.

Proposition 6.15 (Spectral calculus and order). *Let $A \in \mathbf{R}^{d \times d}$ be symmetric, and write its spectral decomposition as*

$$A = \sum_{j=1}^d \lambda_j u_j u_j^\top,$$

where $\lambda_1, \dots, \lambda_d \in \mathbf{R}$ are the eigenvalues and $u_1, \dots, u_d$ is an orthonormal basis of eigenvectors. For a function $f : \mathbf{R} \rightarrow \mathbf{R}$, define

$$f(A) = \sum_{j=1}^d f(\lambda_j) u_j u_j^\top.$$

Then the following hold.

(a) If $f(x) \leq g(x)$ for every x in the spectrum of A , then

$$f(A) \preceq g(A).$$

In particular, if $\|A\|_{\text{op}} \leq K$ and $f(x) \leq g(x)$ for all $x \in [-K, K]$, then $f(A) \preceq g(A)$.

(b) If A is positive definite, then $\log(A)$ is well-defined by the same spectral rule:

$$\log(A) = \sum_{j=1}^d \log(\lambda_j) u_j u_j^{\top}.$$

It satisfies

$$e^{\log(A)} = A.$$

If H is symmetric, then e^H is positive definite and

$$\log(e^H) = H.$$

Moreover, if $0 \prec A \preceq B$, then

$$\log(A) \preceq \log(B).$$

(c) If $A \preceq B$ and f is increasing on an interval containing the spectra of both A and B , then

$$\text{Tr}(f(A)) \leq \text{Tr}(f(B)).$$

In particular,

$$A \preceq B \quad \implies \quad \text{Tr}(e^A) \leq \text{Tr}(e^B).$$

Proof. Part (a) is immediate from the spectral decomposition:

$$g(A) - f(A) = \sum_{j=1}^d (g(\lambda_j) - f(\lambda_j)) u_j u_j^{\top} \succeq 0.$$

Part (b) is a classical theorem in matrix analysis: the scalar logarithm is not only increasing on $(0, \infty)$, but *operator-monotone* on the positive definite cone.

For part (c), the variational characterization of eigenvalues implies that

$$\lambda_j(A) \leq \lambda_j(B), \quad j = 1, \dots, d,$$

when the eigenvalues are ordered nonincreasingly. Hence

$$\text{Tr}(f(A)) = \sum_{j=1}^d f(\lambda_j(A)) \leq \sum_{j=1}^d f(\lambda_j(B)) = \text{Tr}(f(B)).$$

□

Remark 6.16. Part (b) is special and worth pausing over. The matrix logarithm is defined only on the positive definite cone, and on that cone it is operator-monotone. One should *not* expect this from an arbitrary increasing scalar function. For instance, the scalar function $x \mapsto e^x$ is increasing on $\mathbf{R}$, but in general

$$A \preceq B \quad \text{does not imply} \quad e^A \preceq e^B,$$

unless A and B commute. The same warning applies to x^2 . What remains true without commutativity is the weaker trace monotonicity from part (c).

The next result is one of the first indications that $\text{Tr}(e^\cdot)$ is the right noncommutative substitute for the scalar exponential.

Theorem 6.17 (Golden–Thompson inequality). *For symmetric matrices $H, K \in \mathbf{R}^{d \times d}$,*

$$\text{Tr}(e^{H+K}) \leq \text{Tr}(e^H e^K).$$

If H and K commute, then $e^{H+K} = e^H e^K$ and we recover equality. In general, commutativity fails, and Golden–Thompson says that after taking trace we still retain a one-sided substitute for the scalar identity. This already points in the correct direction, but it is not enough for a clean iteration over many independent summands. There is no equally simple n -term product formula that behaves well under conditioning. In the Tropp-style story of matrix concentration, Golden–Thompson explains why trace exponentials are natural, while Lieb’s theorem is the result that actually makes the iteration work.

Proposition 6.18 (Lieb’s concavity theorem). *Fix a symmetric matrix $H \in \mathbf{R}^{d \times d}$. Then the map*

$$A \longmapsto \text{Tr}(e^{H+\log(A)})$$

is concave on the cone of positive definite matrices. Equivalently, for every $\alpha \in [0, 1]$ and every positive definite A, B ,

$$\text{Tr}(e^{H+\log(\alpha A+(1-\alpha)B)}) \geq \alpha \text{Tr}(e^{H+\log A}) + (1-\alpha) \text{Tr}(e^{H+\log B}).$$

Remark 6.19. Lieb’s theorem is deep. Unlike the surrounding probabilistic argument, its proof belongs to matrix analysis and is decidedly nontrivial once $d > 1$. For matrix concentration it plays the same structural role that convexity and Jensen’s inequality play in one dimension: it is the mechanism that allows conditioning to enter the proof in a stable way.

The form we actually use is the following corollary.

Corollary 6.20 (A Jensen-type consequence of Lieb). *Let H be a deterministic symmetric matrix, and let Z be a random symmetric matrix such that $\mathbb{E}e^Z$ is finite. Then*

$$\mathbb{E} \text{Tr}(e^{H+Z}) \leq \text{Tr}(e^{H+\log(\mathbb{E}e^Z)}).$$

Proof. Set

$$\psi(A) = \text{Tr}(e^{H+\log A}),$$

defined for positive definite A . By Theorem 6.18, the function ψ is concave. Since e^Z is positive definite almost surely, Jensen’s inequality gives

$$\mathbb{E} \psi(e^Z) \leq \psi(\mathbb{E}e^Z).$$

Finally, because $\log(e^Z) = Z$ for symmetric Z , we obtain

$$\mathbb{E} \text{Tr}(e^{H+Z}) = \mathbb{E} \psi(e^Z) \leq \psi(\mathbb{E}e^Z) = \text{Tr}(e^{H+\log(\mathbb{E}e^Z)}).$$

□

The next proposition is the Tropp-style “matrix Laplace transform” bound. It is the correct noncommutative replacement for the scalar identity

$$\mathbb{E}e^{\theta \sum_i X_i} = \prod_i \mathbb{E}e^{\theta X_i}.$$

Proposition 6.21 (Matrix Laplace transform method). *Let $X_1, \dots, X_n$ be independent symmetric random matrices, and set*

$$S = \sum_{i=1}^n X_i.$$

Then for every $t \in \mathbf{R}$ and every $\theta > 0$,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq e^{-\theta t} \operatorname{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E} e^{\theta X_i} \right) \right).$$

Consequently,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq \inf_{\theta > 0} \left\{ e^{-\theta t} \operatorname{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E} e^{\theta X_i} \right) \right) \right\}.$$

Proof. Fix $\theta > 0$. By Markov's inequality,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) = \mathbb{P}(e^{\theta \lambda_{\max}(S)} \geq e^{\theta t}) \leq e^{-\theta t} \mathbb{E} e^{\theta \lambda_{\max}(S)}.$$

Since the eigenvalues of $e^{\theta S}$ are $e^{\theta \lambda_j(S)}$, we have

$$e^{\theta \lambda_{\max}(S)} = \lambda_{\max}(e^{\theta S}) \leq \operatorname{Tr}(e^{\theta S}),$$

and therefore

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq e^{-\theta t} \mathbb{E} \operatorname{Tr}(e^{\theta S}).$$

It remains to bound $\mathbb{E} \operatorname{Tr}(e^{\theta S})$. We condition on the last summand and apply Theorem 6.20 with

$$H = \theta \sum_{i=1}^{n-1} X_i, \quad Z = \theta X_n.$$

Because X_n is independent of $X_1, \dots, X_{n-1}$, this yields

$$\begin{aligned} \mathbb{E} \operatorname{Tr}(e^{\theta S}) &= \mathbb{E}_{1, \dots, n-1} \mathbb{E}_n \operatorname{Tr} \left(e^{\theta \sum_{i=1}^{n-1} X_i + \theta X_n} \right) \\ &\leq \mathbb{E}_{1, \dots, n-1} \operatorname{Tr} \left(\exp \left(\theta \sum_{i=1}^{n-1} X_i + \log \mathbb{E} e^{\theta X_n} \right) \right). \end{aligned}$$

Now the last term has the same form as before, except that one random summand has been replaced by the deterministic matrix $\log \mathbb{E} e^{\theta X_n}$. Repeating the argument for $X_{n-1}, X_{n-2}, \dots, X_1$, we arrive at

$$\mathbb{E} \operatorname{Tr}(e^{\theta S}) \leq \operatorname{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E} e^{\theta X_i} \right) \right).$$

Substituting this into the previous bound proves the claim. $\square$

6.6 Proof of Theorem 6.14

The only remaining ingredient is a one-step bound on the matrix MGF of a bounded summand. This is the matrix analogue of the scalar estimate used in the proof of Bernstein's inequality.

Lemma 6.22 (A matrix MGF bound for bounded summands). *Let X be a symmetric random matrix such that $\mathbb{E}X = 0$ and $\|X\|_{\text{op}} \leq K$ almost surely. Then for every $\theta \in (0, 3/K)$,*

$$\mathbb{E}e^{\theta X} \preceq \exp(g(\theta) \mathbb{E}[X^2]), \quad g(\theta) = \frac{\theta^2/2}{1 - \theta K/3}.$$

Proof. The scalar Bernstein argument gives the inequality

$$e^{\theta x} \leq 1 + \theta x + \frac{\theta^2/2}{1 - \theta K/3} x^2, \quad x \in [-K, K], \quad \theta \in (0, 3/K).$$

Applying Theorem 6.15(a) through the spectral calculus, we obtain

$$e^{\theta X} \preceq I + \theta X + g(\theta) X^2.$$

Taking expectations and using $\mathbb{E}X = 0$,

$$\mathbb{E}e^{\theta X} \preceq I + g(\theta) \mathbb{E}[X^2].$$

Since $\mathbb{E}[X^2] \succeq 0$, we may use the scalar inequality $1 + u \leq e^u$ through the spectral calculus once more to deduce

$$I + g(\theta) \mathbb{E}[X^2] \preceq \exp(g(\theta) \mathbb{E}[X^2]).$$

This proves the claim. □

Proof of Theorem 6.14. Let

$$S = \sum_{i=1}^n X_i.$$

Fix $\theta \in (0, 3/K)$. By Theorem 6.21,

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq e^{-\theta t} \text{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E}e^{\theta X_i} \right) \right).$$

By Theorem 6.22,

$$\mathbb{E}e^{\theta X_i} \preceq \exp(g(\theta) \mathbb{E}[X_i^2]), \quad g(\theta) = \frac{\theta^2/2}{1 - \theta K/3}.$$

Both sides are positive definite, so Theorem 6.15(b) gives

$$\log \mathbb{E}e^{\theta X_i} \preceq g(\theta) \mathbb{E}[X_i^2].$$

Summing over i and applying Theorem 6.15(c) with $f(x) = e^x$, we obtain

$$\text{Tr} \left(\exp \left(\sum_{i=1}^n \log \mathbb{E}e^{\theta X_i} \right) \right) \leq \text{Tr} \left(\exp \left(g(\theta) \sum_{i=1}^n \mathbb{E}[X_i^2] \right) \right) = \text{Tr}(e^{g(\theta)V}).$$

If $\mu_1, \dots, \mu_d$ are the eigenvalues of V , then $\mu_j \leq v$ for all j , so

$$\text{Tr}(e^{g(\theta)V}) = \sum_{j=1}^d e^{g(\theta)\mu_j} \leq d e^{g(\theta)v}.$$

Therefore

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq d \exp(-\theta t + g(\theta)v).$$

Choosing

$$\theta = \frac{t}{v + Kt/3}$$

when $v > 0$ (and arguing by continuity if $v = 0$) gives

$$\mathbb{P}(\lambda_{\max}(S) \geq t) \leq d \exp\left(-\frac{t^2/2}{v + Kt/3}\right).$$

This proves the one-sided estimate.

For the operator norm, observe that

$$\|S\|_{\text{op}} = \max\{\lambda_{\max}(S), \lambda_{\max}(-S)\}.$$

Since the matrices $-X_i$ satisfy the same assumptions and have the same variance proxy V , we may apply the one-sided bound to both S and $-S$, and then use the union bound:

$$\mathbb{P}(\|S\|_{\text{op}} \geq t) \leq 2d \exp\left(-\frac{t^2/2}{v + Kt/3}\right).$$

Finally, set $x = \log(2d/\delta)$. The inequality

$$2d \exp\left(-\frac{t^2/2}{v + Kt/3}\right) \leq \delta$$

is equivalent to

$$t^2 \geq 2x \left(v + \frac{Kt}{3}\right).$$

Solving this quadratic inequality in t gives

$$t \geq \frac{Kx}{3} + \sqrt{\frac{K^2 x^2}{9} + 2vx}.$$

Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, we further obtain the simpler sufficient bound

$$t \geq \sqrt{2vx} + \frac{2Kx}{3}.$$

This proves the high-probability form. □

6.7 Two Useful Extensions

The matrix Bernstein theorem is already quite flexible in its symmetric form. Two standard consequences are worth recording.

Controlling $\mathbb{E}\|S\|_{\text{op}}$. Integrating the tail bound from Theorem 6.14 yields

$$\mathbb{E} \left\| \sum_{i=1}^n X_i \right\|_{\text{op}} \lesssim \sqrt{v \log(2d)} + K \log(2d).$$

Thus the same two scales appear in expectation as in the high-probability bound: a square-root variance term and a linear boundedness term. Only the dimension enters logarithmically.

Rectangular matrices and Hermitian dilation. If $A \in \mathbf{R}^{d_1 \times d_2}$ is rectangular, the standard trick is to embed it into the symmetric block matrix

$$\mathcal{H}(A) = \begin{bmatrix} 0 & A \\ A^\top & 0 \end{bmatrix} \in \mathbf{R}^{(d_1+d_2) \times (d_1+d_2)}.$$

Its eigenvalues are $\pm s_1(A), \dots, \pm s_r(A)$, where $s_j(A)$ are the singular values of A , so

$$\lambda_{\max}(\mathcal{H}(A)) = \|A\|_{\text{op}}.$$

Moreover,

$$\mathcal{H}(A)^2 = \begin{bmatrix} AA^\top & 0 \\ 0 & A^\top A \end{bmatrix}.$$

This reduces the rectangular case to the symmetric one.

Corollary 6.23 (Rectangular matrix Bernstein). *Let $A_1, \dots, A_n$ be independent mean-zero random matrices in $\mathbf{R}^{d_1 \times d_2}$, and assume that*

$$\|A_i\|_{\text{op}} \leq K \quad \text{almost surely for all } i.$$

Define

$$v = \max \left\{ \left\| \sum_{i=1}^n \mathbb{E}[A_i A_i^\top] \right\|_{\text{op}}, \left\| \sum_{i=1}^n \mathbb{E}[A_i^\top A_i] \right\|_{\text{op}} \right\}.$$

Then for every $t \geq 0$,

$$\mathbb{P} \left(\left\| \sum_{i=1}^n A_i \right\|_{\text{op}} \geq t \right) \leq (d_1 + d_2) \exp \left(-\frac{t^2/2}{v + Kt/3} \right).$$

Proof. Apply the one-sided part of Theorem 6.14 to the symmetric matrices $\mathcal{H}(A_i)$. They are independent, mean-zero, symmetric, and satisfy

$$\|\mathcal{H}(A_i)\|_{\text{op}} = \|A_i\|_{\text{op}} \leq K.$$

Also,

$$\sum_{i=1}^n \mathbb{E}[\mathcal{H}(A_i)^2] = \begin{bmatrix} \sum_{i=1}^n \mathbb{E}[A_i A_i^\top] & 0 \\ 0 & \sum_{i=1}^n \mathbb{E}[A_i^\top A_i] \end{bmatrix},$$

whose operator norm is exactly v . Finally,

$$\lambda_{\max} \left(\sum_{i=1}^n \mathcal{H}(A_i) \right) = \left\| \sum_{i=1}^n A_i \right\|_{\text{op}}.$$

The result follows immediately. □

7 Gaussian Processes and Comparison Inequalities

We now turn to a family of results that is especially useful when the object of interest is not a single random variable, but the supremum of a random process. The guiding example is the *canonical Gaussian process*

$$X_t = \langle g, t \rangle, \quad t \in T \subseteq \mathbf{R}^d,$$

where $g \sim \mathcal{N}(0, I_d)$. Even this basic example already leads to nontrivial questions: how large can $\sup_{t \in T} X_t$ be, and how is it related to the geometry of the index set T ?

A recurrent theme is that for Gaussian processes the relevant geometry is encoded by the size of the increments. This leads to powerful comparison principles such as Slepian's lemma and the Sudakov–Fernique theorem, and from there to entropy bounds such as Sudakov minoration.

7.1 Gaussian processes and the canonical metric

Definition 7.1 (Gaussian process). Let T be an index set. A random family $(X_t)_{t \in T}$ is called a *Gaussian process* if for every finite subset

$$T_0 = \{t_1, \dots, t_m\} \subseteq T,$$

the random vector

$$(X_{t_1}, \dots, X_{t_m})$$

is multivariate Gaussian.

Definition 7.2 (Canonical metric). Let $(X_t)_{t \in T}$ be a Gaussian process. Its *canonical metric* is

$$d_X(s, t) = (\mathbb{E}(X_t - X_s)^2)^{1/2}, \quad s, t \in T.$$

Strictly speaking, d_X is in general only a pseudometric: it may happen that $d_X(s, t) = 0$ for distinct points $s \neq t$. In all entropy statements below, this causes no difficulty; equivalently, one may identify points at distance zero and work on the resulting metric quotient.

For many questions involving $\sup_{t \in T} X_t$, the canonical metric is the right quantity to look at. Indeed, adding a common Gaussian variable to all coordinates does not change the increments at all, and it also does not change the supremum except by a random shift. Thus the process is largely controlled by its increment structure rather than by its pointwise variances alone.

7.2 Slepian and Sudakov–Fernique inequalities

In many applications, one wants to compare the random quantities

$$\sup_{t \in T} X_t \quad \text{and} \quad \sup_{t \in T} Y_t$$

for two Gaussian processes $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$.

Remark 7.3 (Making T finite). To avoid measurability issues, one may think of

$$\mathbb{E} \sup_{t \in T} X_t$$

as shorthand for

$$\sup_{T_0 \subseteq T, |T_0| < \infty} \mathbb{E} \max_{t \in T_0} X_t.$$

Thus it is enough to prove the comparison inequalities below for finite index sets T ; the general separable case then follows by approximation. Thus, in the comparison theorems below, we will state the results for finite index sets.

Theorem 7.4 (Slepian inequality). *Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be two mean-zero Gaussian processes. Assume that for all $s, t \in T$,*

$$\mathbb{E}X_t^2 = \mathbb{E}Y_t^2 \quad \text{and} \quad \mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2.$$

Then for every $\tau \in \mathbf{R}$,

$$\mathbb{P} \left\{ \sup_{t \in T} X_t \geq \tau \right\} \leq \mathbb{P} \left\{ \sup_{t \in T} Y_t \geq \tau \right\}.$$

Consequently,

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t.$$

We prove Slepian inequality by Gaussian interpolation. The first step is Stein's identity in dimension one.

Lemma 7.5 (Stein's identity in dimension one). *Let $X \sim N(0, 1)$. Then for every differentiable function $f : \mathbf{R} \rightarrow \mathbf{R}$,*

$$\mathbb{E}[Xf(X)] = \mathbb{E}f'(X),$$

assuming both expectations exist and are finite.

Proof. Let

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

be the standard Gaussian density. Then

$$\mathbb{E}[Xf(X)] = \int_{\mathbf{R}} xf(x)\varphi(x) dx.$$

Since $\varphi'(x) = -x\varphi(x)$, integration by parts gives

$$\int_{\mathbf{R}} xf(x)\varphi(x) dx = - \int_{\mathbf{R}} f(x)\varphi'(x) dx = \int_{\mathbf{R}} f'(x)\varphi(x) dx = \mathbb{E}f'(X),$$

where the boundary term vanishes because φ decays exponentially. □

Lemma 7.6 (Multivariate Stein identity). *Let $X \sim N(0, \Sigma)$ in $\mathbf{R}^n$. Then for every differentiable function $f : \mathbf{R}^n \rightarrow \mathbf{R}$,*

$$\mathbb{E}[Xf(X)] = \Sigma \mathbb{E}\nabla f(X),$$

assuming all expectations exist and are finite. Equivalently, for every $i = 1, \dots, n$,

$$\mathbb{E}[X_i f(X)] = \sum_{j=1}^n \Sigma_{ij} \mathbb{E}[\partial_j f(X)].$$

Proof. Write $X = AZ$, where $Z \sim N(0, I_n)$ and $AA^\top = \Sigma$. Then

$$X_i = \sum_{k=1}^n A_{ik} Z_k,$$

so

$$\mathbb{E}[X_i f(X)] = \sum_{k=1}^n A_{ik} \mathbb{E}[Z_k f(AZ)].$$

Conditioning on all coordinates of Z except Z_k , and applying Theorem 7.5 to the k -th coordinate, we obtain

$$\mathbb{E}[Z_k f(AZ)] = \mathbb{E}[\partial_k(f(AZ))].$$

By the chain rule,

$$\partial_k(f(AZ)) = \sum_{j=1}^n A_{jk} \partial_j f(X).$$

Therefore

$$\mathbb{E}[X_i f(X)] = \sum_{k=1}^n A_{ik} \sum_{j=1}^n A_{jk} \mathbb{E}[\partial_j f(X)] = \sum_{j=1}^n (AA^\top)_{ij} \mathbb{E}[\partial_j f(X)] = \sum_{j=1}^n \Sigma_{ij} \mathbb{E}[\partial_j f(X)].$$

□

Lemma 7.7 (Gaussian interpolation). *Consider two independent Gaussian random vectors*

$$X \sim N(0, \Sigma^X), \quad Y \sim N(0, \Sigma^Y).$$

Define the interpolation Gaussian vector

$$Z(u) := \sqrt{u} X + \sqrt{1-u} Y, \quad u \in [0, 1].$$

Then for every twice differentiable function $f : \mathbf{R}^n \rightarrow \mathbf{R}$,

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{1}{2} \sum_{i,j=1}^n (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E}[\partial_{ij} f(Z(u))],$$

assuming all expectations exist and are finite.

Proof. Using the chain rule,

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \sum_{i=1}^n \mathbb{E} \left[\partial_i f(Z(u)) \frac{dZ_i(u)}{du} \right].$$

Since

$$\frac{dZ_i(u)}{du} = \frac{X_i}{2\sqrt{u}} - \frac{Y_i}{2\sqrt{1-u}},$$

we obtain

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{1}{2} \sum_{i=1}^n \mathbb{E} \left[\partial_i f(Z(u)) \left(\frac{X_i}{\sqrt{u}} - \frac{Y_i}{\sqrt{1-u}} \right) \right].$$

Let us compute the part containing X_i . Condition on Y , and define

$$g_i(x) := \partial_i f(\sqrt{u} x + \sqrt{1-u} Y).$$

Then

$$\frac{1}{\sqrt{u}} \mathbb{E}[X_i \partial_i f(Z(u)) \mid Y] = \frac{1}{\sqrt{u}} \mathbb{E}[X_i g_i(X) \mid Y].$$

Applying Theorem 7.6 to the Gaussian vector X , we get

$$\frac{1}{\sqrt{u}} \mathbb{E}[X_i g_i(X) \mid Y] = \frac{1}{\sqrt{u}} \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E}[\partial_j g_i(X) \mid Y].$$

By the chain rule,

$$\partial_j g_i(x) = \sqrt{u} \partial_{ij} f(\sqrt{u} x + \sqrt{1-u} Y).$$

Therefore

$$\frac{1}{\sqrt{u}} \mathbb{E}[X_i \partial_i f(Z(u)) \mid Y] = \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E}[\partial_{ij} f(Z(u)) \mid Y].$$

Taking expectation in Y , we obtain

$$\frac{1}{\sqrt{u}} \mathbb{E}[X_i \partial_i f(Z(u))] = \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E}[\partial_{ij} f(Z(u))].$$

Similarly,

$$\frac{1}{\sqrt{1-u}} \mathbb{E}[Y_i \partial_i f(Z(u))] = \sum_{j=1}^n \Sigma_{ij}^Y \mathbb{E}[\partial_{ij} f(Z(u))].$$

Substituting these identities into the derivative formula above proves the result. $\square$

Lemma 7.8 (Slepian inequality, functional form). *Consider two mean-zero Gaussian random vectors X and Y in $\mathbf{R}^n$. Assume that for all $i, j = 1, \dots, n$,*

$$\mathbb{E}X_i^2 = \mathbb{E}Y_i^2 \quad \text{and} \quad \mathbb{E}(X_i - X_j)^2 \leq \mathbb{E}(Y_i - Y_j)^2.$$

Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be twice differentiable and satisfy

$$\partial_{ij} f(x) \geq 0 \quad \text{for all } x \in \mathbf{R}^n \text{ and all } i \neq j.$$

Then

$$\mathbb{E}f(X) \geq \mathbb{E}f(Y),$$

assuming both expectations exist and are finite.

Proof. Let Σ^X and Σ^Y be the covariance matrices of X and Y . Since

$$\mathbb{E}(X_i - X_j)^2 = \Sigma_{ii}^X + \Sigma_{jj}^X - 2\Sigma_{ij}^X,$$

and similarly for Y , the assumptions imply that

$$\Sigma_{ii}^X = \Sigma_{ii}^Y \quad \text{and} \quad \Sigma_{ij}^X \geq \Sigma_{ij}^Y \quad \text{for all } i \neq j.$$

Since the conclusion depends only on the marginal laws of X and Y , we may take them independent. Applying Theorem 7.7 to

$$Z(u) = \sqrt{u} X + \sqrt{1-u} Y,$$

we get

$$\frac{d}{du} \mathbb{E}f(Z(u)) = \frac{1}{2} \sum_{i,j=1}^n (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E}[\partial_{ij} f(Z(u))].$$

The diagonal terms vanish because $\Sigma_{ii}^X = \Sigma_{ii}^Y$, so

$$\frac{d}{du} \mathbb{E}f(Z(u)) = \frac{1}{2} \sum_{i \neq j} (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E}[\partial_{ij} f(Z(u))].$$

Each term in this sum is nonnegative, hence

$$\frac{d}{du} \mathbb{E}f(Z(u)) \geq 0.$$

Therefore $u \mapsto \mathbb{E}f(Z(u))$ is increasing, and so

$$\mathbb{E}f(X) = \mathbb{E}f(Z(1)) \geq \mathbb{E}f(Z(0)) = \mathbb{E}f(Y).$$

□

Proof of Theorem 7.4. By Theorem 7.3, it is enough to consider finite T . Relabeling if needed, write

$$T = \{1, \dots, n\}, \quad X = (X_1, \dots, X_n), \quad Y = (Y_1, \dots, Y_n).$$

Fix $\tau \in \mathbf{R}$. For $\varepsilon > 0$, choose a twice differentiable, non-increasing function

$$h_\varepsilon : \mathbf{R} \rightarrow [0, 1]$$

such that

$$h_\varepsilon(x) = 1 \quad \text{for } x \leq \tau - \varepsilon, \quad h_\varepsilon(x) = 0 \quad \text{for } x \geq \tau.$$

Then $h_\varepsilon(x) \rightarrow \mathbf{1}_{(-\infty, \tau)}(x)$ pointwise as $\varepsilon \downarrow 0$.

Define

$$f_\varepsilon(x) := \prod_{i=1}^n h_\varepsilon(x_i), \quad x = (x_1, \dots, x_n) \in \mathbf{R}^n.$$

Then $f_\varepsilon(x) \rightarrow \mathbf{1}_{\{\max_{i \leq n} x_i < \tau\}}$ pointwise as $\varepsilon \downarrow 0$. Moreover, for $i \neq j$,

$$\partial_{ij} f_\varepsilon(x) = h'_\varepsilon(x_i) h'_\varepsilon(x_j) \prod_{k \notin \{i, j\}} h_\varepsilon(x_k) \geq 0,$$

because $h'_\varepsilon \leq 0$ and $0 \leq h_\varepsilon \leq 1$. Thus Theorem 7.8 gives

$$\mathbb{E}f_\varepsilon(X) \geq \mathbb{E}f_\varepsilon(Y).$$

Letting $\varepsilon \downarrow 0$ and using dominated convergence, we obtain

$$\mathbb{P} \left\{ \max_{i \leq n} X_i < \tau \right\} \geq \mathbb{P} \left\{ \max_{i \leq n} Y_i < \tau \right\}.$$

Equivalently,

$$\mathbb{P} \left\{ \max_{i \leq n} X_i \geq \tau \right\} \leq \mathbb{P} \left\{ \max_{i \leq n} Y_i \geq \tau \right\}.$$

This proves the tail comparison. The expectation comparison follows immediately. □

Theorem 7.9 (Sudakov–Fernique). *Let $X = (X_t)_{t \in T}$ and $Y = (Y_t)_{t \in T}$ be centered Gaussian processes. Assume that*

$$\mathbb{E}[(X_t - X_s)^2] \leq \mathbb{E}[(Y_t - Y_s)^2] \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \max_{t \in T} X_t \leq \mathbb{E} \max_{t \in T} Y_t.$$

Remark 7.10. We will use Theorem 7.9 repeatedly, but we will not prove it here. Compared with Theorem 7.4, the point is that one no longer needs the equality of pointwise variances. Only the increment structure matters.

7.3 Packing, covering, and Sudakov minoration

We now connect Gaussian suprema to the metric entropy of the index set. Recall the definition. Let T be a subset of a metric space (E, d) , and let $\varepsilon > 0$. The covering number $\mathcal{N}(T, d, \varepsilon)$ is the smallest number of closed d -balls of radius ε in E whose union covers T .

An ε -packing is a subset $P \subseteq T$ such that

$$d(s, t) > \varepsilon \quad \text{for all distinct } s, t \in P.$$

Also recall that we have

$$\mathcal{N}(T, d, \varepsilon) \leq \mathcal{P}(T, d, \varepsilon).$$

Theorem 7.11 (Sudakov minoration). *Let $(X_t)_{t \in T}$ be a centered separable Gaussian process, and let*

$$d_X(s, t) = (\mathbb{E}(X_t - X_s)^2)^{1/2}$$

be its canonical metric. Then for every $\varepsilon > 0$,

$$\varepsilon \sqrt{\log \mathcal{N}(T, d_X, \varepsilon)} \leq c \mathbb{E} \sup_{t \in T} X_t,$$

where $c > 0$ is an absolute constant.

Proof. Fix $\varepsilon > 0$. If $\mathcal{N}(T, d_X, \varepsilon) = \infty$, then $\mathcal{P}(T, d_X, \varepsilon) = \infty$ as well. Hence for every integer $m \geq 1$, there exists an ε -packing

$$P_{\varepsilon, m} = \{t_1, \dots, t_m\} \subseteq T.$$

Applying the argument below to $P_{\varepsilon, m}$ shows that

$$\mathbb{E} \sup_{t \in T} X_t \geq c \varepsilon \sqrt{\log m}$$

for all m , and therefore $\mathbb{E} \sup_{t \in T} X_t = +\infty$. In this case the desired inequality is trivial. We may therefore assume that $\mathcal{N}(T, d_X, \varepsilon) < \infty$.

Since $\mathcal{N}(T, d_X, \varepsilon) \leq \mathcal{P}(T, d_X, \varepsilon)$, we can choose a finite ε -packing

$$P_\varepsilon = \{t_1, \dots, t_m\} \subseteq T$$

with

$$m \geq \mathcal{N}(T, d_X, \varepsilon).$$

Of course,

$$\mathbb{E} \sup_{t \in T} X_t \geq \mathbb{E} \max_{1 \leq j \leq m} X_{t_j}.$$

Now let $g_1, \dots, g_m \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, independent of everything else, and define a new Gaussian process

$$Y_{t_j} = \frac{\varepsilon}{\sqrt{2}} g_j, \quad j = 1, \dots, m.$$

For distinct i, j ,

$$\mathbb{E}(Y_{t_i} - Y_{t_j})^2 = \frac{\varepsilon^2}{2} \mathbb{E}(g_i - g_j)^2 = \varepsilon^2.$$

On the other hand, because P_ε is an ε -packing,

$$\mathbb{E}(X_{t_i} - X_{t_j})^2 = d_X(t_i, t_j)^2 > \varepsilon^2.$$

Therefore

$$\mathbb{E}(Y_{t_i} - Y_{t_j})^2 \leq \mathbb{E}(X_{t_i} - X_{t_j})^2 \quad \text{for all } i, j,$$

and Theorem 7.9 yields

$$\mathbb{E} \max_{1 \leq j \leq m} Y_{t_j} \leq \mathbb{E} \max_{1 \leq j \leq m} X_{t_j} \leq \mathbb{E} \sup_{t \in T} X_t.$$

Finally,

$$\mathbb{E} \max_{1 \leq j \leq m} Y_{t_j} = \frac{\varepsilon}{\sqrt{2}} \mathbb{E} \max_{1 \leq j \leq m} g_j.$$

A standard estimate for the maximum of m independent standard Gaussians gives

$$\mathbb{E} \max_{1 \leq j \leq m} g_j \geq c_0 \sqrt{\log m}$$

for an absolute constant $c_0 > 0$. Hence

$$\mathbb{E} \sup_{t \in T} X_t \geq \frac{c_0}{\sqrt{2}} \varepsilon \sqrt{\log m} \geq \frac{c_0}{\sqrt{2}} \varepsilon \sqrt{\log \mathcal{N}(T, d_X, \varepsilon)}.$$

This proves the theorem. □

7.4 Canonical Gaussian process: covering-number examples

A particularly important example is the canonical Gaussian process on a set $T \subseteq \mathbf{R}^d$:

$$X_t = \langle g, t \rangle, \quad t \in T, \quad g \sim \mathcal{N}(0, I_d).$$

Its canonical metric is exactly the Euclidean metric. Indeed,

$$d_X(s, t)^2 = \mathbb{E}(\langle g, t - s \rangle)^2 = \|t - s\|_2^2.$$

Thus Theorem 7.11 translates bounds on Gaussian suprema into covering-number bounds in Euclidean space.

Example 7.12 (An ellipsoid). Let

$$T = \Sigma^{1/2} B_2^d = \{\Sigma^{1/2} u : \|u\|_2 \leq 1\},$$

where $\Sigma \succeq 0$. Then

$$\sup_{t \in T} \langle g, t \rangle = \sup_{\|u\|_2 \leq 1} \langle \Sigma^{1/2} g, u \rangle = \|\Sigma^{1/2} g\|_2.$$

Therefore

$$\mathbb{E} \sup_{t \in T} \langle g, t \rangle = \mathbb{E} \|\Sigma^{1/2} g\|_2 \leq \sqrt{\mathbb{E} \|\Sigma^{1/2} g\|_2^2} = \sqrt{\text{Tr}(\Sigma)},$$

where we used Jensen's inequality. Applying Theorem 7.11, we obtain

$$\log \mathcal{N}(\Sigma^{1/2} B_2^d, \|\cdot\|_2, \varepsilon) \leq \frac{c \text{Tr}(\Sigma)}{\varepsilon^2}.$$

Example 7.13 (The ℓ_1 ball). Let

$$T = B_1^d = \{x \in \mathbf{R}^d : \|x\|_1 \leq 1\}.$$

Then

$$\sup_{t \in B_1^d} \langle g, t \rangle = \|g\|_\infty,$$

so

$$\mathbb{E} \sup_{t \in B_1^d} \langle g, t \rangle = \mathbb{E} \|g\|_\infty.$$

By Theorem 3.25 applied to the $2d$ centered Gaussian variables

$$g_1, \dots, g_d, -g_1, \dots, -g_d,$$

we get

$$\mathbb{E} \|g\|_\infty = \mathbb{E} \max_{1 \leq i \leq d} |g_i| \leq \sqrt{2 \log(2d)}.$$

Therefore Theorem 7.11 yields

$$\log \mathcal{N}(B_1^d, \|\cdot\|_2, \varepsilon) \leq \frac{c \log(2d)}{\varepsilon^2}.$$

This estimate is very effective when ε is not too small. For smaller ε , a volumetric argument gives a complementary bound. Indeed, the inclusions

$$B_1^d \subseteq B_2^d \subseteq \sqrt{d} B_1^d$$

imply

$$B_1^d + \frac{\varepsilon}{2} B_2^d \subseteq \left(1 + \frac{\varepsilon \sqrt{d}}{2}\right) B_1^d.$$

Hence

$$\begin{aligned} \mathcal{N}(B_1^d, \|\cdot\|_2, \varepsilon) &\leq \frac{\text{Vol}(B_1^d + \frac{\varepsilon}{2} B_2^d)}{\text{Vol}((\varepsilon/2) B_2^d)} \\ &\leq \frac{\text{Vol}\left(\left(1 + \frac{\varepsilon \sqrt{d}}{2}\right) B_1^d\right)}{\text{Vol}((\varepsilon/2) B_2^d)} \\ &= \left(1 + \frac{\varepsilon \sqrt{d}}{2}\right)^d \left(\frac{2}{\varepsilon}\right)^d \frac{\text{Vol}(B_1^d)}{\text{Vol}(B_2^d)}. \end{aligned}$$

Using the standard bound

$$\frac{\text{Vol}(B_1^d)}{\text{Vol}(B_2^d)} \leq \left(\frac{c}{\sqrt{d}}\right)^d,$$

we arrive at

$$\mathcal{N}(B_1^d, \|\cdot\|_2, \varepsilon) \leq \left(c \left(1 + \frac{2}{\varepsilon \sqrt{d}}\right)\right)^d.$$

Equivalently,

$$\log \mathcal{N}(B_1^d, \|\cdot\|_2, \varepsilon) \leq d \log \left(c \left(1 + \frac{2}{\varepsilon \sqrt{d}}\right)\right).$$

Combining the two estimates, we get

$$\log \mathcal{N}(B_1^d, \|\cdot\|_2, \varepsilon) \leq \min \left\{ d \log \left(c \left(1 + \frac{2}{\varepsilon \sqrt{d}}\right)\right), \frac{c \log(2d)}{\varepsilon^2} \right\}.$$

In particular, when $\varepsilon \asymp d^{-1/2}$, the volumetric bound is of order d , whereas the Sudakov bound is of order $d \log d$. Thus the volumetric argument is better in the very small-scale regime, while Sudakov minoration becomes more informative at larger scales.

7.5 Dual Sudakov minoration

Sudakov minoration bounds the entropy of a set in terms of the Gaussian supremum over that same set. There is a dual statement in which one covers the Euclidean ball by translates of a polar body.

Definition 7.14 (Polar set). Let $T \subseteq \mathbf{R}^d$ be convex and symmetric, meaning $T = -T$. Its *polar* is

$$T^\circ = \left\{ y \in \mathbf{R}^d : \sup_{x \in T} \langle x, y \rangle \leq 1 \right\}.$$

A few standard examples are useful to keep in mind:

- $(B_2^d)^\circ = B_2^d$;
- if $1 \leq p \leq \infty$ and q is the conjugate exponent, then

$$(B_p^d)^\circ = B_q^d;$$

- if T is an ellipsoid with semi-axes $a_1, \dots, a_d > 0$, then T° is an ellipsoid with semi-axes $a_1^{-1}, \dots, a_d^{-1}$.

We will also use the set-based covering-number notation

$$\mathcal{N}(A, K),$$

which denotes the minimum number of translates of K needed to cover A . Thus

$$\mathcal{N}(T, \|\cdot\|_2, \varepsilon) = \mathcal{N}(T, \varepsilon B_2^d).$$

Theorem 7.15 (Dual Sudakov minoration). *Let $T \subseteq \mathbf{R}^d$ be a symmetric convex body. Then for every $\varepsilon > 0$,*

$$\varepsilon \sqrt{\log \mathcal{N}(B_2^d, \varepsilon T^\circ)} \leq c \mathbb{E} \sup_{t \in T} \langle g, t \rangle,$$

where $g \sim \mathcal{N}(0, I_d)$ and $c > 0$ is an absolute constant.

We will not prove Theorem 7.15 here. The important point is that it gives entropy bounds for coverings of the Euclidean ball by a body dual to the one defining the Gaussian supremum.

Example 7.16 (A Euclidean-ball covering estimate). Let

$$T = \Sigma^{1/2} B_2^d, \quad \Sigma \succ 0.$$

By Theorem 7.12,

$$\mathbb{E} \sup_{t \in T} \langle g, t \rangle \leq \sqrt{\text{Tr}(\Sigma)}.$$

On the other hand,

$$T^\circ = \left\{ y \in \mathbf{R}^d : \sup_{\|u\|_2 \leq 1} \langle \Sigma^{1/2} u, y \rangle \leq 1 \right\} = \left\{ y \in \mathbf{R}^d : \|\Sigma^{1/2} y\|_2 \leq 1 \right\}.$$

Thus covering B_2^d by translates of εT° is the same as covering B_2^d by balls of radius ε for the metric

$$d_\Sigma(s, t) = \|\Sigma^{1/2}(s - t)\|_2.$$

Indeed,

$$s \in t + \varepsilon T^\circ \iff \|\Sigma^{1/2}(s - t)\|_2 \leq \varepsilon.$$

Therefore Theorem 7.15 yields

$$\log \mathcal{N}(B_2^d, d_\Sigma, \varepsilon) \leq \frac{c \text{Tr}(\Sigma)}{\varepsilon^2}.$$

7.6 Gaussian width

The quantity that appears repeatedly in the previous discussion deserves its own name.

Definition 7.17 (Gaussian width). Let $T \subseteq \mathbf{R}^d$ be nonempty, and let $g \sim \mathcal{N}(0, I_d)$. The *Gaussian width* of T is

$$W(T) = \mathbb{E} \sup_{t \in T} \langle g, t \rangle \in [0, \infty].$$

Geometrically, $W(T)$ is the expected support function of T in a random Gaussian direction. It is a very natural size parameter in high-dimensional probability: it appears in comparison theorems, in entropy estimates, in random matrix bounds, and later in learning-theoretic applications.

Proposition 7.18 (Basic properties of Gaussian width). *Let $T, K \subseteq \mathbf{R}^d$ be nonempty sets, let Q be an orthogonal matrix, and let $y \in \mathbf{R}^d$. Then:*

1. $W(T) < \infty$ if and only if T is bounded;
2. $W(QT + y) = W(T)$;
3. $W(T + K) = W(T) + W(K)$, where

$$T + K = \{t + k : t \in T, k \in K\};$$

4. for every $\alpha \in \mathbf{R}$,

$$W(\alpha T) = |\alpha| W(T);$$

- 5.

$$W(T) = \frac{1}{2} W(T - T) = \frac{1}{2} \mathbb{E} \sup_{x, y \in T} \langle g, x - y \rangle;$$

6. if T is finite, then

$$W(T) \leq \sqrt{2 \log |T|} \operatorname{diam}(T),$$

where

$$\operatorname{diam}(T) = \sup_{x, y \in T} \|x - y\|_2.$$

Proof. (1) If T is bounded, then

$$\sup_{t \in T} \langle g, t \rangle \leq \|g\|_2 \sup_{t \in T} \|t\|_2,$$

so $W(T) < \infty$.

Conversely, suppose T is unbounded. Choose $t_k \in T$ with $\|t_k\|_2 \rightarrow \infty$, and write $u_k = t_k / \|t_k\|_2$. By compactness of the unit sphere, some subsequence u_{k_ℓ} converges to a unit vector u . On the event $\langle g, u \rangle > 0$, which has probability $1/2$, we have $\langle g, u_{k_\ell} \rangle > \langle g, u \rangle / 2$ for all large ℓ , and hence

$$\langle g, t_{k_\ell} \rangle = \|t_{k_\ell}\|_2 \langle g, u_{k_\ell} \rangle \rightarrow \infty.$$

Therefore $\sup_{t \in T} \langle g, t \rangle = +\infty$ with positive probability, so $W(T) = +\infty$.

(2) We have

$$W(QT + y) = \mathbb{E} \sup_{t \in T} \langle g, Qt + y \rangle = \mathbb{E} \sup_{t \in T} (\langle Q^\top g, t \rangle + \langle g, y \rangle).$$

Since $Q^\top g \sim \mathcal{N}(0, I_d)$ and $\mathbb{E} \langle g, y \rangle = 0$, this gives $W(QT + y) = W(T)$.

(3) For every fixed g ,

$$\sup_{z \in T+K} \langle g, z \rangle = \sup_{t \in T, k \in K} \langle g, t+k \rangle = \sup_{t \in T} \langle g, t \rangle + \sup_{k \in K} \langle g, k \rangle.$$

Taking expectations proves the claim.

(4) If $\alpha \geq 0$, then

$$W(\alpha T) = \alpha W(T).$$

If $\alpha < 0$, then

$$W(\alpha T) = |\alpha| \mathbb{E} \sup_{t \in T} \langle -g, t \rangle = |\alpha| W(T),$$

since $-g$ has the same law as g .

(5) For every fixed g ,

$$\sup_{x, y \in T} \langle g, x-y \rangle = \sup_{x \in T} \langle g, x \rangle + \sup_{y \in T} \langle -g, y \rangle.$$

Taking expectations and using the symmetry of g gives

$$W(T - T) = 2W(T).$$

(6) Fix $t_0 \in T$. Then

$$W(T) = \mathbb{E} \max_{t \in T} \langle g, t \rangle = \mathbb{E} \max_{t \in T} \langle g, t - t_0 \rangle,$$

because $\mathbb{E} \langle g, t_0 \rangle = 0$. For each $t \in T$, the random variable $\langle g, t - t_0 \rangle$ is centered Gaussian with variance

$$\|t - t_0\|_2^2 \leq \text{diam}(T)^2.$$

Applying Theorem 3.25 to the family $\{\langle g, t - t_0 \rangle : t \in T\}$, we obtain

$$W(T) \leq \sqrt{2 \log |T|} \text{diam}(T).$$

□

Example 7.19 (Basic Gaussian width computations). Let $g \sim \mathcal{N}(0, I_d)$. Then:

1. For the Euclidean ball,

$$W(B_2^d) = \mathbb{E} \sup_{\|t\|_2 \leq 1} \langle g, t \rangle = \mathbb{E} \|g\|_2 \leq \sqrt{\mathbb{E} \|g\|_2^2} = \sqrt{d}.$$

2. For an ellipsoid,

$$W(\Sigma^{1/2} B_2^d) = \mathbb{E} \sup_{\|u\|_2 \leq 1} \langle \Sigma^{1/2} g, u \rangle = \mathbb{E} \|\Sigma^{1/2} g\|_2 \leq \sqrt{\text{Tr}(\Sigma)}.$$

3. For the ℓ_1 ball,

$$W(B_1^d) = \mathbb{E} \|g\|_\infty \leq \sqrt{2 \log(2d)}.$$

4. For the ℓ_∞ ball,

$$W(B_\infty^d) = \mathbb{E} \sup_{\|t\|_\infty \leq 1} \langle g, t \rangle = \mathbb{E} \|g\|_1 = \sum_{i=1}^d \mathbb{E} |g_i| = d \sqrt{\frac{2}{\pi}}.$$

7.7 A useful consequence: Gaussian contraction

We end with a very useful corollary of Theorem 7.9. It says that applying coordinatewise 1-Lipschitz maps can only decrease Gaussian width.

Corollary 7.20 (Gaussian contraction inequality). *Let $T \subseteq \mathbf{R}^d$, let $g = (g_1, \dots, g_d) \sim \mathcal{N}(0, I_d)$, and let*

$$\phi_1, \dots, \phi_d : \mathbf{R} \rightarrow \mathbf{R}$$

be 1-Lipschitz functions. Then

$$\mathbb{E} \sup_{t \in T} \sum_{i=1}^d g_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^d g_i t_i = W(T).$$

Proof. Define two centered Gaussian processes on T by

$$X_t = \sum_{i=1}^d g_i \phi_i(t_i), \quad Y_t = \sum_{i=1}^d g_i t_i.$$

For any $s, t \in T$,

$$\mathbb{E}(X_t - X_s)^2 = \sum_{i=1}^d (\phi_i(t_i) - \phi_i(s_i))^2 \leq \sum_{i=1}^d (t_i - s_i)^2 = \mathbb{E}(Y_t - Y_s)^2,$$

since each ϕ_i is 1-Lipschitz. By the Sudakov–Fernique theorem,

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t.$$

The right-hand side equals $\mathbb{E} \sup_{t \in T} \langle g, t \rangle = W(T)$ by definition, which proves the claim. $\square$

8 Generic chaining

We now return to Gaussian processes from a more structural point of view. In the previous section we saw that the size of

$$\mathbb{E} \sup_{t \in T} X_t$$

is governed by the geometry of the index set through the increment metric. Sudakov minoration gave a lower bound in terms of covering numbers. The next question is the upper bound.

A natural first answer is Dudley’s entropy integral. It is already a multiscale argument, and in that sense it is already a chaining bound. But from the conceptual point of view, the more fundamental object is Talagrand’s γ_2 -functional. So we will organize the material in the following order: first we prove the generic chaining upper bound in terms of γ_2 , and only then we recover Dudley’s entropy integral as a corollary.

Historically, Dudley’s entropy bound was one of the first systematic upper bounds of this type. Fernique then introduced the majorizing-measure point of view, and Talagrand proved that the resulting geometric quantity gives the correct order of $\mathbb{E} \sup_{t \in T} X_t$ for Gaussian processes. This majorizing measure theorem is one of the landmarks of the modern theory of Gaussian processes.

Throughout this section, the upper bounds will only use the increment condition

$$\|X_t - X_s\|_{\psi_2} \leq L d(s, t), \quad s, t \in T,$$

for some metric d on T and some constant $L > 0$. So exact Gaussianity is not needed for the upper half of the theory. The genuinely Gaussian step is the matching lower bound.

A canonical example to keep in mind is

$$X_t = \langle g, t \rangle, \quad t \in T \subseteq \mathbf{R}^d,$$

where $g \sim \mathcal{N}(0, I_d)$. Then

$$X_t - X_s = \langle g, t - s \rangle \sim \mathcal{N}(0, \|t - s\|_2^2),$$

and therefore

$$\|X_t - X_s\|_{\psi_2} \lesssim \|t - s\|_2.$$

So the canonical Gaussian process has sub-Gaussian increments with respect to the Euclidean metric.

8.1 Admissible sequences and the γ_2 functional

The generic chaining functional is built from multiscale approximations whose sizes are prescribed in advance.

Definition 8.1 (Admissible sequence). Let T be a nonempty set. A sequence of subsets

$$T_0, T_1, T_2, \dots \subseteq T$$

is called *admissible* if

$$|T_0| = 1 \quad \text{and} \quad |T_k| \leq 2^{2^k} \quad \text{for all } k \geq 1.$$

Definition 8.2 (Talagrand's γ_2 functional). Let (T, d) be a metric space. We define

$$\gamma_2(T, d) = \inf_{(T_k) \text{ admissible}} \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k),$$

where

$$d(t, T_k) = \inf_{s \in T_k} d(t, s).$$

Two simple facts follow immediately from the definition. First, for every $c \geq 0$,

$$\gamma_2(T, cd) = c \gamma_2(T, d).$$

Second, if $d_1 \leq d_2$ pointwise on $T \times T$, then

$$\gamma_2(T, d_1) \leq \gamma_2(T, d_2).$$

The particular cardinality bound 2^{2^k} is not accidental. At scale k , the proof will compare points in T_k with points in T_{k+1} . The number of such pairs is at most

$$|T_k| |T_{k+1}| \leq 2^{2^k} \cdot 2^{2^{k+1}} = 2^{3 \cdot 2^k},$$

so its logarithm is of order 2^k . After taking a square root, the natural probabilistic price is exactly $2^{k/2}$. This is why the geometry of γ_2 is perfectly matched to the sub-Gaussian union-bound computation.

8.2 The generic chaining upper bound

We now prove the main upper bound. As promised, the proof uses only sub-Gaussian increments.

Theorem 8.3 (Generic chaining upper bound). *Let $(X_t)_{t \in T}$ be a centered process such that*

$$\|X_t - X_s\|_{\psi_2} \leq L d(s, t) \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL \gamma_2(T, d),$$

where $C > 0$ is an absolute constant.

Proof. We prove the result first when T is finite.

Choose an admissible sequence $(T_k)_{k \geq 0}$. Since T is finite, by enlarging the sequence if necessary we may assume that

$$T_k = T \quad \text{for all sufficiently large } k.$$

Let $T_0 = \{t_0\}$. For each $t \in T$ and each $k \geq 0$, choose a point $\pi_k(t) \in T_k$ such that

$$d(t, \pi_k(t)) = d(t, T_k).$$

Because $T_k = T$ for all large k , we have $\pi_k(t) = t$ eventually. Therefore, for every $t \in T$,

$$X_t - X_{t_0} = \sum_{k=0}^{\infty} (X_{\pi_{k+1}(t)} - X_{\pi_k(t)}),$$

with only finitely many nonzero terms.

Now fix $u \geq 1$. Consider the event

$$\Omega_u = \bigcap_{k=0}^{\infty} \bigcap_{a \in T_k, b \in T_{k+1}} \left\{ |X_b - X_a| \leq C_1 L u 2^{k/2} d(a, b) \right\},$$

where $C_1 > 0$ is a constant to be chosen.

For a fixed k , $a \in T_k$, and $b \in T_{k+1}$, the increment $X_b - X_a$ is centered and sub-Gaussian, with

$$\|X_b - X_a\|_{\psi_2} \leq L d(a, b).$$

Hence, by Theorem 2.13(1),

$$\mathbb{P}\left(|X_b - X_a| > C_1 L u 2^{k/2} d(a, b)\right) \leq 2 \exp(-c C_1^2 u^2 2^k)$$

for an absolute constant $c > 0$.

On the other hand,

$$|T_k| |T_{k+1}| \leq 2^{2^k} \cdot 2^{2^{k+1}} \leq 2^{2^{k+2}}.$$

Therefore, by the union bound,

$$\mathbb{P}(\Omega_u^c) \leq 2 \sum_{k=0}^{\infty} 2^{2^{k+2}} \exp(-c C_1^2 u^2 2^k).$$

Choose C_1 so large that

$$cC_1^2 \geq 4 \log 2 + 1.$$

Then for every $u \geq 1$,

$$2^{2^{k+2}} \exp(-cC_1^2 u^2 2^k) \leq \exp(-u^2 2^k),$$

and consequently

$$\mathbb{P}(\Omega_u^c) \leq 2 \sum_{k=0}^{\infty} e^{-u^2 2^k} \leq 4e^{-u^2}.$$

Now work on the event Ω_u . For every $t \in T$,

$$\begin{aligned} X_t - X_{t_0} &= \sum_{k=0}^{\infty} (X_{\pi_{k+1}(t)} - X_{\pi_k(t)}) \\ &\leq \sum_{k=0}^{\infty} |X_{\pi_{k+1}(t)} - X_{\pi_k(t)}| \\ &\leq C_1 L u \sum_{k=0}^{\infty} 2^{k/2} d(\pi_{k+1}(t), \pi_k(t)). \end{aligned}$$

By the triangle inequality,

$$d(\pi_{k+1}(t), \pi_k(t)) \leq d(t, \pi_{k+1}(t)) + d(t, \pi_k(t)) = d(t, T_{k+1}) + d(t, T_k).$$

Hence

$$\begin{aligned} \sum_{k=0}^{\infty} 2^{k/2} d(\pi_{k+1}(t), \pi_k(t)) &\leq \sum_{k=0}^{\infty} 2^{k/2} (d(t, T_k) + d(t, T_{k+1})) \\ &\leq 2 \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k). \end{aligned}$$

Therefore, on Ω_u ,

$$\sup_{t \in T} (X_t - X_{t_0}) \leq 2C_1 L u \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k).$$

Set

$$A = \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k).$$

We have shown that

$$\mathbb{P}\left(\sup_{t \in T} (X_t - X_{t_0}) > 2C_1 L u A\right) \leq 4e^{-u^2}, \quad u \geq 1.$$

Since $t = t_0$ gives $X_t - X_{t_0} = 0$, the random variable

$$Z = \sup_{t \in T} (X_t - X_{t_0})$$

is nonnegative. Using the tail-integral formula,

$$\begin{aligned} \mathbb{E}Z &= \int_0^{\infty} \mathbb{P}(Z > s) ds \\ &\leq 2C_1 L A + \int_{2C_1 L A}^{\infty} \mathbb{P}(Z > s) ds. \end{aligned}$$

With the substitution $s = 2C_1 LA u$, the tail estimate gives

$$\mathbb{E}Z \leq 2C_1 LA + 2C_1 LA \int_1^\infty 4e^{-u^2} du \leq C_2 LA$$

for an absolute constant $C_2 > 0$.

Finally,

$$\mathbb{E} \sup_{t \in T} X_t = \mathbb{E} X_{t_0} + \mathbb{E} \sup_{t \in T} (X_t - X_{t_0}) = \mathbb{E} Z,$$

because $\mathbb{E} X_{t_0} = 0$. Thus

$$\mathbb{E} \sup_{t \in T} X_t \leq C_2 LA.$$

Since this holds for every admissible sequence, taking the infimum over all admissible sequences yields

$$\mathbb{E} \sup_{t \in T} X_t \leq C_2 L \gamma_2(T, d).$$

For separable processes, one passes to the limit over increasing finite subsets in the standard way. $\square$

Remark 8.4. At this stage, the main upper bound is already proved. Nothing beyond sub-Gaussian increment control was used. This is one of the reasons generic chaining is so robust: the upper half of the theory is not specifically Gaussian.

8.3 Dudley's entropy integral as a corollary

We now compare γ_2 with covering numbers. This is the step that turns the abstract chaining functional into the familiar entropy integral.

Proposition 8.5 (Comparing γ_2 with entropy). *Let (T, d) be a bounded metric space, and write*

$$D = \sup_{s, t \in T} d(s, t).$$

Then

$$\gamma_2(T, d) \leq C \int_0^D \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon,$$

where $C > 0$ is an absolute constant.

Proof. If $\mathcal{N}(T, d, \varepsilon) = \infty$ for some $\varepsilon > 0$, then the right-hand side is $+\infty$, and there is nothing to prove. So we may assume that the relevant covering numbers are finite.

For each $k \geq 0$, define

$$\varepsilon_k = \inf \{ \varepsilon > 0 : \mathcal{N}(T, d, \varepsilon) \leq 2^{2^k} \}.$$

Since $\mathcal{N}(T, d, D) = 1$, the quantity ε_k is well-defined, and $\varepsilon_k \downarrow 0$.

Choose $T_k \subseteq T$ to be an ε_k -net with

$$|T_k| \leq 2^{2^k}.$$

Then (T_k) is admissible, and therefore

$$\gamma_2(T, d) \leq \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k) \leq \sum_{k=0}^{\infty} 2^{k/2} \varepsilon_k.$$

It remains to compare the last sum with the integral. By the definition of ε_k , whenever $\varepsilon < \varepsilon_k$,

$$\mathcal{N}(T, d, \varepsilon) > 2^{2^k}.$$

Hence for every $\varepsilon \in [\varepsilon_{k+1}, \varepsilon_k)$,

$$\sqrt{\log \mathcal{N}(T, d, \varepsilon)} \geq \sqrt{\log(2^{2^k})} = \sqrt{\log 2} 2^{k/2}.$$

Therefore

$$\int_{\varepsilon_{k+1}}^{\varepsilon_k} \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon \geq \sqrt{\log 2} 2^{k/2} (\varepsilon_k - \varepsilon_{k+1}).$$

Summing over $k \geq 0$, we obtain

$$\int_0^D \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon \geq c \sum_{k=0}^{\infty} 2^{k/2} (\varepsilon_k - \varepsilon_{k+1})$$

for an absolute constant $c > 0$.

Now observe that

$$\sum_{k=0}^{\infty} 2^{k/2} (\varepsilon_k - \varepsilon_{k+1}) = \varepsilon_0 + \sum_{k=1}^{\infty} (2^{k/2} - 2^{(k-1)/2}) \varepsilon_k.$$

Since

$$2^{k/2} - 2^{(k-1)/2} = (1 - 2^{-1/2}) 2^{k/2},$$

the right-hand side is bounded below by a constant multiple of

$$\sum_{k=0}^{\infty} 2^{k/2} \varepsilon_k.$$

Combining the previous estimates gives

$$\sum_{k=0}^{\infty} 2^{k/2} \varepsilon_k \leq C \int_0^D \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon,$$

and hence

$$\gamma_2(T, d) \leq C \int_0^D \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon.$$

□

Corollary 8.6 (Dudley's entropy integral). *Let $(X_t)_{t \in T}$ be a centered process such that*

$$\|X_t - X_s\|_{\psi_2} \leq L d(s, t) \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL \int_0^D \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon, \quad D = \sup_{s, t \in T} d(s, t),$$

where $C > 0$ is an absolute constant.

Proof. Combine Theorem 8.3 with Theorem 8.5. □

Remark 8.7. In this order of presentation, Dudley's entropy integral appears as a clean corollary of generic chaining rather than as the starting point. This is conceptually useful: the geometric object of real interest is γ_2 , while the entropy integral is a convenient upper estimate for it.

8.4 Talagrand's majorizing measure theorem

The upper bound from Theorem 8.3 is already quite general. What makes the Gaussian case special is that there is also a matching lower bound.

Theorem 8.8 (Talagrand's majorizing measure theorem). *Let $(X_t)_{t \in T}$ be a centered Gaussian process, and let*

$$d_X(s, t) = (\mathbb{E}(X_t - X_s)^2)^{1/2}$$

be its canonical metric. Then there exist absolute constants $c, C > 0$ such that

$$c \gamma_2(T, d_X) \leq \mathbb{E} \sup_{t \in T} X_t \leq C \gamma_2(T, d_X).$$

Remark 8.9. The upper bound is exactly the generic chaining estimate we have just proved. The lower bound is the deep part. Fernique introduced the majorizing-measure viewpoint, and Talagrand proved that it gives the correct answer up to absolute constants. This theorem is one of the central structural results in the theory of Gaussian processes.

It is also useful to keep in mind how it fits with the entropy bounds from the previous section. The upper half contains Dudley's bound after the comparison in Theorem 8.5. The lower half contains Sudakov minoration as a rough entropy shadow. So γ_2 sits between the two classical entropy bounds and refines both of them.

8.5 A comparison principle

The majorizing measure theorem immediately yields a very useful comparison theorem. It says that once a process has sub-Gaussian increments dominated by those of a Gaussian process, its supremum is controlled by the corresponding Gaussian supremum.

Corollary 8.10 (Talagrand's comparison theorem). *Let T be an index set. Let $(X_t)_{t \in T}$ be a centered process, and let $(Y_t)_{t \in T}$ be a centered Gaussian process. Assume that*

$$\|X_t - X_s\|_{\psi_2} \leq L (\mathbb{E}(Y_t - Y_s)^2)^{1/2} \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL \mathbb{E} \sup_{t \in T} Y_t,$$

where $C > 0$ is an absolute constant.

Proof. Define

$$d_Y(s, t) = (\mathbb{E}(Y_t - Y_s)^2)^{1/2}.$$

By Theorem 8.3,

$$\mathbb{E} \sup_{t \in T} X_t \leq C_1 L \gamma_2(T, d_Y).$$

Since Y is Gaussian with canonical metric d_Y , Theorem 8.8 gives

$$\gamma_2(T, d_Y) \leq C_2 \mathbb{E} \sup_{t \in T} Y_t.$$

Combining the two inequalities proves the claim. □

A particularly important special case is the canonical Gaussian process.

Corollary 8.11 (Sub-Gaussian increments and Gaussian width). *Let $T \subseteq \mathbf{R}^d$, and let $(X_t)_{t \in T}$ be a centered process such that*

$$\|X_t - X_s\|_{\psi_2} \leq L \|t - s\|_2 \quad \text{for all } s, t \in T.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CL W(T),$$

where $W(T)$ is the Gaussian width from Theorem 7.17.

Proof. Apply Theorem 8.10 with

$$Y_t = \langle g, t \rangle, \quad g \sim \mathcal{N}(0, I_d).$$

Then

$$(\mathbb{E}(Y_t - Y_s)^2)^{1/2} = \|t - s\|_2,$$

and

$$\mathbb{E} \sup_{t \in T} Y_t = W(T).$$

□

8.6 Examples

We conclude with two examples. The first is a general finite-set estimate. The second shows, in a concrete way, why γ_2 can be much smaller than Dudley's entropy integral.

Proposition 8.12 (Finite sets). *Let T be a finite metric space. Then*

$$\gamma_2(T, d) \leq C \sqrt{\log |T|} \sup_{s, t \in T} d(s, t),$$

where $C > 0$ is an absolute constant.

Proof. Write

$$T = \{t_1, \dots, t_m\}, \quad D = \sup_{s, t \in T} d(s, t).$$

For each $k \geq 0$, define

$$T_k = \{t_1, \dots, t_{m \wedge 2^{2^k}}\}.$$

Then (T_k) is admissible. If $2^{2^k} \geq m$, then $T_k = T$, so $d(t, T_k) = 0$ for all $t \in T$. Otherwise $d(t, T_k) \leq D$ for all $t \in T$. Therefore

$$\gamma_2(T, d) \leq D \sum_{k: 2^{2^k} < m} 2^{k/2}.$$

If k_0 is the largest index with $2^{2^{k_0}} < m$, then $2^{k_0} \asymp \log m$, and hence

$$\sum_{k=0}^{k_0} 2^{k/2} \leq C 2^{k_0/2} \leq C \sqrt{\log m}.$$

This proves the claim. □

Example 8.13 (A set for which Dudley is loose). For each $d \geq 1$, define

$$T_d = \left\{ \frac{e_i}{\sqrt{1 + \log i}} : 1 \leq i \leq d \right\} \subseteq \mathbf{R}^d.$$

Consider the canonical Gaussian process

$$X_t = \langle g, t \rangle, \quad g \sim \mathcal{N}(0, I_d).$$

Then

$$\sup_{t \in T_d} X_t = \max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}}.$$

We first show that the Gaussian supremum is uniformly bounded in d . For $u > 0$,

$$\mathbb{P} \left(\max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} \geq u \right) \leq \sum_{i=1}^d \mathbb{P} \left(g_i \geq u \sqrt{1 + \log i} \right).$$

Using the Gaussian Chernoff bound,

$$\mathbb{P}(g_i \geq a) \leq e^{-a^2/2}, \quad a \geq 0,$$

we get

$$\mathbb{P} \left(\max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} \geq u \right) \leq e^{-u^2/2} \sum_{i=1}^d i^{-u^2/2}.$$

For $u \geq 3$, the series on the right is bounded by an absolute constant, uniformly in d . Hence

$$\mathbb{P} \left(\max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} \geq u \right) \leq C e^{-u^2/2}, \quad u \geq 3.$$

Integrating the tail, we conclude that

$$\sup_{d \geq 1} \mathbb{E} \max_{1 \leq i \leq d} \frac{g_i}{\sqrt{1 + \log i}} < \infty.$$

Since $X_t = \langle g, t \rangle$ is Gaussian, Theorem 8.8 therefore implies

$$\sup_{d \geq 1} \gamma_2(T_d, \|\cdot\|_2) < \infty.$$

Now let us look at Dudley's entropy integral. Fix $\varepsilon \in (0, \frac{1}{2}]$, and set

$$m = \left\lfloor \exp \left(\frac{1}{4\varepsilon^2} \right) \right\rfloor.$$

Assume $m \leq d$. If $i \neq j$ and $1 \leq i, j \leq m$, then

$$\left\| \frac{e_i}{\sqrt{1 + \log i}} - \frac{e_j}{\sqrt{1 + \log j}} \right\|_2^2 = \frac{1}{1 + \log i} + \frac{1}{1 + \log j} \geq \frac{2}{1 + \log m}.$$

Since $\varepsilon \leq \frac{1}{2}$, we have $1 \leq \frac{1}{4\varepsilon^2}$, so

$$1 + \log m \leq 1 + \frac{1}{4\varepsilon^2} \leq \frac{1}{2\varepsilon^2}.$$

Therefore

$$\left\| \frac{e_i}{\sqrt{1 + \log i}} - \frac{e_j}{\sqrt{1 + \log j}} \right\|_2^2 \geq 4\varepsilon^2,$$

which means that the first m points form a 2ε -packing. By Theorem 6.3,

$$\mathcal{N}(T_d, \|\cdot\|_2, \varepsilon) \geq m.$$

Hence

$$\log \mathcal{N}(T_d, \|\cdot\|_2, \varepsilon) \geq \frac{c}{\varepsilon^2}$$

whenever $m \leq d$, that is, whenever

$$\varepsilon \geq \frac{c'}{\sqrt{\log d}}.$$

Consequently, for all sufficiently large d ,

$$\int_0^{1/2} \sqrt{\log \mathcal{N}(T_d, \|\cdot\|_2, \varepsilon)} \, d\varepsilon \geq c \int_{c'/\sqrt{\log d}}^{1/2} \frac{d\varepsilon}{\varepsilon} \geq c'' \log \log d.$$

So Dudley's entropy integral diverges with d , while $\gamma_2(T_d, \|\cdot\|_2)$ remains uniformly bounded.

This is a very instructive example. The entropy is large at many scales, but different points are responsible for different scales. Dudley's bound pays the worst possible price at every scale. The γ_2 -functional is more flexible, because the supremum stays inside the multiscale sum. That is exactly the feature that makes generic chaining sharper.

8.7 A logarithmic comparison between γ_2 and Dudley's sum in $\mathbf{R}^d$

Besides Talagrand's functional

$$\gamma_2(T, d) = \inf_{(T_k)} \sup_{\text{admissible}} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k),$$

it is sometimes convenient to consider the more rigid quantity obtained by swapping the supremum and the sum:

$$\mathcal{D}(T, d) = \inf_{(T_k)} \sum_{k=0}^{\infty} 2^{k/2} \sup_{t \in T} d(t, T_k).$$

One may think of $\mathcal{D}(T, d)$ as a ‘‘Dudley sum’’ attached to admissible sequences. By definition,

$$\gamma_2(T, d) \leq \mathcal{D}(T, d),$$

since the supremum of a sum is never larger than the sum of suprema. The next example shows that in finite-dimensional Euclidean geometry the converse only loses a logarithmic factor in the ambient dimension.

Proposition 8.14 (Dudley sum versus generic chaining in $\mathbf{R}^d$). *Let $T \subseteq \mathbf{R}^d$ be nonempty and bounded, and let $d(s, t) = \|s - t\|_2$. Then*

$$\gamma_2(T, \|\cdot\|_2) \leq \mathcal{D}(T, \|\cdot\|_2) \leq C \log(2d) \gamma_2(T, \|\cdot\|_2),$$

where $C > 0$ is an absolute constant.

Proof. The lower bound has already been noted, so we prove the upper bound. For $k = 0$, define

$$e_0 = \inf \left\{ \sup_{t \in T} \|t - S\|_2 : S \subseteq T, |S| = 1 \right\},$$

and for $k \geq 1$, define the entropy numbers

$$e_k = \inf \left\{ \sup_{t \in T} \|t - S\|_2 : S \subseteq T, |S| \leq 2^{2^k} \right\}.$$

Here, as usual, $\|t - S\|_2 = \inf_{s \in S} \|t - s\|_2$. This special treatment of $k = 0$ matches the usual admissible-sequence convention

$$|T_0| = 1.$$

Some references allow $|T_0| \leq 2$ instead; this only changes the resulting quantities by absolute constants. Since the cardinality restriction is independent at each scale, we may optimize each level separately, and therefore

$$\mathcal{D}(T, \|\cdot\|_2) = \sum_{k=0}^{\infty} 2^{k/2} e_k.$$

Also, for every admissible sequence (T_k) and every k ,

$$\sup_{t \in T} \sum_{j=0}^{\infty} 2^{j/2} \|t - T_j\|_2 \geq 2^{k/2} \sup_{t \in T} \|t - T_k\|_2,$$

so after taking the infimum over admissible sequences we get

$$\gamma_2(T, \|\cdot\|_2) \geq 2^{k/2} e_k \quad \text{for every } k \geq 0.$$

Fix an integer

$$m = \left\lceil 2 \log_2(2d) \right\rceil.$$

Then

$$\sum_{k=0}^m 2^{k/2} e_k \leq (m+1) \gamma_2(T, \|\cdot\|_2) \leq C \log(2d) \gamma_2(T, \|\cdot\|_2).$$

So it remains to bound the tail $\sum_{k>m} 2^{k/2} e_k$.

Set

$$D = \sup_{s, t \in T} \|s - t\|_2.$$

A standard volumetric estimate for Euclidean coverings gives

$$\mathcal{N}(T, \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2D}{\varepsilon}\right)^d, \quad \varepsilon > 0.$$

Therefore, if $\varepsilon > 0$ satisfies

$$\left(1 + \frac{2D}{\varepsilon}\right)^d \leq 2^{2^k},$$

then $e_k \leq \varepsilon$. Solving this inequality yields

$$e_k \leq C_1 D \min\left\{1, 2^{-2^k/d}\right\} \quad \text{for all } k \geq 0,$$

with an absolute constant $C_1 > 0$. On the other hand, since T_0 must consist of a single point, any admissible sequence satisfies

$$\sup_{t \in T} \|t - T_0\|_2 \geq \frac{D}{2},$$

and hence

$$\gamma_2(T, \|\cdot\|_2) \geq e_0 \geq \frac{D}{2}.$$

Using the bound on e_k and the choice of m , we conclude that

$$\sum_{k>m} 2^{k/2} e_k \leq C_1 D \sum_{k>m} 2^{k/2} 2^{-2^k/d} \leq C_2 D \leq 2C_2 \gamma_2(T, \|\cdot\|_2),$$

where we used that for $k > m$ the factor $2^{-2^k/d}$ decays super-exponentially. Combining the bounds for the initial part and the tail proves the claim. $\square$

Remark 8.15. The quantity $\mathcal{D}(T, \|\cdot\|_2)$ is the discrete analogue of Dudley's entropy integral. So the proposition may be read as saying that in $\mathbf{R}^d$, generic chaining and Dudley's multiscale bound are never more than logarithmically far apart.

The logarithmic loss may really appear. For the set $T = B_1^d \subseteq \mathbf{R}^d$, the Gaussian width satisfies

$$W(B_1^d) = \mathbb{E}\|g\|_\infty, \quad g \sim \mathcal{N}(0, I_d),$$

and standard upper and lower bounds for the maximum of Gaussian coordinates show that this quantity is of order $\sqrt{\log d}$. Therefore, by the majorizing measure theorem, $\gamma_2(B_1^d, \|\cdot\|_2)$ is also of order $\sqrt{\log d}$. On the other hand, the entropy estimate for the ℓ_1 ball implies that Dudley's integral is of order $(\log d)^{3/2}$. So in this example the gap is indeed of order $\log d$.

8.8 Sudakov minoration revisited via majorizing measures

The majorizing measure theorem also gives a very short route to Sudakov's lower bound.

Example 8.16 (Sudakov minoration from the majorizing measure theorem). Let $T \subseteq \mathbf{R}^d$ and let $g \sim \mathcal{N}(0, I_d)$. Then for every $\varepsilon > 0$,

$$\varepsilon \sqrt{\log \mathcal{N}(T, \|\cdot\|_2, \varepsilon)} \leq C \mathbb{E} \sup_{t \in T} \langle g, t \rangle,$$

where $C > 0$ is an absolute constant.

Proof. Set

$$M = \mathcal{N}(T, \|\cdot\|_2, \varepsilon).$$

If $M = 1$, then T is covered by one ball of radius ε , so the left-hand side vanishes and there is nothing to prove. Assume henceforth that $M \geq 2$. Choose $k \geq 0$ so that

$$2^{2^k} < M \leq 2^{2^{k+1}}.$$

Consider any admissible sequence $(T_j)_{j \geq 0}$. Since $|T_k| \leq 2^{2^k} < M$, the set T_k cannot be an ε -net of T . Therefore there exists $t \in T$ such that

$$\|t - T_k\|_2 > \varepsilon.$$

Hence

$$\sup_{u \in T} \sum_{j=0}^{\infty} 2^{j/2} \|u - T_j\|_2 \geq 2^{k/2} \sup_{u \in T} \|u - T_k\|_2 > 2^{k/2} \varepsilon.$$

Taking the infimum over admissible sequences yields

$$\gamma_2(T, \|\cdot\|_2) \geq 2^{k/2} \varepsilon.$$

Now we use the upper side of the dyadic sandwich. From

$$M \leq 2^{2^{k+1}}$$

we get

$$\log_2 M \leq 2^{k+1}, \quad \text{hence} \quad 2^{k/2} \geq 2^{-1/2} \sqrt{\log_2 M} \geq c_0 \sqrt{\log M}$$

for an absolute constant $c_0 > 0$. Therefore

$$\gamma_2(T, \|\cdot\|_2) \geq c \varepsilon \sqrt{\log \mathcal{N}(T, \|\cdot\|_2, \varepsilon)}$$

for an absolute constant $c > 0$. Applying the majorizing measure theorem to the canonical Gaussian process $X_t = \langle g, t \rangle$ finishes the proof. $\square$

9 Empirical processes, symmetrization, and VC classes

We now turn to empirical process theory. The main object is the random fluctuation of an empirical average around its mean, uniformly over a class of test functions. The simplest motivating example is still the Kolmogorov–Smirnov statistic, but the same ideas work for much richer classes.

9.1 Empirical processes and a first look at the KS statistic

Definition 9.1 (Empirical process). Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P$ on $\mathcal{X}$, and let $\mathcal{F}$ be a class of measurable functions $f : \mathcal{X} \rightarrow \mathbf{R}$. The collection

$$\{(P_n - P)f : f \in \mathcal{F}\}, \quad P_n f = \frac{1}{n} \sum_{i=1}^n f(X_i), \quad Pf = \mathbb{E}f(X),$$

is called the *empirical process indexed by $\mathcal{F}$* .

Definition 9.2 (Glivenko–Cantelli class). A class $\mathcal{F}$ is called *Glivenko–Cantelli* if

$$\sup_{f \in \mathcal{F}} |(P_n - P)f| \rightarrow 0 \quad \text{almost surely.}$$

A classical example is the class of threshold indicators

$$\mathcal{F}_{\text{KS}} = \{x \mapsto \mathbf{1}(x \leq t) : t \in \mathbf{R}\}.$$

In this case,

$$\sup_{f \in \mathcal{F}_{\text{KS}}} |(P_n - P)f| = \sup_{t \in \mathbf{R}} |F_n(t) - F(t)|,$$

where

$$F(t) = P(X \leq t), \quad F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq t).$$

So the KS statistic is simply the supremum of an empirical process over a one-parameter class.

9.2 Symmetrization

One of the basic tools in empirical process theory is to compare the unknown centered process $(P_n - P)f$ to a Rademacher process built from the observed sample.

Lemma 9.3 (Symmetrization). *Let $\mathcal{F}$ be a class of measurable functions for which the expectations below are finite, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent random signs, independent of $X_1, \dots, X_n$. Then*

$$\mathbb{E} \sup_{f \in \mathcal{F}} (Pf - P_n f) \leq 2 \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i),$$

$$\mathbb{E} \sup_{f \in \mathcal{F}} (P_n f - Pf) \leq 2 \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i),$$

and consequently

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - Pf| \leq 2 \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|.$$

Proof. We prove the first inequality. Let $X'_1, \dots, X'_n$ be an independent copy of $X_1, \dots, X_n$. Then

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} (Pf - P_n f) &= \mathbb{E} \sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n \mathbb{E} f(X'_i) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right) \\ &\leq \mathbb{E} \mathbb{E}_{X'} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X'_i) - f(X_i)). \end{aligned}$$

Since

$$(f(X'_i) - f(X_i))_{i=1}^n \stackrel{d}{=} (\varepsilon_i (f(X'_i) - f(X_i)))_{i=1}^n,$$

we may insert independent signs and obtain

$$\begin{aligned} \mathbb{E} \mathbb{E}_{X'} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X'_i) - f(X_i)) &= \mathbb{E} \mathbb{E}_{X'} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(X'_i) - f(X_i)) \\ &\leq \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) + \mathbb{E} \mathbb{E}_{X'} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X'_i) \\ &= 2 \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i). \end{aligned}$$

The second one-sided inequality follows in the same way, or by applying the first one to the class $-\mathcal{F}$. The absolute-value version is proved by running the same argument with absolute values kept throughout: after introducing the ghost sample, one symmetrizes the difference $P_n - P'_n$, inserts Rademacher signs, and then uses the triangle inequality inside the absolute value. $\square$

9.3 De-symmetrization

Symmetrization tells us that the empirical process is controlled by the Rademacher process. A partial converse is also true. For noncentered classes one pays only a small remainder term.

Proposition 9.4 (A simple de-symmetrization bound). *Let $\mathcal{F}$ be a class of measurable functions for which the expectations below are finite. Then*

$$\mathbb{E}\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| + \frac{1}{\sqrt{n}} \sup_{f \in \mathcal{F}} |P f|.$$

In particular, if every function in the class is centered, then

$$\mathbb{E}\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f|.$$

Proof. Write

$$\frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) = \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(X_i) - P f) + P f \cdot \frac{1}{n} \sum_{i=1}^n \varepsilon_i.$$

Taking the supremum over $f \in \mathcal{F}$, then expectations, gives

$$\mathbb{E}\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \leq \mathbb{E}\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(X_i) - P f) \right| + \sup_{f \in \mathcal{F}} |P f| \cdot \mathbb{E} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \right|.$$

Since $\mathbb{E} |n^{-1} \sum_i \varepsilon_i| \leq n^{-1/2}$, it remains to bound the first term. Set

$$\tilde{\mathcal{F}} = \{x \mapsto f(x) - P f : f \in \mathcal{F}\}.$$

Every $\tilde{f} \in \tilde{\mathcal{F}}$ is centered. Let $X'_1, \dots, X'_n$ be an independent copy of $X_1, \dots, X_n$, and write

$$P'_n \tilde{f} = \frac{1}{n} \sum_{i=1}^n \tilde{f}(X'_i).$$

For fixed $X_{1:n}$ and $\varepsilon_{1:n}$,

$$\frac{1}{n} \sum_{i=1}^n \varepsilon_i \tilde{f}(X_i) = \mathbb{E}_{X'} \left[\frac{1}{n} \sum_{i=1}^n \varepsilon_i (\tilde{f}(X_i) - \tilde{f}(X'_i)) \mid X_{1:n}, \varepsilon_{1:n} \right],$$

because $P \tilde{f} = 0$. By Jensen's inequality,

$$\mathbb{E}\mathbb{E}_\varepsilon \sup_{\tilde{f} \in \tilde{\mathcal{F}}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \tilde{f}(X_i) \right| \leq \mathbb{E}\mathbb{E}_{X'} \mathbb{E}_\varepsilon \sup_{\tilde{f} \in \tilde{\mathcal{F}}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (\tilde{f}(X_i) - \tilde{f}(X'_i)) \right|.$$

Now fix a sign vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$ and define

$$(\tilde{X}_i, \tilde{X}'_i) = \begin{cases} (X_i, X'_i), & \varepsilon_i = 1, \\ (X'_i, X_i), & \varepsilon_i = -1. \end{cases}$$

Since each pair (X_i, X'_i) is exchangeable and the pairs are independent across i , the joint law of

$$((\tilde{X}_i, \tilde{X}'_i))_{i=1}^n$$

is the same as that of

$$((X_i, X'_i))_{i=1}^n.$$

Moreover,

$$\varepsilon_i(\tilde{f}(X_i) - \tilde{f}(X'_i)) = \tilde{f}(\tilde{X}_i) - \tilde{f}(\tilde{X}'_i).$$

Therefore

$$\mathbb{E}\mathbb{E}_{X'}\mathbb{E}_\varepsilon \sup_{\tilde{f} \in \tilde{\mathcal{F}}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i(\tilde{f}(X_i) - \tilde{f}(X'_i)) \right| = \mathbb{E} \sup_{\tilde{f} \in \tilde{\mathcal{F}}} |P_n \tilde{f} - P'_n \tilde{f}|.$$

Finally, by the triangle inequality,

$$\mathbb{E} \sup_{\tilde{f} \in \tilde{\mathcal{F}}} |P_n \tilde{f} - P'_n \tilde{f}| \leq \mathbb{E} \sup_{\tilde{f} \in \tilde{\mathcal{F}}} |P_n \tilde{f}| + \mathbb{E} \sup_{\tilde{f} \in \tilde{\mathcal{F}}} |P'_n \tilde{f}| = 2 \mathbb{E} \sup_{\tilde{f} \in \tilde{\mathcal{F}}} |P_n \tilde{f}|.$$

Since $P\tilde{f} = 0$, this equals

$$2 \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f|.$$

This proves the claim. $\square$

9.4 A first rough bound for the KS statistic

The class $\mathcal{F}_{\text{KS}}$ is so simple that one can already get a nontrivial estimate from symmetrization and a maximal inequality alone.

Proposition 9.5 (A first rough KS bound). *For every distribution function F ,*

$$\mathbb{E} \sup_{t \in \mathbf{R}} |F_n(t) - F(t)| \leq C \sqrt{\frac{\log(n+1)}{n}},$$

where $C > 0$ is an absolute constant.

Proof. By Theorem 9.3,

$$\mathbb{E} \sup_{t \in \mathbf{R}} |F_n(t) - F(t)| \leq 2 \mathbb{E} \mathbb{E}_\varepsilon \sup_{t \in \mathbf{R}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbf{1}(X_i \leq t) \right|.$$

Now condition on $X_1, \dots, X_n$. As t varies, the vector

$$(\mathbf{1}(X_1 \leq t), \dots, \mathbf{1}(X_n \leq t)) \in \{0, 1\}^n$$

can take at most $n+1$ different values: only the gaps between the order statistics matter. Therefore we are taking the supremum over at most $n+1$ random variables. For each fixed t ,

$$\frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbf{1}(X_i \leq t)$$

is centered and sub-Gaussian with variance proxy at most $1/n$, since

$$\frac{1}{n^2} \sum_{i=1}^n \mathbf{1}(X_i \leq t)^2 \leq \frac{1}{n}.$$

Applying the maximal inequality for finitely many sub-Gaussian random variables to the $2(n+1)$ variables corresponding to both signs yields

$$\mathbb{E}_\varepsilon \sup_{t \in \mathbf{R}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbf{1}(X_i \leq t) \right| \leq C \sqrt{\frac{\log(n+1)}{n}}.$$

Taking expectation in $X_{1:n}$ gives the result. $\square$

Remark 9.6 (Kolmogorov–Smirnov asymptotic law). The bound in Theorem 9.5 is only a first consistency statement. If F is continuous, then in fact

$$\sqrt{n} \sup_{t \in \mathbf{R}} |F_n(t) - F(t)| \xrightarrow{d} \sup_{0 \leq u \leq 1} |B(u)|,$$

where B is a standard Brownian bridge. We will not prove this classical result here; it serves only as a reminder that the true behavior of the KS statistic is of order $n^{-1/2}$, with no logarithmic loss.

9.5 From expectation to high probability

The next proposition lets us pass from an expectation bound to a high-probability bound whenever the class is uniformly bounded.

Proposition 9.7 (Bounded differences for empirical processes). *Assume that every $f \in \mathcal{F}$ satisfies $\|f\|_\infty \leq l$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \leq \mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| + l \sqrt{\frac{2 \log(1/\delta)}{n}}.$$

Proof. Define

$$g(x_1, \dots, x_n) = \sup_{f \in \mathcal{F}} |P f - \frac{1}{n} \sum_{i=1}^n f(x_i)|.$$

If only the i -th coordinate is changed from x_i to x'_i , then

$$\begin{aligned} & |g(x_1, \dots, x_n) - g(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)| \\ & \leq \sup_{f \in \mathcal{F}} \frac{1}{n} |f(x_i) - f(x'_i)| \leq \frac{2l}{n}. \end{aligned}$$

So g satisfies the bounded differences property with constants $2l/n$. McDiarmid's inequality therefore gives

$$\mathbb{P}(g(X_{1:n}) - \mathbb{E}g(X_{1:n}) \geq t) \leq \exp\left(-\frac{nt^2}{2l^2}\right).$$

Taking $t = l\sqrt{2 \log(1/\delta)/n}$ yields the claim. $\square$

9.6 Growth functions and VC dimension

We now specialize to classes of events. Let $\mathcal{A}$ be a collection of measurable subsets of $\mathcal{X}$. A natural quantity of interest is

$$\sup_{A \in \mathcal{A}} |P_n(A) - P(A)|, \quad P_n(A) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \in A).$$

The point is that for indicator classes the complexity is purely combinatorial. It is measured by the number of binary patterns that the class can realize on a finite sample.

Definition 9.8 (Growth function / shattering function). For $n \in \mathbf{N}$, define

$$S_{\mathcal{A}}(n) = \max_{x_1, \dots, x_n \in \mathcal{X}} \left| \left\{ (\mathbf{1}(x_1 \in A), \dots, \mathbf{1}(x_n \in A)) : A \in \mathcal{A} \right\} \right|.$$

In words, $S_{\mathcal{A}}(n)$ is the largest number of distinct binary patterns that $\mathcal{A}$ can realize on n points.

A few elementary observations are immediate:

$$S_{\mathcal{A}}(n) \leq 2^n, \quad |\mathcal{A}| < \infty \implies S_{\mathcal{A}}(n) \leq |\mathcal{A}|,$$

and for the threshold class $\{(-\infty, t] : t \in \mathbf{R}\}$ one has $S_{\mathcal{A}}(n) = n + 1$. The last fact already suggests that one can have a very large infinite class while still realizing only a small number of patterns on n points.

The next proposition is the direct analogue of the finite-class bound, with the cardinality of the class replaced by the shattering function.

Proposition 9.9 (Uniform law via the shattering function). *Let $\mathcal{A}$ be a class of measurable subsets of $\mathcal{X}$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\sup_{A \in \mathcal{A}} |P_n(A) - P(A)| \leq 2\sqrt{\frac{2 \log(2S_{\mathcal{A}}(n))}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Proof. Fix the sample $X_1, \dots, X_n$ and let

$$M(X_{1:n}) = \left| \left\{ (\mathbf{1}(X_1 \in A), \dots, \mathbf{1}(X_n \in A)) : A \in \mathcal{A} \right\} \right|.$$

Conditional on $X_{1:n}$, the class $\mathcal{A}$ induces only $M(X_{1:n})$ distinct vectors in $\{0, 1\}^n$. For each such vector $v = (v_1, \dots, v_n)$, the Rademacher sum

$$\frac{1}{n} \sum_{i=1}^n \varepsilon_i v_i$$

is centered and sub-Gaussian with variance proxy at most $1/n$. Hence, by symmetrization and the maximal inequality for finite sub-Gaussian families,

$$\begin{aligned} \mathbb{E} \sup_{A \in \mathcal{A}} |P_n(A) - P(A)| &\leq 2 \mathbb{E} \mathbb{E}_{\varepsilon} \sup_{A \in \mathcal{A}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbf{1}(X_i \in A) \right| \\ &\leq 2 \mathbb{E} \sqrt{\frac{2 \log(2M(X_{1:n}))}{n}} \\ &\leq 2 \sqrt{\frac{2 \log(2S_{\mathcal{A}}(n))}{n}}, \end{aligned}$$

where we used $M(X_{1:n}) \leq S_{\mathcal{A}}(n)$ for every realization of the sample.

Now define

$$G(X_1, \dots, X_n) = \sup_{A \in \mathcal{A}} |P_n(A) - P(A)|.$$

If one sample point is changed, then $P_n(A)$ changes by at most $1/n$ for every A , while $P(A)$ is unchanged. Therefore G satisfies the bounded-differences property with constants $1/n, \dots, 1/n$. McDiarmid's inequality gives

$$\mathbb{P}(G - \mathbb{E}G \geq t) \leq e^{-2nt^2}, \quad t \geq 0.$$

Taking $t = \sqrt{\log(1/\delta)/(2n)}$ and combining with the expectation bound finishes the proof. $\square$

The polynomial regime is described by the Vapnik–Chervonenkis dimension.

Definition 9.10 (VC dimension). The *VC dimension* of $\mathcal{A}$ is the largest integer d such that

$$S_{\mathcal{A}}(d) = 2^d.$$

Equivalently, it is the size of the largest finite set shattered by $\mathcal{A}$.

Let us record two standard examples. For intervals on the line, the VC dimension is 2: two points can be shattered, but for three ordered points $x_1 < x_2 < x_3$ the pattern $(1, 0, 1)$ cannot be realized by a single interval. For half-spaces in $\mathbf{R}^p$, the VC dimension is $p + 1$. The upper bound is a consequence of the following convex-geometric theorem.

Theorem 9.11 (Radon). *Let $x_1, \dots, x_{p+2} \in \mathbf{R}^p$. Then there exists a partition*

$$\{1, \dots, p+2\} = I \sqcup J, \quad I, J \neq \emptyset,$$

such that

$$\text{conv}\{x_i : i \in I\} \cap \text{conv}\{x_j : j \in J\} \neq \emptyset.$$

Corollary 9.12 (Half-spaces in $\mathbf{R}^p$). *Let $\mathcal{A}$ be the class of affine half-spaces in $\mathbf{R}^p$. Then*

$$\text{VC}(\mathcal{A}) = p + 1.$$

Proof. For the lower bound, choose $p + 1$ affinely independent points $x_1, \dots, x_{p+1}$. For every subset $I \subseteq [p + 1]$, there exists an affine functional ℓ such that

$$\ell(x_i) = 1 \quad (i \in I), \quad \ell(x_i) = -1 \quad (i \notin I).$$

Hence the half-space $\{x : \ell(x) \geq 0\}$ realizes exactly the labels indexed by I . So $p + 1$ affinely independent points are shattered.

For the upper bound, take any $p + 2$ points in $\mathbf{R}^p$. By Radon's theorem, they can be partitioned into two disjoint groups whose convex hulls intersect. If the labeling corresponding to that partition were realizable by a half-space, then one convex hull would lie entirely inside the half-space and the other entirely outside it. Since half-spaces are convex, this would force the two convex hulls to be disjoint, a contradiction. Thus no set of size $p + 2$ can be shattered. $\square$

We now prove the fundamental combinatorial bound. The proof below follows the shifting method from the lecture, but we write it in a slightly more systematic way.

Theorem 9.13 (Sauer–Shelah–Vapnik–Chervonenkis). *If $\mathcal{A}$ has VC dimension d , then for every $n \geq d$,*

$$S_{\mathcal{A}}(n) \leq \sum_{i=0}^d \binom{n}{i} \leq \left(\frac{en}{d}\right)^d.$$

Proof. Fix points $x_1, \dots, x_n \in \mathcal{X}$ and consider the set family

$$\mathcal{C} = \{\{i \in [n] : x_i \in A\} : A \in \mathcal{A}\} \subseteq 2^{[n]}.$$

Then $|\mathcal{C}|$ is exactly the number of patterns realized by $\mathcal{A}$ on $x_1, \dots, x_n$. It is therefore enough to show that every family $\mathcal{C} \subseteq 2^{[n]}$ with VC dimension at most d satisfies

$$|\mathcal{C}| \leq \sum_{i=0}^d \binom{n}{i}.$$

For $i \in [n]$ and $E \subseteq [n]$, define the i -th *shift* by

$$S_i(E) = \begin{cases} E \setminus \{i\}, & \text{if } i \in E \text{ and } E \setminus \{i\} \notin \mathcal{C}, \\ E, & \text{otherwise.} \end{cases}$$

Applying this to all members of $\mathcal{C}$ gives a new family

$$S_i(\mathcal{C}) = \{S_i(E) : E \in \mathcal{C}\}.$$

Two basic facts are needed.

(1) *Cardinality is preserved.* If $E \neq F$ in $\mathcal{C}$, then $S_i(E) = S_i(F)$ can happen only if, say, $i \in E$, $i \notin F$, and $E \setminus \{i\} = F$. But then E would not be shifted, because $E \setminus \{i\} \in \mathcal{C}$. Hence the map $E \mapsto S_i(E)$ is injective on $\mathcal{C}$, so

$$|S_i(\mathcal{C})| = |\mathcal{C}|.$$

(2) *Shattering does not increase under shifting.* If a set $B \subseteq [n]$ is shattered by $S_i(\mathcal{C})$, then it is shattered by $\mathcal{C}$. Indeed, if $i \notin B$, then shifting does not affect the trace on B , so every pattern on B realized by $S_i(\mathcal{C})$ is also realized by $\mathcal{C}$. Now assume $i \in B$. If $H \subseteq B$ contains i , choose $F \in S_i(\mathcal{C})$ such that $F \cap B = H$. Since shift operations never create the coordinate i , the set F itself belongs to $\mathcal{C}$, and therefore H is realized by $\mathcal{C}$. If $H \subseteq B$ does not contain i , then $H \cup \{i\}$ is realized by some $F \in S_i(\mathcal{C})$. Again $i \in F$, so $F \in \mathcal{C}$; moreover, because F survived the shift, we must also have $F \setminus \{i\} \in \mathcal{C}$. Finally,

$$(F \setminus \{i\}) \cap B = H.$$

Thus every pattern on B is realized already by $\mathcal{C}$.

Starting from $\mathcal{C}$, apply shifts repeatedly in arbitrary order until no shift changes the family. This procedure must terminate, because whenever a shift is nontrivial, the quantity

$$\sum_{E \in \mathcal{C}} |E|$$

strictly decreases, while cardinality is preserved. Let $\mathcal{C}^*$ denote the terminal family. Then

$$|\mathcal{C}^*| = |\mathcal{C}|, \quad \text{VC}(\mathcal{C}^*) \leq \text{VC}(\mathcal{C}) \leq d.$$

Since $\mathcal{C}^*$ is stable under every shift, it is downward closed: if $E \in \mathcal{C}^*$ and $i \in E$, then necessarily

$$E \setminus \{i\} \in \mathcal{C}^*,$$

otherwise the i -th shift would modify E . Repeating this argument shows that every subset of E also belongs to $\mathcal{C}^*$. Therefore, if $E \in \mathcal{C}^*$ had cardinality at least $d + 1$, then all subsets of E would belong to $\mathcal{C}^*$, so E itself would be shattered, contradicting $\text{VC}(\mathcal{C}^*) \leq d$. Hence every member of $\mathcal{C}^*$ has size at most d . Consequently,

$$|\mathcal{C}| = |\mathcal{C}^*| \leq \sum_{i=0}^d \binom{n}{i}.$$

Since this holds for every choice of $x_1, \dots, x_n$, taking the maximum over all samples gives

$$S_{\mathcal{A}}(n) \leq \sum_{i=0}^d \binom{n}{i}.$$

It remains to prove the second inequality. If $d = 0$, the claim is trivial. If $d = n$, then

$$\sum_{i=0}^n \binom{n}{i} = 2^n \leq e^n = \left(\frac{en}{d}\right)^d.$$

So assume $1 \leq d < n$. For every $\mu \in (0, 1]$ and every $0 \leq i \leq d$, we have $\mu^{-i} \leq \mu^{-d}$, hence

$$\sum_{i=0}^d \binom{n}{i} \leq \mu^{-d} \sum_{i=0}^d \binom{n}{i} \mu^i \leq \mu^{-d} (1 + \mu)^n.$$

Choose

$$\mu = \frac{d}{n} \leq 1.$$

Then

$$\sum_{i=0}^d \binom{n}{i} \leq \left(\frac{n}{d}\right)^d \left(1 + \frac{d}{n}\right)^n \leq \left(\frac{n}{d}\right)^d e^d = \left(\frac{en}{d}\right)^d,$$

where we used $(1 + u/m)^m \leq e^u$ with $u = d$ and $m = n$. This completes the proof. $\square$

Remark 9.14. Combining Theorem 9.9 with Theorem 9.13 yields the preliminary bound

$$\mathbb{E} \sup_{A \in \mathcal{A}} |P_n(A) - P(A)| \lesssim \sqrt{\frac{d \log(en/d)}{n}}.$$

This is already enough to deduce a uniform law of large numbers for finite-VC classes. We record it only as a first pass. The logarithm comes from the crude counting argument through the growth function, and it will disappear once we combine Dudley's entropy method with a sharper covering-number estimate.

9.7 A Dudley covering bound for VC classes

We now prove the key covering estimate and then use it to remove the extra logarithm from the VC uniform law.

Lemma 9.15 (Empirical covering numbers for binary VC classes). *Let $\mathcal{F} \subseteq \{0, 1\}^{\mathcal{X}}$ be a class of binary-valued functions with VC dimension d . Then there exists an absolute constant $c > 0$ such that for every empirical measure P_n and every $\varepsilon \in (0, 1)$,*

$$\mathcal{N}(\mathcal{F}, L_2(P_n), \varepsilon) \leq \left(\frac{c}{\varepsilon}\right)^{4d}.$$

Equivalently,

$$\sup_{x_1, \dots, x_n \in \mathcal{X}} \mathcal{N}(\mathcal{F}, L_2(P_n), \varepsilon) \leq \left(\frac{c}{\varepsilon}\right)^{4d}.$$

Proof. Fix points $x_1, \dots, x_n \in \mathcal{X}$ and let

$$V = \{(f(x_1), \dots, f(x_n)) : f \in \mathcal{F}\} \subseteq \{0, 1\}^n.$$

Shattering a set of coordinates in V is the same as shattering the corresponding sample points by $\mathcal{F}$, so V has VC dimension at most d .

We will bound packing numbers, since covering numbers are always bounded by packing numbers. On V consider the Hamming-type metric

$$\rho(u, v) = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{1}(u_i \neq v_i)\right)^{1/2}.$$

This is exactly the $L_2(P_n)$ distance between the corresponding binary functions. Let $V_0 \subseteq V$ be an ε -packing, so whenever $u, v \in V_0$ are distinct,

$$\rho(u, v) > \varepsilon.$$

Equivalently,

$$\sum_{i=1}^n \mathbf{1}(u_i \neq v_i) > n\varepsilon^2.$$

For distinct $u, v \in V_0$, define

$$A_{u,v} = \{i \in [n] : u_i \neq v_i\}.$$

Then $|A_{u,v}| > n\varepsilon^2$.

Now let $Y_1, \dots, Y_k$ be i.i.d. random variables uniformly distributed on $[n] = \{1, \dots, n\}$. For any fixed pair $u \neq v$,

$$\mathbb{P}(Y_1 \notin A_{u,v}) \leq 1 - \varepsilon^2,$$

and therefore

$$\mathbb{P}(Y_j \notin A_{u,v} \text{ for all } j = 1, \dots, k) \leq (1 - \varepsilon^2)^k \leq e^{-k\varepsilon^2}.$$

Applying the union bound over all unordered pairs in V_0 , we obtain

$$\begin{aligned} & \mathbb{P}(\forall u, v \in V_0, u \neq v, \exists j \in [k] \text{ such that } Y_j \in A_{u,v}) \\ & \geq 1 - \binom{|V_0|}{2} e^{-k\varepsilon^2}. \end{aligned}$$

Choose

$$k = \left\lceil \frac{2 \log |V_0|}{\varepsilon^2} \right\rceil + 1.$$

Then

$$\binom{|V_0|}{2} e^{-k\varepsilon^2} \leq \frac{|V_0|^2}{2} e^{-2 \log |V_0|} e^{-\varepsilon^2} < 1,$$

so the right-hand side is strictly positive. Hence there exists a deterministic choice $y_1, \dots, y_k \in [n]$ such that every two distinct vectors in V_0 differ on at least one of these coordinates. Hence the projection map

$$\pi : V_0 \rightarrow \{0, 1\}^k, \quad \pi(u) = (u_{y_1}, \dots, u_{y_k}),$$

is injective. Let

$$V' = \pi(V_0) \subseteq \{0, 1\}^k.$$

Then $|V'| = |V_0|$. Moreover, V' still has VC dimension at most d , because a shattered set of coordinates in V' would correspond to a shattered set of coordinates in V . By Theorem 9.13,

$$|V_0| = |V'| \leq \sum_{i=0}^d \binom{k}{i} \leq \left(\frac{ek}{d} \right)^d.$$

Set $M = |V_0|$. If $\log M < \varepsilon^2$, then $M < e$ and the desired bound is trivial after enlarging the constant. So assume $\log M \geq \varepsilon^2$. Then

$$k = \left\lceil \frac{2 \log M}{\varepsilon^2} \right\rceil + 1 \leq \frac{4 \log M}{\varepsilon^2},$$

and hence

$$M \leq \left(\frac{4e \log M}{d\varepsilon^2} \right)^d.$$

Taking logarithms gives

$$\log M \leq d \log \left(\frac{4e}{\varepsilon^2} \right) + d \log \left(\frac{\log M}{d} \right).$$

Using the elementary inequality $\log x \leq x/e$ for $x > 0$, we obtain

$$d \log \left(\frac{\log M}{d} \right) \leq \frac{\log M}{e}.$$

Therefore

$$\log M \leq \frac{e}{e-1} d \log \left(\frac{4e}{\varepsilon^2} \right),$$

which implies

$$M \leq \left(\frac{c}{\varepsilon} \right)^{4d}$$

for an absolute constant $c > 0$. Since every ε -packing has size at most $(c/\varepsilon)^{4d}$, the same is true for the covering number. $\square$

9.8 Removing the logarithm in the VC uniform law

We now combine symmetrization with chaining. The point is that, conditionally on the sample, the indexed Rademacher process has sub-Gaussian increments with respect to the empirical L_2 metric. This lets us feed the empirical covering lemma into Dudley's integral.

Theorem 9.16 (Expected uniform deviation for VC classes). *Let $\mathcal{F} \subseteq \{0, 1\}^{\mathcal{X}}$ have VC dimension d , and let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P$. Then*

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq C \sqrt{\frac{d}{n}},$$

where $C > 0$ is an absolute constant.

Proof. By Theorem 9.3,

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq 2 \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|.$$

Fix the sample $X_1, \dots, X_n$. Define the process

$$X_f = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i f(X_i), \quad f \in \mathcal{F}.$$

For any $f, g \in \mathcal{F}$, the increment $X_f - X_g$ is a Rademacher sum, so for every $\lambda \in \mathbf{R}$,

$$\mathbb{E}_\varepsilon e^{\lambda(X_f - X_g)} \leq \exp \left(\frac{\lambda^2}{2} \cdot \frac{1}{n} \sum_{i=1}^n (f(X_i) - g(X_i))^2 \right) = \exp \left(\frac{\lambda^2}{2} \|f - g\|_{L_2(P_n)}^2 \right).$$

Thus, conditionally on $X_{1:n}$, the process $(X_f)_{f \in \mathcal{F}}$ is centered and sub-Gaussian with respect to the metric $\|f - g\|_{L_2(P_n)}$. By Dudley's entropy integral,

$$\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} X_f \leq C \gamma_2(\mathcal{F}, L_2(P_n)) \leq C \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), \varepsilon)} d\varepsilon,$$

where we used $\|f - g\|_{L_2(P_n)} \leq 1$ for binary-valued functions. Applying Theorem 9.15, we obtain

$$\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} X_f \leq C \int_0^1 \sqrt{4d \log(c/\varepsilon)} \, d\varepsilon \leq C' \sqrt{d},$$

where $C' > 0$ is absolute because the integral

$$\int_0^1 \sqrt{\log(c/\varepsilon)} \, d\varepsilon$$

is finite. Therefore

$$\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \leq \frac{2}{\sqrt{n}} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} X_f \leq C'' \sqrt{\frac{d}{n}},$$

where we used symmetry to bound the expectation of the supremum of the absolute value by twice the one-sided supremum. Taking expectation in $X_{1:n}$ and returning to the symmetrization step finishes the proof. $\square$

9.9 Three consequences

We now record three immediate corollaries. The first is the high-probability uniform law for VC classes. The second is a DKW inequality with nonoptimal constants. The third is the corresponding excess-risk bound for empirical risk minimization in binary classification.

Corollary 9.17 (Uniform convergence for VC classes). *Let $\mathcal{A}$ be a class of measurable subsets of $\mathcal{X}$ with VC dimension d . Then with probability at least $1 - \delta$,*

$$\sup_{A \in \mathcal{A}} |P_n(A) - P(A)| \leq C \sqrt{\frac{d}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}},$$

where $C > 0$ is an absolute constant.

Proof. Apply Theorem 9.16 to the indicator class

$$\mathcal{F}_{\mathcal{A}} = \{\mathbf{1}(\cdot \in A) : A \in \mathcal{A}\},$$

which has the same VC dimension as $\mathcal{A}$. Now define

$$g(X_1, \dots, X_n) = \sup_{A \in \mathcal{A}} |P_n(A) - P(A)|.$$

If one sample point is changed, then g changes by at most $1/n$, because every indicator takes values in $\{0, 1\}$ and only one term in the empirical average is affected. So McDiarmid's inequality yields

$$\mathbb{P}(g - \mathbb{E}g \geq t) \leq e^{-2nt^2}.$$

Take $t = \sqrt{\log(1/\delta)/(2n)}$ and use Theorem 9.16. $\square$

Remark 9.18. This direct bounded-differences argument is slightly sharper than invoking Theorem 9.7 with $l = 1$. The latter proposition uses the general estimate $2l/n$ for the effect of changing one sample point, whereas for indicator classes one can see directly that the optimal bound here is $1/n$.

Corollary 9.19 (A DKW inequality with nonoptimal constants). *Let F be a distribution function on $\mathbf{R}$ and F_n the empirical cdf. Then with probability at least $1 - \delta$,*

$$\sup_{t \in \mathbf{R}} |F_n(t) - F(t)| \leq \frac{C}{\sqrt{n}} + \sqrt{\frac{\log(1/\delta)}{2n}} \leq C' \sqrt{\frac{\log(2/\delta)}{n}},$$

where $C, C' > 0$ are absolute constants.

Proof. Apply Theorem 9.17 to the class of half-lines

$$\mathcal{A} = \{(-\infty, t] : t \in \mathbf{R}\}.$$

This class has VC dimension 1. □

Remark 9.20. The sharp Dvoretzky–Kiefer–Wolfowitz inequality has a better constant:

$$\sup_{t \in \mathbf{R}} |F_n(t) - F(t)| \leq \sqrt{\frac{\log(2/\delta)}{2n}}$$

with probability at least $1 - \delta$. Our argument does not aim at sharp constants; its value is that it fits into the general VC framework.

9.10 Application: binary classification with a VC class

We now illustrate the previous uniform convergence bound in statistical learning.

Definition 9.21 (Binary classification setup). Let $\mathcal{F}$ be a class of classifiers $f : \mathcal{X} \rightarrow \{0, 1\}$. Suppose that

$$(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{i.i.d.}}{\sim} P$$

on $\mathcal{X} \times \{0, 1\}$. For $f \in \mathcal{F}$, define the population risk

$$R(f) = \mathbb{P}(f(X) \neq Y)$$

and the empirical risk

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(f(X_i) \neq Y_i).$$

If $f^* \in \arg \min_{f \in \mathcal{F}} R(f)$ and

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} R_n(f),$$

then the *excess risk* of empirical risk minimization is

$$R(\hat{f}) - R(f^*).$$

The relevant loss class is

$$\mathcal{G} = \{(x, y) \mapsto \mathbf{1}(f(x) \neq y) : f \in \mathcal{F}\}.$$

A simple but important observation is that $\mathcal{G}$ has the same VC dimension as $\mathcal{F}$. Indeed, for any fixed sample $((x_i, y_i))_{i=1}^n$, the vector

$$(\mathbf{1}(f(x_1) \neq y_1), \dots, \mathbf{1}(f(x_n) \neq y_n))$$

is obtained from

$$(f(x_1), \dots, f(x_n))$$

by either leaving a coordinate unchanged or flipping the corresponding bit, depending on whether $y_i = 0$ or $y_i = 1$. So the number of patterns does not change.

Corollary 9.22 (Uniform convergence and excess risk for ERM). *Assume that $\mathcal{F}$ has VC dimension d . Then with probability at least $1 - \delta$, simultaneously for all $f \in \mathcal{F}$,*

$$|R_n(f) - R(f)| \leq C\sqrt{\frac{d}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Consequently, with probability at least $1 - \delta$,

$$R(\hat{f}) - R(f^*) \leq 2C\sqrt{\frac{d}{n}} + \sqrt{\frac{2\log(1/\delta)}{n}}.$$

In particular, the same bound controls the generalization error $|R_n(\hat{f}) - R(\hat{f})|$.

Proof. Apply Theorem 9.17 to the loss class $\mathcal{G}$ on the sample space $\mathcal{X} \times \{0, 1\}$. This yields the stated uniform bound. For the excess risk, note that by optimality of $\hat{f}$ for the empirical risk,

$$R_n(\hat{f}) - R_n(f^*) \leq 0.$$

Therefore,

$$\begin{aligned} R(\hat{f}) - R(f^*) &= (R(\hat{f}) - R_n(\hat{f})) + (R_n(\hat{f}) - R_n(f^*)) + (R_n(f^*) - R(f^*)) \\ &\leq 2 \sup_{f \in \mathcal{F}} |R_n(f) - R(f)|. \end{aligned}$$

Insert the uniform deviation bound. □

10 Nonparametric classes, contraction, and bracketing

We now take a step beyond VC classes. The point of view is still the same: the size of an empirical process is governed by the geometry of the index class. For general function classes the relevant geometry is no longer combinatorial, but metric. This leads naturally to entropy integrals, contraction inequalities, and bracketing.

10.1 A motivating example: Lipschitz functions

Let $\mathcal{F}$ denote the class of all 1-Lipschitz functions $f : [0, 1] \rightarrow [0, 1]$. We are interested in the Rademacher average

$$\mathbb{E}_{X, \varepsilon} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i),$$

where $X_1, \dots, X_n$ are i.i.d. random variables in $[0, 1]$. To use Dudley's entropy bound, we need a covering-number estimate.

Since

$$\|f - g\|_{L_2(P_n)} \leq \|f - g\|_{L_\infty},$$

it is enough to bound the L_∞ covering numbers. Draw an ε -grid on the unit square $[0, 1]^2$. If f is 1-Lipschitz, then between two consecutive vertical grid lines it cannot cross more than one horizontal strip. So on each such interval one can choose a horizontal segment of the grid that stays within distance ε of the graph of f . Piecing these segments together yields a step function g such that

$$\|f - g\|_{L_\infty} \leq \varepsilon.$$

Thus the collection of all such step functions forms an ε -net of $\mathcal{F}$. A crude counting argument gives

$$\log \mathcal{N}(\mathcal{F}, L_\infty, \varepsilon) \lesssim \frac{1}{\varepsilon} \log \left(\frac{1}{\varepsilon} \right), \quad 0 < \varepsilon \leq 1.$$

Since the same bound then holds for $L_2(P_n)$, Dudley's entropy integral yields

$$\mathbb{E}_{X, \varepsilon} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \lesssim \frac{1}{\sqrt{n}} \int_0^1 \sqrt{\frac{1}{\varepsilon} \log \left(\frac{1}{\varepsilon} \right)} d\varepsilon.$$

The integral is finite, so we obtain

$$\mathbb{E}_{X, \varepsilon} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \lesssim \frac{1}{\sqrt{n}}.$$

The striking point is that we get the same $n^{-1/2}$ rate uniformly over the whole Lipschitz class.

10.2 A truncated Dudley bound for Rademacher averages

The previous example illustrates the easy regime, where the entropy integral converges at the origin. For more complicated classes the integral may diverge at small scales. A convenient remedy is to stop the chaining at a positive scale and bound the remaining local oscillation directly. For our purposes, it is enough to record the following corollary.

Corollary 10.1 (Truncated Dudley bound for Rademacher averages). *Let $\mathcal{F} \subseteq [0, 1]^\mathcal{X}$ and fix the sample $X_1, \dots, X_n$. Then for every $\eta \in (0, 1]$,*

$$\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \leq \eta + \frac{C}{\sqrt{n}} \int_{\eta/4}^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), u)} du,$$

where $C > 0$ is an absolute constant.

Proof. Condition on the sample and define

$$Z_f = \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i), \quad f \in \mathcal{F}.$$

Fix a minimal η -net $\mathcal{F}_\eta \subseteq \mathcal{F}$ in $L_2(P_n)$, so that

$$|\mathcal{F}_\eta| = \mathcal{N}(\mathcal{F}, L_2(P_n), \eta).$$

For each $f \in \mathcal{F}$ choose $\pi(f) \in \mathcal{F}_\eta$ such that

$$\|f - \pi(f)\|_{L_2(P_n)} \leq \eta.$$

Then

$$\sup_{f \in \mathcal{F}} Z_f \leq \sup_{f \in \mathcal{F}} (Z_f - Z_{\pi(f)}) + \sup_{g \in \mathcal{F}_\eta} Z_g.$$

By Cauchy-Schwarz,

$$|Z_f - Z_g| = \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(X_i) - g(X_i)) \right| \leq \|f - g\|_{L_2(P_n)},$$

so the local term is bounded deterministically by η :

$$\sup_{f \in \mathcal{F}} (Z_f - Z_{\pi(f)}) \leq \sup_{\|f - g\|_{L_2(P_n)} \leq \eta} |Z_f - Z_g| \leq \eta.$$

It remains to control the supremum over the finite net $\mathcal{F}_\eta$. For $f, g \in \mathcal{F}_\eta$, Hoeffding's lemma gives

$$\mathbb{E}_\varepsilon e^{\lambda(Z_f - Z_g)} \leq \exp\left(\frac{\lambda^2}{2n} \|f - g\|_{L_2(P_n)}^2\right), \quad \lambda \in \mathbf{R}.$$

Thus $(Z_f)_{f \in \mathcal{F}_\eta}$ is centered and sub-Gaussian with respect to the metric

$$d_n(f, g) = \frac{\|f - g\|_{L_2(P_n)}}{\sqrt{n}}.$$

Applying the generic chaining upper bound and then the comparison between γ_2 and Dudley's entropy integral, we obtain

$$\mathbb{E}_\varepsilon \sup_{g \in \mathcal{F}_\eta} Z_g \leq C \gamma_2(\mathcal{F}_\eta, d_n) \leq C \int_0^{1/\sqrt{n}} \sqrt{\log \mathcal{N}(\mathcal{F}_\eta, d_n, \varepsilon)} d\varepsilon.$$

Changing variables $u = \sqrt{n} \varepsilon$ yields

$$\mathbb{E}_\varepsilon \sup_{g \in \mathcal{F}_\eta} Z_g \leq \frac{C}{\sqrt{n}} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}_\eta, L_2(P_n), u)} du.$$

Now split the integral at $\eta/2$. For $0 < u < \eta/2$, we have

$$\mathcal{N}(\mathcal{F}_\eta, L_2(P_n), u) \leq |\mathcal{F}_\eta| = \mathcal{N}(\mathcal{F}, L_2(P_n), \eta),$$

so

$$\int_0^{\eta/2} \sqrt{\log \mathcal{N}(\mathcal{F}_\eta, L_2(P_n), u)} du \leq \frac{\eta}{2} \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), \eta)}.$$

Since $u \mapsto \mathcal{N}(\mathcal{F}, L_2(P_n), u)$ is nonincreasing,

$$\frac{\eta}{2} \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), \eta)} \leq 2 \int_{\eta/4}^{\eta/2} \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), u)} du.$$

For $u \geq \eta/2$, we simply use $\mathcal{F}_\eta \subseteq \mathcal{F}$:

$$\mathcal{N}(\mathcal{F}_\eta, L_2(P_n), u) \leq \mathcal{N}(\mathcal{F}, L_2(P_n), u).$$

Hence

$$\mathbb{E}_\varepsilon \sup_{g \in \mathcal{F}_\eta} Z_g \leq \frac{C}{\sqrt{n}} \int_{\eta/4}^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), u)} du.$$

Combining this with the local bound completes the proof. $\square$

Remark 10.2. This is the only truncated form we will need. Conceptually, it is obtained by stopping the process at a coarse η -net, bounding the local oscillation directly, and then applying the already-established γ_2 theory only to the finite net. So one does not need to run a new chaining proof from scratch.

10.3 Polynomial entropy and nonparametric rates

The previous corollary is especially useful for classes whose entropy behaves polynomially in $1/\varepsilon$. This is the regime one usually has in mind for genuinely nonparametric classes.

Definition 10.3 (Polynomial entropy). Let $\mathcal{F} \subseteq [0, 1]^{\mathcal{X}}$. We say that $\mathcal{F}$ has *polynomial entropy exponent* $p > 0$ if there exists $A \geq 1$ such that

$$\sup_Q \log \mathcal{N}(\mathcal{F}, L_2(Q), \varepsilon) \leq \left(\frac{A}{\varepsilon} \right)^p, \quad 0 < \varepsilon \leq 1,$$

where the supremum is over all probability measures Q on $\mathcal{X}$.

Remark 10.4. By the Koltchinskii–Pollard principle, one may equivalently take the supremum over finitely supported measures. So the same entropy bound automatically applies with $Q = P_n$.

Theorem 10.5 (Rates under polynomial entropy). *Assume that $\mathcal{F}$ has polynomial entropy exponent p . Then*

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f| \leq C_{A,p} \begin{cases} n^{-1/2}, & 0 < p < 2, \\ n^{-1/2} \log n, & p = 2, \\ n^{-1/p}, & p > 2, \end{cases}$$

where $C_{A,p} > 0$ depends only on A and p .

Proof. By symmetrization and symmetry of the Rademacher process,

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f| \leq 4 \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i).$$

Now condition on the sample and apply Theorem 10.1: for every $\eta \in (0, 1]$,

$$\mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \leq \eta + \frac{C}{\sqrt{n}} \int_{\eta/4}^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L_2(P_n), u)} \, du.$$

Using the entropy assumption and then taking expectation in the sample, we obtain

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f| \leq C \left(\eta + \frac{A^{p/2}}{\sqrt{n}} \int_{\eta/4}^1 u^{-p/2} \, du \right).$$

If $0 < p < 2$, the integral converges at the origin, so letting $\eta \downarrow 0$ yields

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f| \leq \frac{C_{A,p}}{\sqrt{n}}.$$

If $p = 2$, then

$$\int_{\eta/4}^1 u^{-1} \, du = \log \left(\frac{4}{\eta} \right).$$

Choosing $\eta = n^{-1/2}$ gives

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f| \leq C_A \frac{\log n}{\sqrt{n}}.$$

Finally, assume $p > 2$. Then

$$\int_{\eta/4}^1 u^{-p/2} du \leq C_p \eta^{1-p/2}.$$

Therefore

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f| \leq C_{A,p} \left(\eta + \frac{1}{\sqrt{n}} \eta^{1-p/2} \right).$$

Choosing

$$\eta = n^{-1/p}$$

balances the two terms and yields

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P)f| \leq C_{A,p} n^{-1/p}.$$

□

Remark 10.6. The threshold $p = 2$ marks the transition between the Donsker-type and non-Donsker-type behaviors in this simple entropy scale. For $p < 2$, the ordinary Dudley integral already converges and one keeps the root- n rate. For $p > 2$, the small scales are too rich, and the rate deteriorates to $n^{-1/p}$.

10.4 Contraction for Rademacher averages

The next tool says that Lipschitz transformations do not increase Rademacher averages by more than the Lipschitz constant. We record the one-sided version, which is enough for our applications.

Theorem 10.7 (A one-sided contraction inequality). *Let $\varphi_1, \dots, \varphi_n : \mathbf{R} \rightarrow \mathbf{R}$ be L -Lipschitz functions. Then for every set $T \subseteq \mathbf{R}^n$,*

$$\mathbb{E}_\varepsilon \sup_{t \in T} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \varphi_i(t_i) \leq L \mathbb{E}_\varepsilon \sup_{t \in T} \frac{1}{n} \sum_{i=1}^n \varepsilon_i t_i.$$

Proof. By homogeneity, it is enough to consider the case $L = 1$. Set

$$F_n(T; \varphi_1, \dots, \varphi_n) = \mathbb{E}_\varepsilon \sup_{t \in T} \sum_{i=1}^n \varepsilon_i \varphi_i(t_i).$$

We show that replacing one coordinate map by the identity can only increase this quantity. Indeed, expanding in ε_n only and writing

$$A_t = \sum_{i=1}^{n-1} \varepsilon_i \varphi_i(t_i),$$

we obtain

$$\begin{aligned} F_n(T; \varphi_1, \dots, \varphi_n) &= \frac{1}{2} \mathbb{E} \left[\sup_{t \in T} (A_t + \varphi_n(t_n)) + \sup_{s \in T} (A_s - \varphi_n(s_n)) \right] \\ &\leq \frac{1}{2} \mathbb{E} \sup_{t, s \in T} \left(A_t + A_s + |\varphi_n(t_n) - \varphi_n(s_n)| \right) \\ &\leq \frac{1}{2} \mathbb{E} \sup_{t, s \in T} \left(A_t + A_s + |t_n - s_n| \right). \end{aligned}$$

The quantity inside the last supremum is symmetric in (t, s) , so the absolute value may be removed:

$$\sup_{t,s \in T} (A_t + A_s + |t_n - s_n|) = \sup_{t,s \in T} (A_t + A_s + t_n - s_n).$$

Therefore,

$$\begin{aligned} F_n(T; \varphi_1, \dots, \varphi_n) &\leq \frac{1}{2} \mathbb{E} \left[\sup_{t \in T} (A_t + t_n) + \sup_{s \in T} (A_s - s_n) \right] \\ &= F_n(T; \varphi_1, \dots, \varphi_{n-1}, \text{id}). \end{aligned}$$

Repeating this argument for the coordinates $n, n-1, \dots, 1$ successively replaces every φ_i by the identity map and gives

$$F_n(T; \varphi_1, \dots, \varphi_n) \leq F_n(T; \text{id}, \dots, \text{id}).$$

Dividing by n finishes the proof. $\square$

A standard stronger form, due to Ledoux and Talagrand, allows absolute values and assumes $\varphi_i(0) = 0$. We will not need that version here.

Definition 10.8 (Empirical Rademacher complexity). For a class $\mathcal{F}$ of measurable functions, define

$$\mathfrak{R}_n(\mathcal{F}) = \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i).$$

Corollary 10.9 (A general excess-risk bound). *Let $\mathcal{F}$ be a class of prediction rules, and let $\ell : \mathbf{R} \times \mathcal{Y} \rightarrow \mathbf{R}$ be a loss such that for every $y \in \mathcal{Y}$, the map $u \mapsto \ell(u, y)$ is L -Lipschitz. Define*

$$R(f) = \mathbb{E} \ell(f(X), Y), \quad R_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(X_i), Y_i),$$

and let

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} R_n(f).$$

Then

$$\mathbb{E} [R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f)] \leq 4L \mathfrak{R}_n(\mathcal{F}).$$

Proof. Fix $g \in \mathcal{F}$. Then

$$\begin{aligned} R(\hat{f}) - R(g) &= (R(\hat{f}) - R_n(\hat{f})) + (R_n(\hat{f}) - R_n(g)) + (R_n(g) - R(g)) \\ &\leq \sup_{f \in \mathcal{F}} (R(f) - R_n(f)) + \sup_{f \in \mathcal{F}} (R_n(f) - R(f)), \end{aligned}$$

where we used the defining property of $\hat{f}$ to bound the middle term by 0. Taking the infimum over $g \in \mathcal{F}$ yields

$$R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f) \leq \sup_{f \in \mathcal{F}} (R(f) - R_n(f)) + \sup_{f \in \mathcal{F}} (R_n(f) - R(f)).$$

Taking expectations and applying symmetrization to both terms gives

$$\mathbb{E} [R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f)] \leq 4 \mathbb{E} \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \ell(f(X_i), Y_i).$$

Now apply Theorem 10.7 with

$$\varphi_i(u) = \ell(u, Y_i)$$

and use the L -Lipschitz property. $\square$

Example 10.10 (Hinge loss). Suppose $Y \in \{-1, +1\}$ and define

$$\psi(t) = \max\{0, 1 - t\}.$$

Then ψ is 1-Lipschitz, and therefore

$$\mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \psi(Y_i f(X_i)) \leq \mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i Y_i f(X_i).$$

Since $(\varepsilon_i Y_i)_{i=1}^n$ has the same distribution as $(\varepsilon_i)_{i=1}^n$, the right-hand side equals

$$\mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i).$$

So the excess risk under hinge loss is controlled by the same Rademacher average as the original class.

10.5 Bracketing entropy

Covering numbers based on empirical measures are not the only way to proceed. A different route, often sharper for genuinely infinite-dimensional classes, is to work with brackets.

Definition 10.11 (Bracket). Let $u, l : \mathcal{X} \rightarrow \mathbf{R}$ be measurable functions with $l(x) \leq u(x)$ for all $x \in \mathcal{X}$. The set

$$[l, u] = \{f : \mathcal{X} \rightarrow \mathbf{R} : l(x) \leq f(x) \leq u(x) \text{ for all } x \in \mathcal{X}\}$$

is called a *bracket*.

Definition 10.12 (Bracketing number). Fix a probability measure P on $\mathcal{X}$ and $q \geq 1$. A bracket $[l, u]$ is an ε -bracket in $L_q(P)$ if

$$\|u - l\|_{L_q(P)} \leq \varepsilon.$$

The minimal number of such brackets needed to cover $\mathcal{F}$ is denoted by

$$N_{[]}(\mathcal{F}, L_q(P), \varepsilon).$$

The point is that the brackets are built relative to the underlying distribution P itself, so one no longer needs a supremum over empirical measures. We only state the corresponding theorem.

Theorem 10.13 (Bracketing entropy bound). Assume that $\|f\|_{L_\infty(P)} \leq m$ for every $f \in \mathcal{F}$. Then there exists an absolute constant $C > 0$ such that

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq \frac{C}{\sqrt{n}} \int_0^m \sqrt{\log N_{[]}(\mathcal{F}, L_2(P), \varepsilon)} \, d\varepsilon.$$

Remark 10.14. Bracketing is often the right substitute when ordinary covering numbers are too large or when one wants a bound that depends directly on P . There is also a parallel theory based on the Koltchinskii–Pollard entropy

$$\sup_Q N(\mathcal{F}, L_2(Q), \varepsilon),$$

where the supremum is over all probability measures Q . We will not develop that direction here.

11 Median-of-means, uniform moment estimation, one-sided lower tails, and a small-ball application

We now turn to a different theme: robust estimation under weak moment assumptions. The guiding principle is that averaging inside blocks and then taking a median across blocks often restores sub-Gaussian behavior even when only a second moment is available. This point of view is useful both for scalar estimation and for genuinely high-dimensional problems.

11.1 Median-of-means for scalar mean estimation

Suppose that $X_1, \dots, X_n$ are i.i.d. real-valued random variables with mean μ and variance $\text{Var}(X_1) \leq \sigma^2$. Fix an integer $K \in \{1, \dots, n\}$, and for simplicity assume that $n = Km$. (If $K \nmid n$, one may discard at most $K - 1$ observations.) Partition $[n]$ into K disjoint blocks $B_1, \dots, B_K$ of common size m , and define the block means

$$\bar{X}_j = \frac{1}{m} \sum_{i \in B_j} X_i, \quad j = 1, \dots, K.$$

The *median-of-means estimator* is

$$\hat{\mu}_{\text{MOM}} = \text{Med}(\bar{X}_1, \dots, \bar{X}_K),$$

where, here and throughout, if $y_{(1)} \leq \dots \leq y_{(K)}$ are the order statistics of $y_1, \dots, y_K$, then

$$\text{Med}(y_1, \dots, y_K) = y_{(\lceil K/2 \rceil)}.$$

Thus Med is the lower median; in particular, it is translation equivariant.

The idea is extremely simple: instead of trusting one global average, we first form several local averages, and then keep only their median. A few atypical blocks can no longer spoil the final estimator.

Proposition 11.1 (Median-of-means under a second-moment assumption). *Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}X_1 = \mu$ and $\text{Var}(X_1) \leq \sigma^2$. Assume that $n = Km$, and fix $\delta \in (0, 1)$. If*

$$K \geq 8 \log(1/\delta),$$

then with probability at least $1 - \delta$,

$$|\hat{\mu}_{\text{MOM}} - \mu| \leq 2\sigma \sqrt{\frac{K}{n}}.$$

In particular, if

$$K = \lceil 8 \log(1/\delta) \rceil,$$

then with probability at least $1 - \delta$,

$$|\hat{\mu}_{\text{MOM}} - \mu| \leq 2\sigma \sqrt{\frac{\lceil 8 \log(1/\delta) \rceil}{n}} \leq 2\sigma \sqrt{\frac{8 \log(1/\delta) + 1}{n}}.$$

Proof. Fix a block B_j . Since $\text{Var}(\bar{X}_j) \leq \sigma^2/m$, Chebyshev's inequality gives

$$\mathbb{P}\left(|\bar{X}_j - \mu| \geq \frac{2\sigma}{\sqrt{m}}\right) \leq \frac{1}{4}.$$

Thus the event

$$G_j = \left\{ |\bar{X}_j - \mu| \leq \frac{2\sigma}{\sqrt{m}} \right\}$$

has probability at least $3/4$.

Now define $N_G = \sum_{j=1}^K \mathbf{1}(G_j)$. Because the blocks are disjoint, the indicators $\mathbf{1}(G_j)$ are independent Bernoulli random variables with means at least $3/4$, so

$$\mathbb{E}N_G \geq \frac{3K}{4}.$$

By Hoeffding's inequality,

$$\mathbb{P}\left(N_G \leq \frac{K}{2}\right) = \mathbb{P}\left(N_G - \mathbb{E}N_G \leq \frac{K}{2} - \mathbb{E}N_G\right) \leq \mathbb{P}\left(N_G - \mathbb{E}N_G \leq -\frac{K}{4}\right) \leq \exp\left(-\frac{2(K/4)^2}{K}\right) = e^{-K/8}.$$

Hence, if $K \geq 8 \log(1/\delta)$, then with probability at least $1 - \delta$ we have $N_G > K/2$.

On this event, more than half of the block means lie in the interval

$$\left[\mu - \frac{2\sigma}{\sqrt{m}}, \mu + \frac{2\sigma}{\sqrt{m}}\right].$$

Therefore their lower median also lies in this interval. Since $m = n/K$, we conclude that

$$|\hat{\mu}_{\text{MOM}} - \mu| \leq \frac{2\sigma}{\sqrt{m}} = 2\sigma\sqrt{\frac{K}{n}},$$

which is the desired bound. $\square$

Remark 11.2. The striking feature of Theorem 11.1 is that when $K \asymp \log(1/\delta)$, the deviation bound has the same $\sqrt{\log(1/\delta)/n}$ form as a sub-Gaussian concentration inequality, even though we only assumed a finite variance. So the median-of-means estimator recovers a sub-Gaussian rate under a much weaker tail assumption than the sample mean.

11.2 A polynomial VC bound via Warren's lemma

In the applications below, the objects being classified will be blocks or observations $y \in \mathcal{Y}$, while the indexing parameter will be a direction $v \in S^{d-1}$ or, more generally, a parameter $\theta \in \Theta \subseteq \mathbf{R}^m$. Thus the polynomial structure will appear in the *parameter*, not in the data point. The next lemma is tailored exactly to this situation.

Lemma 11.3 (Warren's lemma). *Let $\Theta \subseteq \mathbf{R}^m$, let $\mathcal{Y}$ be a set, and consider a class of indicator functions*

$$\mathcal{H} = \{h_\theta : \theta \in \Theta\}, \quad h_\theta(y) = \mathbf{1}(P(y, \theta) \geq 0),$$

where for every fixed $y \in \mathcal{Y}$, the map

$$\theta \mapsto P(y, \theta)$$

is a real polynomial on $\mathbf{R}^m$ of degree at most $r \geq 1$. Then

$$\text{VC}(\mathcal{H}) \leq C m \log(er),$$

where $C > 0$ is an absolute constant.

We will use this lemma as a black box. Its proof belongs to real algebraic geometry, but the message is very simple: if the decision rule is given by the sign of a polynomial in m real parameters of degree r , then the resulting class has VC dimension at most of order $m \log(er)$.

Remark 11.4 (Comparison with half-spaces). It is useful to compare this with the familiar class of affine half-spaces. If

$$h_{(u,b)}(x) = \mathbf{1}(\langle u, x \rangle + b \geq 0), \quad (u, b) \in \mathbf{R}^{d+1},$$

then for each fixed $x \in \mathbf{R}^d$, the map

$$(u, b) \mapsto \langle u, x \rangle + b$$

is an affine polynomial of degree 1 in $d + 1$ parameters. So Warren's lemma gives

$$\text{VC}(\mathcal{H}) \leq C(d + 1),$$

which is consistent with the exact value $d + 1$. Thus half-spaces correspond to the degree-1 case. The point of Warren's lemma is that if we replace affine functions by polynomials of degree r , then the VC dimension increases only by a logarithmic factor in r .

Example 11.5 (One parameter, degree p). Consider a class indexed by a scalar parameter $t \in \mathbf{R}$:

$$h_t(y) = \mathbf{1}(a_0(y) + a_1(y)t + \cdots + a_p(y)t^p \geq 0).$$

Here $m = 1$, and for each fixed y , the decision rule is the sign of a polynomial in t of degree at most p .

Let $y_1, \dots, y_n \in \mathcal{Y}$ be fixed. For each j , either the polynomial

$$t \mapsto a_0(y_j) + a_1(y_j)t + \cdots + a_p(y_j)t^p$$

is identically zero, in which case $h_t(y_j) \equiv 1$, or else it has at most p real roots. Therefore the sign pattern

$$(h_t(y_1), \dots, h_t(y_n))$$

can change only when t crosses one of these roots. Across all n sample points, there are at most np such roots, so $t \in \mathbf{R}$ is split into at most $np + 1$ intervals on which the whole sign pattern is constant. Hence

$$S_{\mathcal{H}}(n) \leq np + 1.$$

In particular, if $\mathcal{H}$ shatters n points, then $2^n \leq np + 1$, so $n \lesssim \log(ep)$. Thus, when there is only one real parameter, the VC dimension is of order $\log(ep)$.

This is exactly what Warren's lemma predicts in the case $m = 1$:

$$\text{VC}(\mathcal{H}) \leq C \log(ep).$$

11.3 Uniform estimation of directional L_p norms

Median-of-means is not limited to first moments. A convenient second example is the estimation of directional L_p norms uniformly over all directions. The point is that, once p is fixed, we may apply the same blockwise idea to the random variables

$$|\langle X, v \rangle|^p, \quad v \in S^{d-1}.$$

The only extra input is a one-step comparison between the p -th and $2p$ -th moments. This assumption is mild when p is fixed. For example, sub-Gaussian one-dimensional marginals satisfy moment comparisons of this type for all p , and many other light-tailed models do as well. For the theorem below, however, we only need the single inequality stated in the assumptions.

Theorem 11.6 (Uniform estimation of directional p -th moments). *Let $p \geq 2$ be an even integer, and let $X_1, \dots, X_n$ be i.i.d. copies of a random vector $X \in \mathbf{R}^d$. Assume that there exists $L \geq 1$ such that the hypercontractivity assumption holds*

$$\|\langle X, v \rangle\|_{L_{2p}} \leq L \|\langle X, v \rangle\|_{L_p} \quad \text{for every } v \in S^{d-1}.$$

Choose

$$k = \lceil C_0(d \log(ep) + \log(1/\delta)) \rceil$$

for a sufficiently large absolute constant $C_0 > 0$, assume that $n = km$, partition $[n]$ into k disjoint blocks $B_1, \dots, B_k$ of size m , and define

$$\text{MOM}_p(v) = \text{Med} \left(\frac{1}{m} \sum_{i \in B_1} |\langle X_i, v \rangle|^p, \dots, \frac{1}{m} \sum_{i \in B_k} |\langle X_i, v \rangle|^p \right).$$

Then there exists an absolute constant $C > 0$ such that with probability at least $1 - \delta$, simultaneously for all $v \in S^{d-1}$,

$$|\text{MOM}_p(v) - \mathbb{E}|\langle X, v \rangle|^p| \leq C L^p \mathbb{E}|\langle X, v \rangle|^p \sqrt{\frac{d \log(ep) + \log(1/\delta)}{n}}.$$

Proof. For $v \in S^{d-1}$, set

$$Y_i(v) = |\langle X_i, v \rangle|^p, \quad \bar{Y}_j(v) = \frac{1}{m} \sum_{i \in B_j} Y_i(v).$$

Then

$$\mathbb{E}Y_i(v) = \mathbb{E}|\langle X, v \rangle|^p,$$

and, by the L_p -to- L_{2p} assumption,

$$\mathbb{E}Y_i(v)^2 = \mathbb{E}|\langle X, v \rangle|^{2p} \leq L^{2p} (\mathbb{E}|\langle X, v \rangle|^p)^2.$$

Hence

$$\text{Var}(Y_i(v)) \leq L^{2p} (\mathbb{E}|\langle X, v \rangle|^p)^2.$$

Applying Chebyshev's inequality to the block mean $\bar{Y}_j(v)$ yields, whenever $\mathbb{E}|\langle X, v \rangle|^p > 0$,

$$\mathbb{P} \left(|\bar{Y}_j(v) - \mathbb{E}|\langle X, v \rangle|^p| \geq 2\sqrt{2} L^p \mathbb{E}|\langle X, v \rangle|^p m^{-1/2} \right) \leq \frac{1}{8}.$$

Now let $Z_j = (X_i)_{i \in B_j} \in (\mathbf{R}^d)^m$ denote the j -th block, and for $z = (x_1, \dots, x_m) \in (\mathbf{R}^d)^m$ define

$$f_v(z) = \mathbf{1} \left(\left(\frac{1}{m} \sum_{\ell=1}^m \langle x_\ell, v \rangle^p - \mathbb{E}|\langle X, v \rangle|^p \right)^2 > \frac{8L^{2p}}{m} (\mathbb{E}|\langle X, v \rangle|^p)^2 \right).$$

Since p is even, we have $|\langle x, v \rangle|^p = \langle x, v \rangle^p$. Therefore, for each fixed $z \in (\mathbf{R}^d)^m$, the map $v \mapsto f_v(z)$ is the indicator of the positivity of a polynomial in v of degree at most $2p$. By Theorem 11.3, the class

$$\mathcal{G} = \{f_v : v \in S^{d-1}\}$$

has VC dimension at most $C_1 d \log(ep)$ for an absolute constant $C_1 > 0$.

If $\mathbb{E}|\langle X, v \rangle|^p = 0$, then $\langle X, v \rangle = 0$ almost surely, so $f_v(Z_1) = 0$ almost surely. If $\mathbb{E}|\langle X, v \rangle|^p > 0$, then the previous Chebyshev bound shows that

$$Pf_v \leq \frac{1}{8},$$

where P denotes the law of one block Z_1 . Hence

$$Pf_v \leq \frac{1}{8} \quad \text{for every } v \in S^{d-1}.$$

Applying the uniform VC inequality from the previous section (namely, Corollary 9.17) to the i.i.d. blocks $Z_1, \dots, Z_k$ and the class $\mathcal{G}$, we obtain that with probability at least $1 - \delta$,

$$\sup_{v \in S^{d-1}} \frac{1}{k} \sum_{j=1}^k f_v(Z_j) \leq \sup_{v \in S^{d-1}} Pf_v + C_2 \sqrt{\frac{d \log(ep) + \log(1/\delta)}{k}} \leq \frac{1}{8} + C_2 \sqrt{\frac{d \log(ep) + \log(1/\delta)}{k}}.$$

By the definition of k , if C_0 is chosen large enough, then the right-hand side is at most $1/4$.

Thus, on this event, for every $v \in S^{d-1}$, at most a quarter of the blocks are bad. In particular, at least $3k/4$ blocks satisfy

$$|\bar{Y}_j(v) - \mathbb{E} |\langle X, v \rangle|^p| \leq 2\sqrt{2} L^p \mathbb{E} |\langle X, v \rangle|^p m^{-1/2}.$$

Since more than half of the blocks are good, their lower median is also good:

$$|\text{MOM}_p(v) - \mathbb{E} |\langle X, v \rangle|^p| \leq 2\sqrt{2} L^p \mathbb{E} |\langle X, v \rangle|^p m^{-1/2} = 2\sqrt{2} L^p \mathbb{E} |\langle X, v \rangle|^p \sqrt{\frac{k}{n}}.$$

Using the choice of k and absorbing constants completes the proof. $\square$

Corollary 11.7 (Uniform estimation of directional L_p norms). *Under the assumptions of Theorem 11.6, define*

$$\eta_n = C L^p \sqrt{\frac{d \log(ep) + \log(1/\delta)}{n}},$$

where $C > 0$ is the constant from the theorem. Then, on the same event of probability at least $1 - \delta$,

$$(1 - \eta_n) \|\langle X, v \rangle\|_{L_p}^p \leq \text{MOM}_p(v) \leq (1 + \eta_n) \|\langle X, v \rangle\|_{L_p}^p \quad \text{for all } v \in S^{d-1}.$$

In particular, whenever $\eta_n < 1$,

$$(1 - \eta_n)^{1/p} \|\langle X, v \rangle\|_{L_p} \leq \text{MOM}_p(v)^{1/p} \leq (1 + \eta_n)^{1/p} \|\langle X, v \rangle\|_{L_p} \quad \text{for all } v \in S^{d-1}.$$

Remark 11.8. The theorem is best read as a *relative* estimate: the error in direction v is proportional to the size of the quantity being estimated in that direction. This is exactly the right scale for moment estimation.

Independent study material

11.4 One-sided lower tails under few moments

Upper-tail concentration inequalities are usually derived from exponential moments. A useful fact is that, for nonnegative random variables, one can obtain a genuinely sub-Gaussian *lower-tail* bound under only a second-moment assumption.

Proposition 11.9 (A one-sided lower tail under two moments). *Let $X_1, \dots, X_n$ be i.i.d. nonnegative random variables such that*

$$\mathbb{E} X_1 = \mu, \quad \mathbb{E} X_1^2 \leq \sigma^2$$

for some $\sigma > 0$. Then for every $t \geq 0$,

$$\mathbb{P}\left(\mu - \frac{1}{n} \sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{nt^2}{2\sigma^2}\right).$$

Proof. Fix $\lambda > 0$. Since $x \geq 0$ and $e^{-u} \leq 1 - u + u^2/2$ for all $u \geq 0$, we have

$$e^{-\lambda x} \leq 1 - \lambda x + \frac{\lambda^2 x^2}{2}, \quad x \geq 0.$$

Applying this with $x = X_1$ and taking expectations gives

$$\mathbb{E}e^{-\lambda X_1} \leq 1 - \lambda\mu + \frac{\lambda^2}{2}\mathbb{E}X_1^2 \leq 1 - \lambda\mu + \frac{\lambda^2\sigma^2}{2} \leq \exp\left(-\lambda\mu + \frac{\lambda^2\sigma^2}{2}\right),$$

where in the last step we used $1 + u \leq e^u$.

Therefore

$$\mathbb{E}e^{\lambda(\mu - X_1)} = e^{\lambda\mu}\mathbb{E}e^{-\lambda X_1} \leq \exp\left(\frac{\lambda^2\sigma^2}{2}\right).$$

By independence,

$$\mathbb{E} \exp\left(\lambda \sum_{i=1}^n (\mu - X_i)\right) \leq \exp\left(\frac{\lambda^2 n \sigma^2}{2}\right).$$

Now apply Chernoff's bound:

$$\begin{aligned} \mathbb{P}\left(\mu - \frac{1}{n} \sum_{i=1}^n X_i \geq t\right) &= \mathbb{P}\left(\sum_{i=1}^n (\mu - X_i) \geq nt\right) \\ &\leq e^{-\lambda nt} \mathbb{E} \exp\left(\lambda \sum_{i=1}^n (\mu - X_i)\right) \\ &\leq \exp\left(-\lambda nt + \frac{\lambda^2 n \sigma^2}{2}\right). \end{aligned}$$

Optimizing over $\lambda > 0$ gives $\lambda = t/\sigma^2$, and hence

$$\mathbb{P}\left(\mu - \frac{1}{n} \sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{nt^2}{2\sigma^2}\right).$$

□

Remark 11.10. Theorem 11.9 is one-sided, and this is essential. Without additional assumptions one should not expect a comparable *upper*-tail bound from only two moments. The nonnegativity of the variables is exactly what allows the simple inequality $e^{-\lambda x} \leq 1 - \lambda x + \lambda^2 x^2/2$.

Another useful lower-tail device is Paley–Zygmund's inequality. Unlike Chernoff's method, it gives a lower bound on the probability that a nonnegative random variable is not too small compared to its mean.

Lemma 11.11 (Paley–Zygmund inequality). *Let $Z \geq 0$ be a random variable with $0 < \mathbb{E}Z^2 < \infty$. Then for every $\theta \in (0, 1)$,*

$$\mathbb{P}(Z \geq \theta \mathbb{E}Z) \geq (1 - \theta)^2 \frac{(\mathbb{E}Z)^2}{\mathbb{E}Z^2}.$$

Proof. We write ◻

$$\mathbb{E}Z = \mathbb{E}[Z \mathbf{1}(Z \geq \theta \mathbb{E}Z)] + \mathbb{E}[Z \mathbf{1}(Z < \theta \mathbb{E}Z)].$$

The second term is bounded by $\theta \mathbb{E}Z$, while the first term is bounded by Cauchy–Schwarz:

$$\mathbb{E}[Z \mathbf{1}(Z \geq \theta \mathbb{E}Z)] \leq (\mathbb{E}Z^2)^{1/2} \mathbb{P}(Z \geq \theta \mathbb{E}Z)^{1/2}.$$

Therefore

$$(1 - \theta)\mathbb{E}Z \leq (\mathbb{E}Z^2)^{1/2} \mathbb{P}(Z \geq \theta \mathbb{E}Z)^{1/2}.$$

Squaring both sides yields the claim. ◻

A convenient consequence is a uniform small-ball estimate under a fourth-moment assumption.

Corollary 11.12 (A small-ball consequence of a fourth-moment bound). *Let $X \in \mathbf{R}^d$ be a random vector such that*

$$\|\langle X, v \rangle\|_{L_4} \leq L \|\langle X, v \rangle\|_{L_2} \quad \text{for every } v \in S^{d-1}.$$

Then for every $\theta \in (0, 1)$ and every $v \in S^{d-1}$,

$$\mathbb{P}\left(|\langle X, v \rangle| \geq \sqrt{\theta} \|\langle X, v \rangle\|_{L_2}\right) \geq \frac{(1 - \theta)^2}{L^4}.$$

Proof. Fix $v \in S^{d-1}$. If $\|\langle X, v \rangle\|_{L_2} = 0$, then $\langle X, v \rangle = 0$ almost surely, and hence

$$\mathbb{P}\left(|\langle X, v \rangle| \geq \sqrt{\theta} \|\langle X, v \rangle\|_{L_2}\right) = 1.$$

So the claim is trivial in this case.

Assume therefore that $\|\langle X, v \rangle\|_{L_2} > 0$, and apply Theorem 11.11 to the nonnegative random variable

$$Z = \langle X, v \rangle^2.$$

Then

$$\mathbb{E}Z = \|\langle X, v \rangle\|_{L_2}^2, \quad \mathbb{E}Z^2 = \mathbb{E}\langle X, v \rangle^4 \leq L^4 \|\langle X, v \rangle\|_{L_2}^4.$$

Hence

$$\mathbb{P}\left(\langle X, v \rangle^2 \geq \theta \|\langle X, v \rangle\|_{L_2}^2\right) = \mathbb{P}(Z \geq \theta \mathbb{E}Z) \geq (1 - \theta)^2 \frac{(\mathbb{E}Z)^2}{\mathbb{E}Z^2} \geq \frac{(1 - \theta)^2}{L^4},$$

which is exactly the stated inequality. ◻

11.5 Lower singular value of the sample covariance matrix

We now combine the small-ball principle with the VC control provided by Theorem 11.3. This gives a clean lower bound on the empirical quadratic form, uniform over all directions.

Definition 11.13 (Small-ball condition). Let $X \in \mathbf{R}^d$ be a mean-zero random vector with covariance matrix

$$\Sigma = \mathbb{E}[XX^\top].$$

We say that X satisfies the (c_0, β) -small-ball condition if there exist constants $c_0 \in (0, 1)$ and $\beta \in (0, 1]$ such that

$$\mathbb{P}\left(|\langle X, v \rangle| \geq c_0 \sqrt{v^\top \Sigma v}\right) \geq \beta \quad \text{for every } v \in S^{d-1}.$$

Theorem 11.14 (Lower singular value of the sample covariance matrix). *Let $X_1, \dots, X_n$ be i.i.d. mean-zero random vectors in $\mathbf{R}^d$, let*

$$\Sigma = \mathbb{E}[XX^\top], \quad \widehat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top,$$

and assume that X satisfies the (c_0, β) -small-ball condition from Theorem 11.13. Then there exists an absolute constant $C > 0$ such that, with probability at least $1 - \delta$,

$$\lambda_{\min}(\widehat{\Sigma}) \geq c_0^2 \left(\beta - C \sqrt{\frac{d + \log(1/\delta)}{n}} \right) \lambda_{\min}(\Sigma).$$

In particular, if

$$n \geq C \beta^{-2} (d + \log(1/\delta)),$$

then with probability at least $1 - \delta$,

$$\lambda_{\min}(\widehat{\Sigma}) \geq \frac{c_0^2 \beta}{2} \lambda_{\min}(\Sigma).$$

Proof. For $v \in S^{d-1}$, define

$$g_v(x) = \mathbf{1}(|\langle x, v \rangle| \geq c_0 \sqrt{v^\top \Sigma v}) = \mathbf{1}(\langle x, v \rangle^2 - c_0^2 v^\top \Sigma v \geq 0).$$

For each fixed $x \in \mathbf{R}^d$, the function $v \mapsto \langle x, v \rangle^2 - c_0^2 v^\top \Sigma v$ is a polynomial in v of degree 2. Hence, by Theorem 11.3, the class

$$\mathcal{G} = \{g_v : v \in S^{d-1}\}$$

has VC dimension at most $C_1 d$ for an absolute constant $C_1 > 0$.

By the uniform VC bound from the previous section, with probability at least $1 - \delta$,

$$\sup_{v \in S^{d-1}} |P_n g_v - P g_v| \leq C_2 \sqrt{\frac{d + \log(1/\delta)}{n}},$$

where

$$P_n g_v = \frac{1}{n} \sum_{i=1}^n g_v(X_i), \quad P g_v = \mathbb{P}(|\langle X, v \rangle| \geq c_0 \sqrt{v^\top \Sigma v}).$$

By the small-ball assumption,

$$P g_v \geq \beta \quad \text{for every } v \in S^{d-1}.$$

Therefore, on the same event,

$$P_n g_v \geq \beta - C_2 \sqrt{\frac{d + \log(1/\delta)}{n}} \quad \text{for every } v \in S^{d-1}.$$

Now fix $v \in S^{d-1}$. Since

$$\langle X_i, v \rangle^2 \geq c_0^2 v^\top \Sigma v \cdot g_v(X_i),$$

we obtain

$$v^\top \widehat{\Sigma} v = \frac{1}{n} \sum_{i=1}^n \langle X_i, v \rangle^2 \geq c_0^2 v^\top \Sigma v \cdot P_n g_v.$$

Using the previous lower bound on $P_n g_v$, we get

$$v^\top \widehat{\Sigma} v \geq c_0^2 \left(\beta - C_2 \sqrt{\frac{d + \log(1/\delta)}{n}} \right) v^\top \Sigma v \quad \text{for every } v \in S^{d-1}.$$

Set

$$\eta_n = \beta - C_2 \sqrt{\frac{d + \log(1/\delta)}{n}}.$$

If $\eta_n \geq 0$, then $v^\top \Sigma v \geq \lambda_{\min}(\Sigma)$ for every $v \in S^{d-1}$, and therefore

$$v^\top \widehat{\Sigma} v \geq c_0^2 \eta_n \lambda_{\min}(\Sigma) \quad \text{for every } v \in S^{d-1}.$$

Taking the infimum over $v \in S^{d-1}$ yields

$$\lambda_{\min}(\widehat{\Sigma}) \geq c_0^2 \eta_n \lambda_{\min}(\Sigma).$$

If $\eta_n < 0$, then the claimed lower bound is trivial, because $\widehat{\Sigma} \succeq 0$ implies $\lambda_{\min}(\widehat{\Sigma}) \geq 0$, while

$$c_0^2 \eta_n \lambda_{\min}(\Sigma) \leq 0.$$

This proves the first claim.

For the second claim, it is enough to require

$$C_2 \sqrt{\frac{d + \log(1/\delta)}{n}} \leq \frac{\beta}{2},$$

because then $\eta_n \geq \beta/2$ and hence

$$\lambda_{\min}(\widehat{\Sigma}) \geq \frac{c_0^2 \beta}{2} \lambda_{\min}(\Sigma). \quad \square$$

Remark 11.15. If $A \in \mathbf{R}^{n \times d}$ is the data matrix with rows $X_1^\top, \dots, X_n^\top$, then

$$\widehat{\Sigma} = \frac{1}{n} A^\top A.$$

Hence Theorem 11.14 is equivalently a lower bound on the least singular value of $n^{-1/2} A$. In the isotropic case $\Sigma = I_d$, the theorem shows that the empirical covariance is bounded away from 0 as soon as

$$n \gtrsim \beta^{-2} (d + \log(1/\delta)).$$

In particular, if the small-ball parameters c_0 and β are absolute constants, then $\widehat{\Sigma}$ is nondegenerate from below with high probability once $n \gtrsim d + \log(1/\delta)$.

Corollary 11.16 (A fourth-moment version). *Assume that X is mean-zero with covariance Σ , and that*

$$\| \langle X, v \rangle \|_{L_4} \leq L \| \langle X, v \rangle \|_{L_2} \quad \text{for every } v \in S^{d-1}.$$

Then for every $\theta \in (0, 1)$, with probability at least $1 - \delta$,

$$\lambda_{\min}(\widehat{\Sigma}) \geq \theta \left(\frac{(1 - \theta)^2}{L^4} - C \sqrt{\frac{d + \log(1/\delta)}{n}} \right) \lambda_{\min}(\Sigma),$$

where $C > 0$ is an absolute constant. In particular, taking $\theta = 1/2$, if

$$n \geq C L^8 (d + \log(1/\delta)),$$

then with probability at least $1 - \delta$,

$$\lambda_{\min}(\widehat{\Sigma}) \geq \frac{1}{16L^4} \lambda_{\min}(\Sigma).$$

Proof. By Theorem 11.12,

$$\mathbb{P}\left(|\langle X, v \rangle| \geq \sqrt{\theta} \sqrt{v^\top \Sigma v}\right) \geq \frac{(1 - \theta)^2}{L^4} \quad \text{for every } v \in S^{d-1}.$$

Thus X satisfies the small-ball condition with

$$c_0 = \sqrt{\theta}, \quad \beta = \frac{(1 - \theta)^2}{L^4}.$$

The claim now follows directly from Theorem 11.14. □

Additional material

This is one of the nice side paths in the course. The proofs are closely related to the concentration inequalities we have developed so far. They also provide a particularly transparent demonstration of how the techniques of the course fit together: Chebyshev on blocks, bounded differences, symmetrization, contraction, and Euclidean geometry all appear in one place. We first state a uniform version of the median-of-means principle.

Lemma 11.17 (Uniform median-of-means bound). *Assume that $n = Km$, let $\mathcal{F}$ be a class of measurable functions on $\mathcal{X}$, and define*

$$\text{MOM}_K(f) = \text{Med}\left(\frac{1}{m} \sum_{i \in B_1} f(X_i), \dots, \frac{1}{m} \sum_{i \in B_K} f(X_i)\right).$$

Set

$$R_n(\mathcal{F}) = \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|, \quad V(\mathcal{F}) = \sup_{f \in \mathcal{F}} \text{Var}(f(X)).$$

Then there exist absolute constants $C, C_0 > 0$ such that if

$$K = \lceil C_0 \log(2/\delta) \rceil,$$

then with probability at least $1 - \delta$,

$$\sup_{f \in \mathcal{F}} |\text{MOM}_K(f) - Pf| \leq C \left(R_n(\mathcal{F}) + \sqrt{\frac{V(\mathcal{F}) \log(2/\delta)}{n}} \right).$$

Proof. We begin with the lower deviation. Fix $t > 0$, and define

$$G_t = \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \mathbf{1}(Pf - \bar{f}_j \geq t), \quad \bar{f}_j = \frac{1}{m} \sum_{i \in B_j} f(X_i).$$

If $G_t < 1/2$, then for every $f \in \mathcal{F}$ more than half of the block averages satisfy $\bar{f}_j > Pf - t$. Since Med is the lower median, this implies

$$Pf - \text{MOM}_K(f) \leq t.$$

So it is enough to bound G_t .

Introduce the function

$$\phi(u) = (u - 1)\mathbf{1}(1 \leq u \leq 2) + \mathbf{1}(u > 2), \quad u \in \mathbf{R}.$$

Then $0 \leq \phi \leq 1$, $\phi(0) = 0$, the function ϕ is 1-Lipschitz, and

$$\mathbf{1}(u \geq 2) \leq \phi(u) \leq \mathbf{1}(u \geq 1).$$

Therefore

$$G_t \leq U_t = \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \phi\left(\frac{2(Pf - \bar{f}_j)}{t}\right).$$

Write

$$A_t = \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \mathbb{E} \phi\left(\frac{2(Pf - \bar{f}_j)}{t}\right), \quad B_t = U_t - A_t.$$

We first bound A_t . Since $\phi(u) \leq \mathbf{1}(u \geq 1)$,

$$A_t \leq \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \mathbb{P}\left(Pf - \bar{f}_j \geq \frac{t}{2}\right).$$

By Chebyshev's inequality,

$$\mathbb{P}\left(Pf - \bar{f}_j \geq \frac{t}{2}\right) \leq \mathbb{P}\left(|\bar{f}_j - Pf| \geq \frac{t}{2}\right) \leq \frac{4 \operatorname{Var}(f(X))}{mt^2}.$$

Hence

$$A_t \leq \frac{4V(\mathcal{F})K}{nt^2}.$$

We next turn to B_t . Changing one block affects only one summand and changes U_t by at most $1/K$. Since A_t is deterministic, B_t also satisfies the bounded differences property with constants $1/K, \dots, 1/K$. McDiarmid's inequality therefore implies that for every $y > 0$,

$$\mathbb{P}(B_t - \mathbb{E}B_t \geq y) \leq e^{-2Ky^2}.$$

It remains to estimate $\mathbb{E}B_t$. Since $\sup_{\alpha} a_{\alpha} - \sup_{\alpha} b_{\alpha} \leq \sup_{\alpha} (a_{\alpha} - b_{\alpha})$, we have

$$B_t \leq \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \left[\phi\left(\frac{2(Pf - \bar{f}_j)}{t}\right) - \mathbb{E} \phi\left(\frac{2(Pf - \bar{f}_j)}{t}\right) \right].$$

By symmetrization at the block level,

$$\begin{aligned} \mathbb{E}B_t &\leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \varepsilon_j \phi\left(\frac{2(Pf - \bar{f}_j)}{t}\right) \\ &\leq \frac{4}{t} \mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \varepsilon_j (Pf - \bar{f}_j), \end{aligned}$$

where we used the contraction inequality and the fact that $\phi(0) = 0$.

Now let $X'_1, \dots, X'_n$ be an independent copy of $X_1, \dots, X_n$, and let $\bar{f}'_j$ denote the corresponding block averages. Since $Pf = \mathbb{E}' \bar{f}'_j$ for every j , Jensen's inequality gives

$$\mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \varepsilon_j (Pf - \bar{f}_j) \leq \mathbb{E} \mathbb{E}' \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \varepsilon_j (\bar{f}'_j - \bar{f}_j).$$

Let $b(i) \in \{1, \dots, K\}$ denote the index of the block containing i . Then

$$\frac{1}{K} \sum_{j=1}^K \varepsilon_j (\bar{f}'_j - \bar{f}_j) = \frac{1}{n} \sum_{i=1}^n \varepsilon_{b(i)} (f(X'_i) - f(X_i)).$$

Conditioning on the block signs $(\varepsilon_1, \dots, \varepsilon_K)$ and applying standard symmetrization with an independent family of Rademacher signs $(\xi_i)_{i=1}^n$, we obtain

$$\mathbb{E} \mathbb{E}' \sup_{f \in \mathcal{F}} \frac{1}{K} \sum_{j=1}^K \varepsilon_j (\bar{f}'_j - \bar{f}_j) \leq \frac{2}{n} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \varepsilon_{b(i)} \xi_i f(X_i) \right|.$$

The random signs $(\varepsilon_{b(i)} \xi_i)_{i=1}^n$ have the same joint law as an i.i.d. Rademacher family, so the last display equals

$$\frac{2}{n} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \varepsilon_i f(X_i) \right| = 2 R_n(\mathcal{F}).$$

Consequently,

$$\mathbb{E} B_t \leq \frac{8 R_n(\mathcal{F})}{t}.$$

Combining the estimates, we have shown that with probability at least $1 - e^{-2Ky^2}$,

$$G_t \leq \frac{4 V(\mathcal{F}) K}{nt^2} + \frac{8 R_n(\mathcal{F})}{t} + y.$$

Choose $y = 1/8$. If

$$t \geq 64 R_n(\mathcal{F}) \quad \text{and} \quad t^2 \geq \frac{32 V(\mathcal{F}) K}{n},$$

then $G_t < 1/2$. Thus, with probability at least $1 - e^{-K/32}$, we have

$$Pf - \text{MOM}_K(f) \leq t \quad \text{for every } f \in \mathcal{F}.$$

Applying the same argument to the class $-\mathcal{F}$ (for which $R_n(-\mathcal{F}) = R_n(\mathcal{F})$ and $V(-\mathcal{F}) = V(\mathcal{F})$) gives, with probability at least $1 - e^{-K/32}$,

$$\text{MOM}_K(f) - Pf \leq t \quad \text{for every } f \in \mathcal{F}.$$

Therefore, by a union bound, both inequalities hold simultaneously with probability at least $1 - 2e^{-K/32}$.

Choosing

$$t = C \left(R_n(\mathcal{F}) + \sqrt{\frac{V(\mathcal{F}) K}{n}} \right)$$

for a sufficiently large absolute constant C , and then using

$$K = \lceil C_0 \log(2/\delta) \rceil$$

with C_0 large enough so that $2e^{-K/32} \leq \delta$, yields

$$\sup_{f \in \mathcal{F}} |\text{MOM}_K(f) - Pf| \leq C \left(R_n(\mathcal{F}) + \sqrt{\frac{V(\mathcal{F}) \log(2/\delta)}{n}} \right)$$

with probability at least $1 - \delta$. □

Remark 11.18. This proof is worth reading slowly. The only place where the median itself enters is the simple implication ◻

$$\frac{1}{K} \sum_{j=1}^K \mathbf{1}(Pf - \bar{f}_j \geq t) < \frac{1}{2} \implies Pf - \text{MOM}_K(f) \leq t.$$

Everything else is a clean empirical-process argument. That is precisely why this lemma is such good study material: it is short, but it showcases several of the main techniques of the course in a very transparent way.

We now apply Theorem 11.17 to the class of linear functionals. This gives a robust mean estimator for random vectors under only a covariance assumption.

Proposition 11.19 (Mean estimation in $\mathbf{R}^d$ under two moments). *Let $X_1, \dots, X_n$ be i.i.d. random vectors in $\mathbf{R}^d$ with mean*

$$\mu = \mathbb{E}X_1$$

and covariance matrix

$$\Sigma = \mathbb{E}[(X_1 - \mu)(X_1 - \mu)^\top].$$

Let

$$B_2^d = \{v \in \mathbf{R}^d : \|v\|_2 \leq 1\},$$

assume that $n = Km$, and choose

$$K = \lceil C_0 \log(2/\delta) \rceil,$$

where $C_0 > 0$ is the constant from Theorem 11.17. For the realized sample, define

$$T(v) = \text{MOM}_K(\langle X, v \rangle), \quad v \in B_2^d.$$

Since T is an order statistic of finitely many linear forms, it is continuous on B_2^d , hence bounded there. Therefore the map

$$\nu \mapsto \sup_{v \in B_2^d} |\langle \nu, v \rangle - T(v)|$$

is continuous and coercive, so it admits a minimizer. Choose any minimizer and denote it by $\hat{\mu}$. Then with probability at least $1 - \delta$,

$$\|\hat{\mu} - \mu\|_2 \leq C \left(\sqrt{\frac{\text{Tr}(\Sigma)}{n}} + \sqrt{\frac{\|\Sigma\|_{\text{op}} \log(2/\delta)}{n}} \right),$$

where $C > 0$ is an absolute constant.

Proof. For $v \in B_2^d$, let $f_v(x) = \langle x - \mu, v \rangle$. Then

$$Pf_v = \mathbb{E} \langle X - \mu, v \rangle = 0.$$

By the triangle inequality and the defining property of $\hat{\mu}$,

$$\begin{aligned} \|\hat{\mu} - \mu\|_2 &= \sup_{v \in B_2^d} |\langle \hat{\mu} - \mu, v \rangle| \\ &\leq \sup_{v \in B_2^d} |\langle \hat{\mu}, v \rangle - T(v)| + \sup_{v \in B_2^d} |\langle \mu, v \rangle - T(v)| \\ &\leq 2 \sup_{v \in B_2^d} |\langle \mu, v \rangle - T(v)|. \end{aligned}$$

For each block B_j ,

$$\frac{1}{m} \sum_{i \in B_j} f_v(X_i) = \frac{1}{m} \sum_{i \in B_j} \langle X_i, v \rangle - \langle \mu, v \rangle.$$

Since the lower median is translation equivariant,

$$T(v) - \langle \mu, v \rangle = \text{MOM}_K(f_v).$$

Hence

$$\|\hat{\mu} - \mu\|_2 \leq 2 \sup_{v \in B_2^d} |\text{MOM}_K(f_v)|.$$

We now apply Theorem 11.17 to the class

$$\mathcal{F} = \{f_v : v \in B_2^d\}.$$

First,

$$V(\mathcal{F}) = \sup_{v \in B_2^d} \text{Var}(\langle X - \mu, v \rangle) = \sup_{v \in B_2^d} v^\top \Sigma v = \|\Sigma\|_{\text{op}}.$$

Second, since the class is symmetric in v ,

$$\begin{aligned} R_n(\mathcal{F}) &= \mathbb{E} \sup_{v \in B_2^d} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \langle X_i - \mu, v \rangle \right| \\ &= \frac{1}{n} \mathbb{E} \left\| \sum_{i=1}^n \varepsilon_i (X_i - \mu) \right\|_2 \\ &\leq \frac{1}{n} \left(\mathbb{E} \left\| \sum_{i=1}^n \varepsilon_i (X_i - \mu) \right\|_2^2 \right)^{1/2} \\ &= \frac{1}{n} \left(\sum_{i=1}^n \mathbb{E} \|X_i - \mu\|_2^2 \right)^{1/2} = \sqrt{\frac{\text{Tr}(\Sigma)}{n}}, \end{aligned}$$

where we used Jensen's inequality and the fact that the mixed terms vanish after averaging over the signs.

Therefore Theorem 11.17 gives, with probability at least $1 - \delta$,

$$\sup_{v \in B_2^d} |\text{MOM}_K(f_v)| \leq C \left(\sqrt{\frac{\text{Tr}(\Sigma)}{n}} + \sqrt{\frac{\|\Sigma\|_{\text{op}} \log(2/\delta)}{n}} \right).$$

Combining this with the earlier bound on $\|\hat{\mu} - \mu\|_2$ finishes the proof. $\square$

Remark 11.20. Theorem 11.19 has the same form as the usual Gaussian benchmark

$$\left\| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right\|_2 \lesssim \sqrt{\frac{\text{Tr}(\Sigma)}{n}} + \sqrt{\frac{\|\Sigma\|_{\text{op}} \log(1/\delta)}{n}},$$

but it only assumes the existence of the covariance matrix. This is one of the cleanest examples of the general message of the course: non-asymptotic rates of Gaussian shape may persist far outside the Gaussian world once the estimator is chosen correctly.

12 Nonparametric least squares and localization

We now return to least squares beyond finite-dimensional linear models. The basic lesson is that the performance of the least-squares estimator is controlled not by the complexity of the whole class, but by the complexity of a *localized* neighborhood around the target function. This point of view is one of the central ideas in modern nonparametric statistics.

Throughout this section, we work in the *fixed-design* setting. Thus $x_1, \dots, x_n \in \mathcal{X}$ are deterministic design points, and we observe

$$Y_i = f^*(x_i) + \sigma w_i, \quad i = 1, \dots, n,$$

where $f^* : \mathcal{X} \rightarrow \mathbf{R}$ is the unknown regression function, $\sigma > 0$ is the noise level, and

$$w_1, \dots, w_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1).$$

We assume that $f^* \in \mathcal{F}$, where $\mathcal{F}$ is the function class over which we perform least squares.

For any function $g : \mathcal{X} \rightarrow \mathbf{R}$, define its empirical norm by

$$\|g\|_n^2 = \frac{1}{n} \sum_{i=1}^n g(x_i)^2.$$

For any $f : \mathcal{X} \rightarrow \mathbf{R}$, we also write

$$\|Y - f\|_n^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - f(x_i))^2.$$

We assume that the constrained least-squares problem $\arg \min_{f \in \mathcal{F}} \|Y - f\|_n^2$ is non-empty, and we let $\hat{f} \in \arg \min_{f \in \mathcal{F}} \|Y - f\|_n^2$ denote any empirical minimizer.

12.1 A basic inequality

The first step is the same as in ordinary least squares: compare the empirical loss of $\hat{f}$ with that of the true function f^* .

Lemma 12.1 (Basic inequality). *For any least-squares estimator $\hat{f} \in \arg \min_{f \in \mathcal{F}} \|Y - f\|_n^2$, one has*

$$\frac{1}{2} \|\hat{f} - f^*\|_n^2 \leq \frac{\sigma}{n} \sum_{i=1}^n w_i (\hat{f}(x_i) - f^*(x_i)).$$

Proof. Since $f^* \in \mathcal{F}$ is feasible and $\hat{f}$ minimizes the empirical loss,

$$\|Y - \hat{f}\|_n^2 \leq \|Y - f^*\|_n^2.$$

Using the model $Y_i = f^*(x_i) + \sigma w_i$, we obtain

$$Y - \hat{f} = (f^* - \hat{f}) + \sigma w.$$

Therefore

$$\|Y - \hat{f}\|_n^2 = \|\hat{f} - f^*\|_n^2 + \frac{2\sigma}{n} \sum_{i=1}^n w_i (f^*(x_i) - \hat{f}(x_i)) + \|Y - f^*\|_n^2.$$

Comparing this expression with $\|Y - f^*\|_n^2$ and rearranging yields

$$\|\hat{f} - f^*\|_n^2 \leq \frac{2\sigma}{n} \sum_{i=1}^n w_i (\hat{f}(x_i) - f^*(x_i)),$$

which is the claimed inequality. $\square$

12.2 Star-shaped classes and localized Gaussian complexity

Definition 12.2 (Star-shaped around $f^\star$). A function class $\mathcal{F}$ is said to be *star-shaped around $f^\star$* if

$$f^\star + \alpha(f - f^\star) \in \mathcal{F} \quad \text{for every } f \in \mathcal{F} \text{ and every } \alpha \in [0, 1].$$

Equivalently, the shifted class

$$\mathcal{F}^\star = \mathcal{F} - \{f^\star\} = \{f - f^\star : f \in \mathcal{F}\}$$

is star-shaped around the origin.

Remark 12.3. If $\mathcal{F}$ is convex and $f^\star \in \mathcal{F}$, then $\mathcal{F}$ is star-shaped around $f^\star$. The star-shaped condition is weaker than convexity, and it is exactly what is needed for the localization argument below.

The complexity quantity that governs least squares is a localized Gaussian width.

Definition 12.4 (Localized Gaussian complexity). For $\delta > 0$, define

$$G_n(\delta; \mathcal{F}^\star) = \mathbb{E}_w \left[\sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq \delta}} \left| \frac{1}{n} \sum_{i=1}^n w_i g(x_i) \right| \right].$$

Since the design points $x_1, \dots, x_n$ are fixed, $G_n(\delta; \mathcal{F}^\star)$ is a deterministic quantity.

The star-shaped property forces the local complexity per unit radius to decrease as the radius increases.

Lemma 12.5 (Monotonicity of localized complexity). *If $\mathcal{F}^\star$ is star-shaped, then the function*

$$\delta \mapsto \frac{G_n(\delta; \mathcal{F}^\star)}{\delta}$$

is non-increasing on $(0, \infty)$. Moreover, the set

$$\left\{ \delta > 0 : \frac{G_n(\delta; \mathcal{F}^\star)}{\delta} \leq \frac{\delta}{2\sigma} \right\}$$

is non-empty.

Proof. (This proof can be skipped at first reading.) Fix $0 < \delta \leq t$. For any $g \in \mathcal{F}^\star$ with $\|g\|_n \leq t$, define

$$\tilde{g} = \frac{\delta}{t} g.$$

Since $\delta/t \in [0, 1]$ and $\mathcal{F}^\star$ is star-shaped, we have $\tilde{g} \in \mathcal{F}^\star$, and

$$\|\tilde{g}\|_n \leq \delta.$$

Therefore

$$\frac{\delta}{t} \left| \frac{1}{n} \sum_{i=1}^n w_i g(x_i) \right| = \left| \frac{1}{n} \sum_{i=1}^n w_i \tilde{g}(x_i) \right| \leq \sup_{\substack{h \in \mathcal{F}^\star \\ \|h\|_n \leq \delta}} \left| \frac{1}{n} \sum_{i=1}^n w_i h(x_i) \right|.$$

Taking the supremum over all $g \in \mathcal{F}^\star$ with $\|g\|_n \leq t$, and then expectation in w , yields

$$\frac{\delta}{t} G_n(t; \mathcal{F}^\star) \leq G_n(\delta; \mathcal{F}^\star),$$

which is equivalent to

$$\frac{G_n(t; \mathcal{F}^*)}{t} \leq \frac{G_n(\delta; \mathcal{F}^*)}{\delta}.$$

This proves monotonicity.

To see that the set of valid radii is non-empty, note that for every $\delta > 0$,

$$G_n(\delta; \mathcal{F}^*) \leq \mathbb{E}_w \left[\sup_{\substack{g \in \mathcal{F}^* \\ \|g\|_n \leq \delta}} \frac{1}{n} \|w\|_2 \left(\sum_{i=1}^n g(x_i)^2 \right)^{1/2} \right] \leq \frac{\mathbb{E} \|w\|_2}{\sqrt{n}} \delta.$$

Hence

$$\frac{G_n(\delta; \mathcal{F}^*)}{\delta} \leq \frac{\mathbb{E} \|w\|_2}{\sqrt{n}}.$$

So every δ satisfying

$$\delta \geq \frac{2\sigma \mathbb{E} \|w\|_2}{\sqrt{n}}$$

is valid. □

Definition 12.6 (Valid radius). Any positive scalar δ_n satisfying

$$\frac{G_n(\delta_n; \mathcal{F}^*)}{\delta_n} \leq \frac{\delta_n}{2\sigma}$$

is called a *valid radius*.

Remark 12.7. The critical inequality compares a localized stochastic term $G_n(\delta; \mathcal{F}^*)$ with the quadratic curvature δ^2/σ coming from least squares. Because $G_n(\delta)/\delta$ is non-increasing, once the inequality holds at one radius, it holds at all larger radii as well.

12.3 A localization theorem for nonparametric least squares

We now state the main result. It shows that the prediction error of least squares is controlled by any valid radius δ_n .

Theorem 12.8 (Localized prediction bound). *Assume that $\mathcal{F}^*$ is star-shaped, and let δ_n be any valid radius in the sense of Theorem 12.6. Then for every $t \geq \delta_n$,*

$$\mathbb{P} \left(\|\hat{f} - f^*\|_n^2 \geq 16t\delta_n \right) \leq \exp \left(-\frac{nt\delta_n}{2\sigma^2} \right).$$

Proof. Write

$$\hat{\Delta} = \hat{f} - f^* \in \mathcal{F}^*.$$

The basic inequality from Theorem 12.1 becomes

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{\sigma}{n} \sum_{i=1}^n w_i \hat{\Delta}(x_i).$$

Fix $u \geq \delta_n$, and define the event

$$A(u) = \left\{ \exists g \in \mathcal{F}^* \text{ with } \|g\|_n \geq u \text{ and } \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(x_i) \right| \geq 2u \|g\|_n \right\}.$$

We claim that

$$\mathbb{P}(A(u)) \leq \exp\left(-\frac{nu^2}{2\sigma^2}\right).$$

To prove this, define

$$Z_n(u) = \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(x_i) \right|.$$

Suppose that $A(u)$ occurs. Then there exists $g \in \mathcal{F}^\star$ with $\|g\|_n \geq u$ such that

$$\left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(x_i) \right| \geq 2u\|g\|_n.$$

Define $\tilde{g} = (u/\|g\|_n)g$. Since $\mathcal{F}^\star$ is star-shaped, $\tilde{g} \in \mathcal{F}^\star$, and $\|\tilde{g}\|_n = u$. Moreover,

$$\left| \frac{\sigma}{n} \sum_{i=1}^n w_i \tilde{g}(x_i) \right| = \frac{u}{\|g\|_n} \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(x_i) \right| \geq 2u^2.$$

Hence

$$A(u) \subseteq \{Z_n(u) \geq 2u^2\}.$$

Next, we show that $Z_n(u)$, viewed as a function of $w = (w_1, \dots, w_n) \in \mathbf{R}^n$, is $\sigma u/\sqrt{n}$ -Lipschitz. Indeed, for any $w, w' \in \mathbf{R}^n$,

$$\begin{aligned} |Z_n(u; w) - Z_n(u; w')| &\leq \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \left| \frac{\sigma}{n} \sum_{i=1}^n (w_i - w'_i) g(x_i) \right| \\ &\leq \frac{\sigma}{n} \|w - w'\|_2 \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \left(\sum_{i=1}^n g(x_i)^2 \right)^{1/2} \\ &= \frac{\sigma}{\sqrt{n}} \|w - w'\|_2 \sup_{\substack{g \in \mathcal{F}^\star \\ \|g\|_n \leq u}} \|g\|_n \\ &\leq \frac{\sigma u}{\sqrt{n}} \|w - w'\|_2. \end{aligned}$$

By Gaussian concentration,

$$\mathbb{P}(Z_n(u) \geq \mathbb{E}Z_n(u) + s) \leq \exp\left(-\frac{ns^2}{2\sigma^2 u^2}\right), \quad s > 0.$$

By definition,

$$\mathbb{E}Z_n(u) = \sigma G_n(u; \mathcal{F}^\star).$$

Since $u \geq \delta_n$ and δ_n is valid, Theorem 12.5 gives

$$\frac{G_n(u; \mathcal{F}^\star)}{u} \leq \frac{G_n(\delta_n; \mathcal{F}^\star)}{\delta_n} \leq \frac{\delta_n}{2\sigma},$$

and therefore

$$\mathbb{E}Z_n(u) = \sigma G_n(u; \mathcal{F}^\star) \leq \frac{u\delta_n}{2} \leq \frac{u^2}{2}.$$

Consequently,

$$\mathbb{P}(A(u)) \leq \mathbb{P}(Z_n(u) \geq 2u^2) \leq \mathbb{P}(Z_n(u) \geq \mathbb{E}Z_n(u) + \tfrac{3}{2}u^2) \leq \exp\left(-\frac{9nu^2}{8\sigma^2}\right) \leq \exp\left(-\frac{nu^2}{2\sigma^2}\right).$$

We now complete the argument. Fix $t \geq \delta_n$, and set

$$u = \sqrt{t\delta_n}.$$

Since $u \geq \delta_n$, the previous bound gives

$$\mathbb{P}(A(u)) \leq \exp\left(-\frac{nt\delta_n}{2\sigma^2}\right).$$

On the complement $A(u)^c$, there are two cases.

If $\|\hat{\Delta}\|_n < u$, then trivially

$$\|\hat{\Delta}\|_n^2 < u^2 \leq 16u^2.$$

If $\|\hat{\Delta}\|_n \geq u$, then, because $\hat{\Delta} \in \mathcal{F}^*$, the event $A(u)^c$ implies

$$\left| \frac{\sigma}{n} \sum_{i=1}^n w_i \hat{\Delta}(x_i) \right| < 2u \|\hat{\Delta}\|_n.$$

Combining this with the basic inequality,

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{\sigma}{n} \sum_{i=1}^n w_i \hat{\Delta}(x_i) \leq \left| \frac{\sigma}{n} \sum_{i=1}^n w_i \hat{\Delta}(x_i) \right| < 2u \|\hat{\Delta}\|_n,$$

so $\|\hat{\Delta}\|_n < 4u$, and hence

$$\|\hat{\Delta}\|_n^2 \leq 16u^2 = 16t\delta_n.$$

Thus, on $A(u)^c$, the inequality

$$\|\hat{f} - f^*\|_n^2 \leq 16t\delta_n$$

always holds. This proves the theorem. $\square$

A convenient consequence is the following simpler high-probability form.

Corollary 12.9 (A simpler high-probability bound). *Under the assumptions of Theorem 12.8, for every $\eta \in (0, 1)$, with probability at least $1 - \eta$,*

$$\|\hat{f} - f^*\|_n^2 \leq 16\delta_n^2 + 32 \frac{\sigma^2 \log(1/\eta)}{n}.$$

Proof. Set

$$t = \max \left\{ \delta_n, \frac{2\sigma^2 \log(1/\eta)}{n\delta_n} \right\}.$$

Then $t \geq \delta_n$, and

$$\frac{nt\delta_n}{2\sigma^2} \geq \log(1/\eta).$$

Hence Theorem 12.8 yields, with probability at least $1 - \eta$,

$$\|\hat{f} - f^*\|_n^2 \leq 16t\delta_n.$$

Finally,

$$16t\delta_n \leq 16\delta_n^2 + 32 \frac{\sigma^2 \log(1/\eta)}{n},$$

which is the desired bound. $\square$

Remark 12.10. Thus, up to constants, the dominant term is the squared critical radius δ_n^2 . Unlike a uniform-convergence argument, which pays for the full complexity of the class $\mathcal{F}$, the localization theorem only pays for the complexity of those functions lying within distance δ_n of f^* .

12.4 Bounding the critical radius via metric entropy

The quantity $G_n(\delta; \mathcal{F}^*)$ is a Gaussian complexity, so it can be upper bounded using the chaining machinery developed earlier.

For $\delta > 0$, define the localized class

$$T_n(\delta) = \{g \in \mathcal{F}^* : \|g\|_n \leq \delta\}.$$

Proposition 12.11 (Dudley bound for the local Gaussian complexity). *There exists an absolute constant $C > 0$ such that*

$$G_n(\delta; \mathcal{F}^*) \leq \frac{C}{\sqrt{n}} \int_0^\delta \sqrt{\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, \varepsilon)} \, d\varepsilon.$$

Proof. For $g \in T_n(\delta)$, define

$$X_g = \frac{1}{n} \sum_{i=1}^n w_i g(x_i).$$

Then $(X_g)_{g \in T_n(\delta)}$ is a centered Gaussian process, and for any $g, h \in T_n(\delta)$,

$$\mathbb{E}(X_g - X_h)^2 = \frac{1}{n^2} \sum_{i=1}^n (g(x_i) - h(x_i))^2 = \frac{1}{n} \|g - h\|_n^2.$$

Since $0 \in T_n(\delta)$, we have

$$\sup_{g \in T_n(\delta)} |X_g| \leq \sup_{g \in T_n(\delta)} X_g + \sup_{g \in T_n(\delta)} (-X_g),$$

and the two suprema have the same expectation by symmetry. Therefore

$$G_n(\delta; \mathcal{F}^*) = \mathbb{E} \sup_{g \in T_n(\delta)} |X_g| \leq 2 \mathbb{E} \sup_{g \in T_n(\delta)} X_g.$$

Applying Dudley's entropy integral to the process X_g , whose canonical metric is

$$d(g, h) = \frac{1}{\sqrt{n}} \|g - h\|_n,$$

gives

$$\mathbb{E} \sup_{g \in T_n(\delta)} X_g \leq \frac{C'}{\sqrt{n}} \int_0^\delta \sqrt{\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, \varepsilon)} \, d\varepsilon.$$

Absorbing the factor 2 into the constant finishes the proof. $\square$

A direct consequence is a sufficient condition for validity.

Corollary 12.12 (A sufficient condition for a valid radius). *Any $\delta_n > 0$ satisfying*

$$\frac{C}{\sqrt{n} \delta_n} \int_0^{\delta_n} \sqrt{\log \mathcal{N}(T_n(\delta_n), \|\cdot\|_n, \varepsilon)} \, d\varepsilon \leq \frac{\delta_n}{2\sigma}$$

is a valid radius.

12.5 Examples

We now illustrate the theorem in two regimes.

Example 12.13 (Fixed-design nonparametric classes with polynomial entropy). Assume that $\mathcal{F}$ is star-shaped around $f^\star$, and that there exist constants $A > 0$ and $p \in (0, 2)$ such that for every $\varepsilon > 0$,

$$\log \mathcal{N}(\mathcal{F}, \|\cdot\|_n, \varepsilon) \leq A \varepsilon^{-p}.$$

Since translation preserves covering numbers,

$$\mathcal{N}(\mathcal{F}^\star, \|\cdot\|_n, \varepsilon) = \mathcal{N}(\mathcal{F}, \|\cdot\|_n, \varepsilon),$$

and since $T_n(\delta) \subseteq \mathcal{F}^\star$, we also have

$$\log \mathcal{N}(T_n(\delta), \|\cdot\|_n, \varepsilon) \leq A \varepsilon^{-p} \quad \text{for all } 0 < \varepsilon \leq \delta.$$

Applying Theorem 12.11,

$$\begin{aligned} G_n(\delta; \mathcal{F}^\star) &\leq \frac{C}{\sqrt{n}} \int_0^\delta \sqrt{A \varepsilon^{-p}} \, d\varepsilon \\ &\leq \frac{C_{A,p}}{\sqrt{n}} \delta^{1-p/2}, \end{aligned}$$

since $p < 2$.

Therefore every $\delta > 0$ satisfying

$$\frac{C_{A,p}}{\sqrt{n}} \delta^{-p/2} \leq \frac{\delta}{2\sigma}$$

is valid. Equivalently, there exists a valid radius δ_n such that

$$\delta_n \leq C_{A,p} \left(\frac{\sigma^2}{n} \right)^{1/(p+2)}.$$

Applying Theorem 12.9, we conclude that with probability at least $1 - \eta$,

$$\|\hat{f} - f^\star\|_n^2 \leq C_{A,p} \left[\left(\frac{\sigma^2}{n} \right)^{2/(p+2)} + \frac{\sigma^2 \log(1/\eta)}{n} \right].$$

12.6 Random-design least squares over convex classes

We now study least squares in the random-design setting. Unlike the fixed-design case, we do not assume that the regression model is correctly specified. Instead, we work with an arbitrary convex class of predictors and compare the empirical least-squares estimator to the best predictor in that class.

Let (X, Y) be a random pair with values in $\mathcal{X} \times \mathbf{R}$, and let

$$(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{i.i.d.}}{\sim} (X, Y).$$

Let $\mathcal{F}$ be a convex class of measurable functions $f : \mathcal{X} \rightarrow \mathbf{R}$, and define the population and empirical risks

$$R(f) = \mathbb{E}(Y - f(X))^2, \quad R_n(f) = \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2.$$

We assume throughout this section that

$$|Y| \leq m \quad \text{and} \quad \sup_{f \in \mathcal{F}} |f(X)| \leq m \quad \text{almost surely}$$

for some constant $m > 0$. We also assume that there exists a population minimizer

$$f^* \in \arg \min_{f \in \mathcal{F}} R(f),$$

and we let

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} R_n(f)$$

be any empirical least-squares estimator.

For functions $g : \mathcal{X} \rightarrow \mathbf{R}$, we write

$$Pg^2 = \mathbb{E}g(X)^2, \quad P_n g^2 = \frac{1}{n} \sum_{i=1}^n g(X_i)^2.$$

When a function depends on both X and Y , the symbols P and P_n are interpreted in the obvious way.

Remark 12.14. We do *not* assume that the model is correctly specified. In particular, we do not assume that

$$Y = f^*(X) + \xi$$

with ξ independent of X . The function f^* is only the best approximation to the regression rule inside the class $\mathcal{F}$.

Our goal is to bound the expected excess risk

$$\mathbb{E}[R(\hat{f}) - R(f^*)].$$

12.7 A deterministic convexity inequality

The first step is a simple but important improvement over the basic inequality $R_n(\hat{f}) \leq R_n(f^*)$. Convexity of the class yields an extra quadratic term.

Proposition 12.15 (Empirical curvature from convexity). *Assume that $\mathcal{F}$ is convex. Then for every $f \in \mathcal{F}$,*

$$R_n(f) - R_n(\hat{f}) \geq P_n(f - \hat{f})^2.$$

Proof. Fix $f \in \mathcal{F}$, and define the path

$$f_t = \hat{f} + t(f - \hat{f}), \quad t \in [0, 1].$$

Since $\mathcal{F}$ is convex, we have $f_t \in \mathcal{F}$ for all $t \in [0, 1]$. Because $\hat{f}$ minimizes R_n over $\mathcal{F}$, the function

$$\phi(t) = R_n(f_t)$$

has a minimum at $t = 0$, and therefore

$$\phi'(0) \geq 0.$$

A direct computation gives

$$\phi'(0) = \frac{2}{n} \sum_{i=1}^n (\hat{f}(X_i) - Y_i)(f(X_i) - \hat{f}(X_i)) = 2P_n[(\hat{f} - Y)(f - \hat{f})].$$

Hence

$$P_n[(\widehat{f} - Y)(f - \widehat{f})] \geq 0.$$

Now expand

$$\begin{aligned} R_n(f) - R_n(\widehat{f}) &= P_n\left((f - Y)^2 - (\widehat{f} - Y)^2\right) \\ &= P_n(f - \widehat{f})^2 + 2P_n[(\widehat{f} - Y)(f - \widehat{f})] \\ &\geq P_n(f - \widehat{f})^2, \end{aligned}$$

which proves the claim. $\square$

12.8 A basic excess-risk decomposition

Set

$$\xi = Y - f^*(X).$$

Since $|Y| \leq m$ and $|f^*(X)| \leq m$, we have

$$|\xi| \leq 2m \quad \text{almost surely.}$$

The next lemma rewrites the excess risk in a form suitable for symmetrization.

Lemma 12.16 (Basic decomposition). *With f^* and $\widehat{f}$ as above,*

$$R(\widehat{f}) - R(f^*) \leq 2(P_n - P)(\xi(\widehat{f} - f^*)) + P(\widehat{f} - f^*)^2 - 2P_n(\widehat{f} - f^*)^2.$$

Proof. By definition,

$$R(\widehat{f}) - R(f^*) = (R(\widehat{f}) - R_n(\widehat{f})) + (R_n(\widehat{f}) - R_n(f^*)) + (R_n(f^*) - R(f^*)).$$

Applying Theorem 12.15 with $f = f^*$, we obtain

$$R_n(\widehat{f}) - R_n(f^*) \leq -P_n(\widehat{f} - f^*)^2.$$

Hence

$$R(\widehat{f}) - R(f^*) \leq (P - P_n)\left((\widehat{f} - Y)^2 - (f^* - Y)^2\right) - P_n(\widehat{f} - f^*)^2.$$

Expanding the square difference,

$$(\widehat{f} - Y)^2 - (f^* - Y)^2 = (\widehat{f} - f^*)^2 + 2(\widehat{f} - f^*)(f^* - Y).$$

Since $f^* - Y = -\xi$, the right-hand side becomes

$$(\widehat{f} - f^*)^2 - 2\xi(\widehat{f} - f^*).$$

Therefore

$$R(\widehat{f}) - R(f^*) \leq (P - P_n)(\widehat{f} - f^*)^2 + 2(P_n - P)(\xi(\widehat{f} - f^*)) - P_n(\widehat{f} - f^*)^2.$$

Finally,

$$(P - P_n)(\widehat{f} - f^*)^2 - P_n(\widehat{f} - f^*)^2 = P(\widehat{f} - f^*)^2 - 2P_n(\widehat{f} - f^*)^2,$$

which yields the claim. $\square$

12.9 Contraction with deterministic offset

We will use the one-sided contraction principle proved earlier, but with a deterministic penalty carried along.

Lemma 12.17 (Contraction with deterministic offset). *Let $T \subseteq \mathbf{R}^n$, let $A : T \rightarrow \mathbf{R}$ be any deterministic function, and let*

$$\varphi_1, \dots, \varphi_n : \mathbf{R} \rightarrow \mathbf{R}$$

be L -Lipschitz functions satisfying $\varphi_i(0) = 0$. Then

$$\mathbb{E}_\varepsilon \sup_{t \in T} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i \varphi_i(t_i) - A(t) \right\} \leq L \mathbb{E}_\varepsilon \sup_{t \in T} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i t_i - \frac{1}{L} A(t) \right\}.$$

Proof. This is proved by exactly the same coordinate-by-coordinate argument as the one-sided contraction inequality from the previous section. The deterministic offset $A(t)$ is simply carried through unchanged. $\square$

12.10 A local offset bound from truncated Dudley

Let

$$\mathcal{G} = \mathcal{F} - f^* = \{f - f^* : f \in \mathcal{F}\}.$$

Since $\mathcal{F}$ is convex and $f^* \in \mathcal{F}$, the class $\mathcal{G}$ is convex and contains 0. Moreover,

$$\sup_{g \in \mathcal{G}} |g(X)| \leq 2m \quad \text{almost surely.}$$

The stochastic terms below involve the offset Rademacher process

$$\sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i) - c P_n g^2 \right\}.$$

The next lemma bounds this process by combining a coarse γ -net argument with the previously established truncated Dudley bound for localized Rademacher averages.

Lemma 12.18 (Offset entropy bound). *Let $\mathcal{G}$ be a class of measurable functions such that $0 \in \mathcal{G}$ and*

$$\sup_{g \in \mathcal{G}} |g(X)| \leq 2m \quad \text{almost surely.}$$

Then for every $c > 0$ and every $0 < \alpha \leq \gamma \leq 2m$,

$$\mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i) - c P_n g^2 \right\} \leq C \left(\alpha + \frac{1}{\sqrt{n}} \int_{\alpha/4}^{\gamma} \sqrt{\log(1 + \mathcal{N}(\mathcal{G}, L_2(P_n), u))} \, du + \frac{\log(1 + \mathcal{N}(\mathcal{G}, L_2(P_n), \gamma))}{cn} \right),$$

where $C > 0$ is an absolute constant.

Proof. Fix the sample $X_1, \dots, X_n$, and write

$$R_n^\varepsilon(g) = \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i).$$

Choose a γ -net $\mathcal{G}_0 \subseteq \mathcal{G}$ in $L_2(P_n)$, and add the zero function if necessary. Then

$$|\mathcal{G}_0| \leq 1 + \mathcal{N}(\mathcal{G}, L_2(P_n), \gamma).$$

For every $g \in \mathcal{G}$, define

$$\pi(g) = \begin{cases} 0, & \text{if } \|g\|_{L_2(P_n)} \leq \gamma, \\ \text{a nearest point to } g \text{ in } \mathcal{G}_0, & \text{if } \|g\|_{L_2(P_n)} > \gamma. \end{cases}$$

Then

$$\|g - \pi(g)\|_{L_2(P_n)} \leq \gamma \quad \text{for every } g \in \mathcal{G}.$$

Moreover, if $\|g\|_{L_2(P_n)} > \gamma$, then

$$\|\pi(g)\|_{L_2(P_n)} \leq \|g\|_{L_2(P_n)} + \|g - \pi(g)\|_{L_2(P_n)} \leq 2\|g\|_{L_2(P_n)}.$$

Hence, for every $g \in \mathcal{G}$,

$$cP_n g^2 \geq \frac{c}{4} P_n \pi(g)^2.$$

Set

$$\mathcal{H} = \{g - \pi(g) : g \in \mathcal{G}\}.$$

Then

$$\sup_{g \in \mathcal{G}} \{R_n^\varepsilon(g) - cP_n g^2\} \leq \sup_{h \in \mathcal{H}} R_n^\varepsilon(h) + \sup_{u \in \mathcal{G}_0} \left\{ R_n^\varepsilon(u) - \frac{c}{4} P_n u^2 \right\}.$$

We bound the two terms separately.

The localized term. Every $h \in \mathcal{H}$ satisfies

$$\|h\|_{L_2(P_n)} \leq \gamma,$$

and

$$|h(X_i)| \leq 4m \quad \text{for every } i = 1, \dots, n \text{ and every } h \in \mathcal{H}.$$

Indeed, the $L_2(P_n)$ bound follows from the construction of $\pi(g)$, and the pointwise bound follows from

$$|h(X_i)| \leq |g(X_i)| + |\pi(g)(X_i)| \leq 2m + 2m = 4m.$$

Define the rescaled class

$$\tilde{\mathcal{H}} = \left\{ \frac{h + 4m}{8m} : h \in \mathcal{H} \right\}.$$

Then every function in $\tilde{\mathcal{H}}$ takes values in $[0, 1]$ on the sample $X_1, \dots, X_n$. Since adding a constant does not change a Rademacher average,

$$\mathbb{E}_\varepsilon \sup_{h \in \mathcal{H}} R_n^\varepsilon(h) = 8m \mathbb{E}_\varepsilon \sup_{\tilde{h} \in \tilde{\mathcal{H}}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \tilde{h}(X_i).$$

Applying Theorem 10.1 to $\tilde{\mathcal{H}}$ with

$$\eta_0 = \frac{\alpha}{4m},$$

we obtain

$$\mathbb{E}_\varepsilon \sup_{h \in \mathcal{H}} R_n^\varepsilon(h) \leq 2\alpha + \frac{C}{\sqrt{n}} \int_{\alpha/2}^{8m} \sqrt{\log \mathcal{N}(\mathcal{H}, L_2(P_n), u)} du.$$

Since $\|h\|_{L_2(P_n)} \leq \gamma$ for all $h \in \mathcal{H}$, the covering number is 1 for all $u \geq \gamma$, and so

$$\mathbb{E}_\varepsilon \sup_{h \in \mathcal{H}} R_n^\varepsilon(h) \leq 2\alpha + \frac{C}{\sqrt{n}} \int_{\alpha/2}^\gamma \sqrt{\log \mathcal{N}(\mathcal{H}, L_2(P_n), u)} du.$$

It remains to compare the entropy of $\mathcal{H}$ to that of $\mathcal{G}$. Write

$$\mathcal{H} = \bigcup_{v \in \mathcal{G}_0} \mathcal{H}_v, \quad \mathcal{H}_v = \{g - v : g \in \mathcal{G}, \pi(g) = v\}.$$

Fix $u > 0$. Take any proper $u/2$ -cover of $\mathcal{G}$ in $L_2(P_n)$. For each $v \in \mathcal{G}_0$, and for each $u/2$ -ball in this cover that intersects the subset

$$\{g \in \mathcal{G} : \pi(g) = v\},$$

choose one representative from the intersection. Translating these representatives by $-v$ gives a proper u -cover of $\mathcal{H}_v$. Hence

$$\mathcal{N}(\mathcal{H}_v, L_2(P_n), u) \leq \mathcal{N}(\mathcal{G}, L_2(P_n), u/2).$$

Therefore

$$\mathcal{N}(\mathcal{H}, L_2(P_n), u) \leq |\mathcal{G}_0| \mathcal{N}(\mathcal{G}, L_2(P_n), u/2) \leq (1 + \mathcal{N}(\mathcal{G}, L_2(P_n), \gamma)) \mathcal{N}(\mathcal{G}, L_2(P_n), u/2).$$

If $u \leq \gamma$, then $u/2 \leq \gamma$, and monotonicity of covering numbers gives

$$1 + \mathcal{N}(\mathcal{G}, L_2(P_n), \gamma) \leq 1 + \mathcal{N}(\mathcal{G}, L_2(P_n), u/2).$$

Thus, for $0 < u \leq \gamma$,

$$\log \mathcal{N}(\mathcal{H}, L_2(P_n), u) \leq 2 \log(1 + \mathcal{N}(\mathcal{G}, L_2(P_n), u/2)).$$

Substituting this into the previous bound and changing variables $v = u/2$, we obtain

$$\mathbb{E}_\varepsilon \sup_{h \in \mathcal{H}} R_n^\varepsilon(h) \leq C\alpha + \frac{C}{\sqrt{n}} \int_{\alpha/4}^\gamma \sqrt{\log(1 + \mathcal{N}(\mathcal{G}, L_2(P_n), v))} dv.$$

The coarse finite-net term. For any $\lambda > 0$, the exponential-max bound yields

$$\begin{aligned} \mathbb{E}_\varepsilon \sup_{u \in \mathcal{G}_0} \left\{ R_n^\varepsilon(u) - \frac{c}{4} P_n u^2 \right\} \\ \leq \frac{1}{\lambda} \log \sum_{u \in \mathcal{G}_0} \mathbb{E}_\varepsilon \exp \left(\lambda R_n^\varepsilon(u) - \frac{\lambda c}{4} P_n u^2 \right). \end{aligned}$$

Since

$$R_n^\varepsilon(u) = \frac{1}{n} \sum_{i=1}^n \varepsilon_i u(X_i),$$

the moment generating function of a Rademacher sum gives

$$\mathbb{E}_\varepsilon e^{\lambda R_n^\varepsilon(u)} \leq \exp \left(\frac{\lambda^2}{2n} P_n u^2 \right).$$

Thus

$$\mathbb{E}_\varepsilon \exp \left(\lambda R_n^\varepsilon(u) - \frac{\lambda c}{4} P_n u^2 \right) \leq \exp \left(\left(\frac{\lambda^2}{2n} - \frac{\lambda c}{4} \right) P_n u^2 \right).$$

Choose

$$\lambda = \frac{cn}{2}.$$

Then

$$\frac{\lambda^2}{2n} - \frac{\lambda c}{4} = 0,$$

so each exponential term is at most 1. Therefore

$$\mathbb{E}_\varepsilon \sup_{u \in \mathcal{G}_0} \left\{ R_n^\varepsilon(u) - \frac{c}{4} P_n u^2 \right\} \leq \frac{2}{cn} \log |\mathcal{G}_0| \leq \frac{2}{cn} \log(1 + \mathcal{N}(\mathcal{G}, L_2(P_n), \gamma)).$$

Combining the two steps proves the lemma. $\square$

12.11 Expected excess risk bound

We now prove the main result of the section.

Theorem 12.19 (Expected excess risk over a convex class). *Assume that $\mathcal{F}$ is convex, that $f^\star \in \arg \min_{f \in \mathcal{F}} R(f)$, and that*

$$|Y| \leq m, \quad \sup_{f \in \mathcal{F}} |f(X)| \leq m \quad \text{almost surely.}$$

Then for every $0 < \alpha \leq \gamma \leq 2m$,

$$\mathbb{E}[R(\hat{f}) - R(f^\star)] \leq Cm \left[\alpha + \frac{1}{\sqrt{n}} \mathbb{E} \int_{\alpha/4}^{\gamma} \sqrt{\log(1 + \mathcal{N}(\mathcal{F}, L_2(P_n), u))} \, du + \frac{m}{n} \mathbb{E} \log(1 + \mathcal{N}(\mathcal{F}, L_2(P_n), \gamma)) \right],$$

where $C > 0$ is an absolute constant.

Proof. Set

$$\mathcal{G} = \mathcal{F} - f^\star.$$

Then $0 \in \mathcal{G}$, $\hat{f} - f^\star \in \mathcal{G}$, and

$$\sup_{g \in \mathcal{G}} |g(X)| = \sup_{f \in \mathcal{F}} |f(X) - f^\star(X)| \leq 2m \quad \text{almost surely.}$$

By Theorem 12.16, with $g = \hat{f} - f^\star$, we have

$$R(\hat{f}) - R(f^\star) \leq 2(P_n - P)(\xi g) + P g^2 - 2P_n g^2.$$

Therefore

$$\begin{aligned} \mathbb{E}[R(\hat{f}) - R(f^\star)] &\leq \underbrace{\mathbb{E} \sup_{g \in \mathcal{G}} \left\{ 2(P_n - P)(\xi g) - \frac{1}{4} P g^2 - \frac{1}{4} P_n g^2 \right\}}_{\text{(I)}} \\ &\quad + \underbrace{\mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{5}{4} P g^2 - \frac{7}{4} P_n g^2 \right\}}_{\text{(II)}}. \end{aligned}$$

We first bound (I). Let $(X'_1, Y'_1), \dots, (X'_n, Y'_n)$ be an independent copy of the sample, and define

$$P'_n h = \frac{1}{n} \sum_{i=1}^n h(X'_i, Y'_i)$$

for any measurable function h of (X, Y) . Also write

$$\xi'_i = Y'_i - f^*(X'_i).$$

Since $P(\xi g) = \mathbb{E}' P'_n(\xi' g)$ and $Pg^2 = \mathbb{E}' P'_n g^2$, Jensen's inequality gives

$$(I) \leq \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ 2(P_n - P'_n)(\xi g) - \frac{1}{4} P'_n g^2 - \frac{1}{4} P_n g^2 \right\}.$$

Now symmetrize:

$$\begin{aligned} (I) &\leq \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{2}{n} \sum_{i=1}^n \varepsilon_i (\xi_i g(X_i) - \xi'_i g(X'_i)) - \frac{1}{4} P'_n g^2 - \frac{1}{4} P_n g^2 \right\} \\ &\leq 2 \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{2}{n} \sum_{i=1}^n \varepsilon_i \xi_i g(X_i) - \frac{1}{4} P_n g^2 \right\}, \end{aligned}$$

where we used $\sup(a + b) \leq \sup a + \sup b$, sign symmetry of the Rademachers, and symmetry between the original and ghost samples.

Condition on $(X_i, Y_i)_{i=1}^n$. For each i , define

$$\varphi_i(u) = \frac{\xi_i}{2m} u.$$

Since $|\xi_i| \leq 2m$, each φ_i is 1-Lipschitz and satisfies $\varphi_i(0) = 0$. Applying Theorem 12.17,

$$\begin{aligned} \mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left\{ \frac{2}{n} \sum_{i=1}^n \varepsilon_i \xi_i g(X_i) - \frac{1}{4} P_n g^2 \right\} &= 4m \mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i \varphi_i(g(X_i)) - \frac{1}{16m} P_n g^2 \right\} \\ &\leq 4m \mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i) - \frac{1}{16m} P_n g^2 \right\}. \end{aligned}$$

Therefore

$$(I) \leq 8m \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i) - \frac{1}{16m} P_n g^2 \right\}.$$

We next bound (II). Since $Pg^2 = \mathbb{E}' P'_n g^2$, Jensen's inequality gives

$$(II) \leq \frac{5}{4} \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ P'_n g^2 - \frac{7}{5} P_n g^2 \right\}.$$

Now use the identity

$$a - \frac{7}{5}b = \frac{6}{5}(a - b) - \frac{1}{5}a - \frac{1}{5}b.$$

Applying it with $a = P'_n g^2$ and $b = P_n g^2$, and then symmetrizing, we obtain

$$\begin{aligned}
(\text{II}) &\leq \frac{5}{4} \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{6}{5} (P'_n - P_n) g^2 - \frac{1}{5} P'_n g^2 - \frac{1}{5} P_n g^2 \right\} \\
&\leq \frac{5}{4} \cdot 2 \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{6}{5} \cdot \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i)^2 - \frac{1}{5} P_n g^2 \right\} \\
&= 3 \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i)^2 - \frac{1}{6} P_n g^2 \right\}.
\end{aligned}$$

Condition again on $(X_i, Y_i)_{i=1}^n$. Since $|g(X_i)| \leq 2m$, the function

$$\psi(u) = \frac{u^2}{4m}, \quad u \in [-2m, 2m],$$

is 1-Lipschitz and satisfies $\psi(0) = 0$. Choose any global 1-Lipschitz extension $\tilde{\psi} : \mathbf{R} \rightarrow \mathbf{R}$ with $\tilde{\psi}(0) = 0$. Since all values $g(X_i)$ lie in $[-2m, 2m]$, we have $\psi(g(X_i)) = \tilde{\psi}(g(X_i))$. Applying Theorem 12.17,

$$\begin{aligned}
\mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i)^2 - \frac{1}{6} P_n g^2 \right\} &= 4m \mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i \tilde{\psi}(g(X_i)) - \frac{1}{24m} P_n g^2 \right\} \\
&\leq 4m \mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i) - \frac{1}{24m} P_n g^2 \right\}.
\end{aligned}$$

Hence

$$(\text{II}) \leq 12m \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i) - \frac{1}{24m} P_n g^2 \right\}.$$

Combining the bounds for (I) and (II), and using

$$\frac{1}{16m} \geq \frac{1}{24m},$$

we obtain

$$\mathbb{E}[R(\hat{f}) - R(f^*)] \leq 20m \mathbb{E} \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(X_i) - \frac{1}{24m} P_n g^2 \right\}.$$

Now apply Theorem 12.18 to the class $\mathcal{G}$ with

$$c = \frac{1}{24m}.$$

Since $\mathcal{G} = \mathcal{F} - f^*$, translation invariance of $L_2(P_n)$ gives

$$\mathcal{N}(\mathcal{G}, L_2(P_n), u) = \mathcal{N}(\mathcal{F}, L_2(P_n), u) \quad \text{for every } u > 0.$$

Therefore

$$\mathbb{E}[R(\hat{f}) - R(f^*)] \leq Cm \left[\alpha + \frac{1}{\sqrt{n}} \mathbb{E} \int_{\alpha/4}^{\gamma} \sqrt{\log(1 + \mathcal{N}(\mathcal{F}, L_2(P_n), u))} du + \frac{m}{n} \mathbb{E} \log(1 + \mathcal{N}(\mathcal{F}, L_2(P_n), \gamma)) \right],$$

which is the desired bound. $\square$

13 Minimax lower bounds

We now begin the lower-bound part of the course. Up to this point, much of our work has had the following form: we propose an estimator and prove that its risk is small. A minimax lower bound asks for the complementary statement. It asks whether any estimator can do substantially better.

When an upper bound and a lower bound match, at least up to numerical constants, we have identified the right statistical rate. The guiding principle in this section is simple: an estimator that is very accurate should also be able to distinguish between well-separated parameters. Thus estimation lower bounds can often be proved by reducing estimation to testing.

13.1 Fano's method

Let $\{\mathbb{P}_\theta : \theta \in \Theta\}$ be a statistical model, where $\mathbb{P}_\theta$ denotes the law of the full observation **Obs**. Let d be a pseudometric on Θ , and let $q > 0$. The minimax risk is

$$\mathcal{R}_n(\Theta, d, q) = \inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_\theta d^q(\hat{\theta}(\text{Obs}), \theta),$$

where the infimum is over all measurable estimators.

A pseudometric is enough for our purposes. It may happen that two distinct parameters have distance zero; then they simply cannot be separated by this particular loss.

The main input is Fano's inequality. We state it in a form that allows an arbitrary reference measure. This flexibility will be useful later in the Yang–Barron argument.

Theorem 13.1 (Fano's inequality). *Let $\mathbb{P}_1, \dots, \mathbb{P}_N$ be probability measures on a measurable space Ψ , with $N \geq 2$. Assume that all $\mathbb{P}_j$ are absolutely continuous with respect to a probability measure $\mathbb{Q}$. Then*

$$\inf_{\hat{J}: \Psi \rightarrow [N]} \frac{1}{N} \sum_{j=1}^N \mathbb{P}_j(\hat{J}(\text{Obs}) \neq j) \geq 1 - \frac{1 + \frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_j \| \mathbb{Q})}{\log N}.$$

Proof. For a decision rule $\hat{J}: \Psi \rightarrow [N]$, write the average probability of correct selection as

$$\frac{1}{N} \sum_{j=1}^N \mathbb{P}_j(\hat{J}(\text{Obs}) = j).$$

Denote $L_j = \frac{d\mathbb{P}_j}{d\mathbb{Q}}$. Then

$$\begin{aligned} \sum_{j=1}^N \mathbb{P}_j(\hat{J}(\text{Obs}) = j) &= \int_{\Psi} \sum_{j=1}^N \mathbf{1}\{\hat{J}(\omega) = j\} L_j(\omega) d\mathbb{Q}(\omega) \\ &\leq \int_{\Psi} \max_{1 \leq j \leq N} L_j(\omega) d\mathbb{Q}(\omega). \end{aligned}$$

Thus the best possible average success probability is bounded above by

$$\frac{1}{N} \mathbb{E}_{\mathbb{Q}} \max_{1 \leq j \leq N} L_j.$$

Set

$$M = \mathbb{E}_{\mathbb{Q}} \max_{1 \leq j \leq N} L_j.$$

Now consider

$$\varphi(x) = x \log x - x + 1, \quad x \geq 0.$$

This function is convex and nonnegative. By Jensen's inequality and the pointwise inequality $\varphi(\max_j a_j) \leq \sum_j \varphi(a_j)$ for nonnegative a_j ,

$$\begin{aligned} \varphi(M) &\leq \mathbb{E}_{\mathbb{Q}} \varphi \left(\max_{1 \leq j \leq N} L_j \right) \leq \sum_{j=1}^N \mathbb{E}_{\mathbb{Q}} \varphi(L_j) \\ &= \sum_{j=1}^N \text{KL}(\mathbb{P}_j \| \mathbb{Q}). \end{aligned}$$

On the other hand, it holds that

$$\varphi(M) = M \log M - M + 1 \geq M \log N - N.$$

Indeed, this is equivalent to

$$M \log(M/N) + N - M + 1 \geq 0,$$

which follows from the scalar inequality $u \log u + 1 - u \geq 0$ with $u = M/N$. Therefore

$$M \log N - N \leq \sum_{j=1}^N \text{KL}(\mathbb{P}_j \| \mathbb{Q}).$$

Dividing by $N \log N$, we find that the average probability of correct selection is at most

$$\frac{1 + \frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_j \| \mathbb{Q})}{\log N}.$$

Taking one minus this upper bound gives the claim. $\square$

13.2 From testing to estimation

We now turn Fano's testing statement into a lower bound for the minimax risk. The argument is short, but it is one of the most useful reductions in the course.

Proposition 13.2 (Fano lower bound for minimax risk). *Let $\theta_1, \dots, \theta_N \in \Theta$, with $N \geq 2$, and set*

$$\Delta = \min_{i \neq j} d(\theta_i, \theta_j).$$

Assume $\Delta > 0$, and let $\mathbb{Q}$ be a probability measure such that each $\mathbb{P}_{\theta_j}$ is absolutely continuous with respect to $\mathbb{Q}$. Then, for every $q > 0$,

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^q(\hat{\theta}(\text{Obs}), \theta) \geq \frac{\Delta^q}{2^q} \left(1 - \frac{1 + \frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_{\theta_j} \| \mathbb{Q})}{\log N} \right).$$

Proof. Fix an arbitrary estimator $\hat{\theta}$. From it, build a nearest-neighbor selector

$$\hat{J}(\text{Obs}) \in \arg \min_{1 \leq j \leq N} d(\hat{\theta}(\text{Obs}), \theta_j),$$

with ties broken in any measurable way.

If the true parameter is θ_j and $\hat{J} \neq j$, then the triangle inequality gives

$$\Delta \leq d(\theta_j, \theta_{\hat{J}}) \leq d(\theta_j, \hat{\theta}) + d(\hat{\theta}, \theta_{\hat{J}}) \leq 2d(\theta_j, \hat{\theta}),$$

because $\theta_{\hat{J}}$ is at least as close to $\hat{\theta}$ as θ_j is. Hence

$$d^q(\hat{\theta}, \theta_j) \geq \left(\frac{\Delta}{2}\right)^q \mathbf{1}\{\hat{J} \neq j\}.$$

Taking expectations under $\mathbb{P}_{\theta_j}$, averaging over j , and applying Theorem 13.1, we get

$$\begin{aligned} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^q(\hat{\theta}, \theta) &\geq \frac{1}{N} \sum_{j=1}^N \mathbb{E}_{\theta_j} d^q(\hat{\theta}, \theta_j) \\ &\geq \left(\frac{\Delta}{2}\right)^q \frac{1}{N} \sum_{j=1}^N \mathbb{P}_{\theta_j}(\hat{J} \neq j) \\ &\geq \left(\frac{\Delta}{2}\right)^q \left(1 - \frac{1 + \frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_{\theta_j} \parallel \mathbb{Q})}{\log N}\right). \end{aligned}$$

Since $\hat{\theta}$ was arbitrary, taking the infimum over estimators proves the result. $\square$

The proposition gives a useful recipe. We need a finite subset of parameters that is:

- (i) well separated in the loss metric;
- (ii) large enough, so that $\log N$ is large;
- (iii) statistically close enough, so that the KL term is not too large.

The lower bound is obtained by balancing these three requirements.

13.3 A first lower-bound example: Gaussian mean estimation

We now return to the basic Gaussian mean problem. Suppose

$$X_1, \dots, X_n \sim N(\theta, \sigma^2 I_d)$$

independently, where $\theta \in \mathbb{R}^d$ is unknown and σ^2 is known. Let $\mathbb{P}_{\theta}$ denote the joint law of the sample. We will show that

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_{\theta} \|\hat{\theta} - \theta\|_2^2 \geq c \frac{d\sigma^2}{n}$$

for a universal constant $c > 0$.

Choose a radius $R > 0$, and take an ε -packing

$$\theta_1, \dots, \theta_N \in B_2^d(R).$$

A standard volume argument gives, for $0 < \varepsilon < R$,

$$N \geq \left(\frac{R}{\varepsilon}\right)^d.$$

The separation is $\Delta \geq \varepsilon$.

Apply Proposition 13.2 with $q = 2$, with $d(\theta, \theta') = \|\theta - \theta'\|_2$, and with reference measure $\mathbb{Q} = \mathbb{P}_0$. For Gaussian shifts,

$$\text{KL}(\mathbb{P}_\theta \| \mathbb{P}_0) = \frac{n\|\theta\|_2^2}{2\sigma^2}.$$

Since $\theta_j \in B_2^d(R)$,

$$\frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_{\theta_j} \| \mathbb{P}_0) \leq \frac{nR^2}{2\sigma^2}.$$

Therefore

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_2^2 \geq \frac{\varepsilon^2}{4} \left(1 - \frac{1 + \frac{nR^2}{2\sigma^2}}{d \log(R/\varepsilon)} \right). \quad (25)$$

Now set

$$R = b\sigma\sqrt{\frac{d}{n}}, \quad \varepsilon = a\sigma\sqrt{\frac{d}{n}},$$

where $0 < a < b$ are absolute constants. Then (25) becomes

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_2^2 \geq \frac{a^2}{4} \frac{d\sigma^2}{n} \left(1 - \frac{1 + b^2 d/2}{d \log(b/a)} \right).$$

For instance, one may take $b = 1$ and then choose $a > 0$ sufficiently small so that the term in parentheses is bounded below by a positive numerical constant, uniformly over $d \geq 1$. Hence

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_2^2 \geq c \frac{d\sigma^2}{n}.$$

On the other hand, the sample mean satisfies

$$\sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \|\bar{X} - \theta\|_2^2 = \frac{d\sigma^2}{n}.$$

Thus the sample mean is minimax-rate optimal.

Remark 13.3. Fano's method is not meant to give sharp numerical constants in this example. It is designed to expose the correct rate. A Bayes-risk argument below gives a cleaner route to the exact constant for Gaussian mean estimation.

13.4 The Yang–Barron entropy method

Fano's method asks us to build many separated parameters whose distributions are still difficult to distinguish. The Yang–Barron idea is to make the reference measure in Fano depend on a covering of the parameter space. This turns metric entropy directly into a minimax lower bound.

Let (Θ, d) be a pseudometric space, and let $\mathbb{P}_\theta$ denote the law of the full observation under parameter θ . For $\varepsilon > 0$, write

$$\mathcal{N}(\varepsilon) = \mathcal{N}(\Theta, d, \varepsilon), \quad \mathcal{P}(\varepsilon) = \mathcal{P}(\Theta, d, \varepsilon)$$

for the covering and packing numbers.

We assume two conditions.

- (i) **KL is controlled by the metric:** for all $\theta, \theta' \in \Theta$,

$$\text{KL}(\mathbb{P}_\theta \| \mathbb{P}_{\theta'}) \leq nd^2(\theta, \theta').$$

- (ii) **Polynomial metric entropy:** there are constants $\alpha > 0$, $c_1, c_2 > 0$, and $\varepsilon_0 > 0$ such that, for all $0 < \varepsilon \leq \varepsilon_0$,

$$\log \mathcal{P}(\varepsilon) \geq c_1 \varepsilon^{-\alpha}, \quad \log \mathcal{N}(\varepsilon) \leq c_2 \varepsilon^{-\alpha}.$$

The first assumption says that nearby parameters are statistically close. The second says that the parameter space has entropy exponent α . The corresponding lower bound on the squared-risk rate will be

$$n^{-2/(2+\alpha)}.$$

13.4.1 A covering-mixture KL bound

Lemma 13.4 (Yang–Barron covering mixture). *Let $\bar{\theta}_1, \dots, \bar{\theta}_M$ be an ε -cover of Θ , with $M = \mathcal{N}(\varepsilon)$. Define the mixture*

$$\mathbb{Q}_\varepsilon = \frac{1}{M} \sum_{\ell=1}^M \mathbb{P}_{\bar{\theta}_\ell}.$$

Under the KL-metric condition above, for every $\theta \in \Theta$,

$$\text{KL}(\mathbb{P}_\theta \| \mathbb{Q}_\varepsilon) \leq n\varepsilon^2 + \log \mathcal{N}(\varepsilon).$$

Proof. Fix $\theta \in \Theta$. Since the points $\bar{\theta}_1, \dots, \bar{\theta}_M$ form an ε -cover, choose $\bar{\theta}_{\ell(\theta)}$ such that

$$d(\theta, \bar{\theta}_{\ell(\theta)}) \leq \varepsilon.$$

As measures,

$$\mathbb{Q}_\varepsilon = \frac{1}{M} \sum_{\ell=1}^M \mathbb{P}_{\bar{\theta}_\ell} \geq \frac{1}{M} \mathbb{P}_{\bar{\theta}_{\ell(\theta)}}.$$

Therefore,

$$\begin{aligned} \text{KL}(\mathbb{P}_\theta \| \mathbb{Q}_\varepsilon) &= \int \log \left(\frac{d\mathbb{P}_\theta}{d\mathbb{Q}_\varepsilon} \right) d\mathbb{P}_\theta \\ &\leq \log M + \int \log \left(\frac{d\mathbb{P}_\theta}{d\mathbb{P}_{\bar{\theta}_{\ell(\theta)}}} \right) d\mathbb{P}_\theta \\ &= \log \mathcal{N}(\varepsilon) + \text{KL}(\mathbb{P}_\theta \| \mathbb{P}_{\bar{\theta}_{\ell(\theta)}}) \\ &\leq \log \mathcal{N}(\varepsilon) + nd^2(\theta, \bar{\theta}_{\ell(\theta)}) \\ &\leq \log \mathcal{N}(\varepsilon) + n\varepsilon^2. \end{aligned}$$

□

Remark 13.5. This is the central Yang–Barron trick. The mixture $\mathbb{Q}_\varepsilon$ is close in KL to every member of the model because every parameter is close to at least one element of the cover. The price for mixing over the cover is exactly $\log \mathcal{N}(\varepsilon)$.

13.4.2 Instantiated minimax lower bound

We now combine the covering-mixture bound with Fano’s inequality.

Theorem 13.6 (Yang–Barron minimax lower rate). *Assume the KL-metric condition and the polynomial entropy condition above. Then there is a constant $c > 0$, depending only on α, c_1, c_2 , such that for all sufficiently large n ,*

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^2(\hat{\theta}(\text{Obs}), \theta) \geq c n^{-2/(2+\alpha)}.$$

Proof. The proof is a balancing argument. Fix a small number $a \in (0, 1)$, to be chosen only in terms of the entropy constants. Let $\varepsilon > 0$, and choose an $a\varepsilon$ -packing

$$\theta_1, \dots, \theta_N \in \Theta.$$

Then

$$\log N \geq \log \mathcal{P}(a\varepsilon) \geq c_1 a^{-\alpha} \varepsilon^{-\alpha},$$

and the separation of the packing is at least $a\varepsilon$.

Apply Proposition 13.2 to this packing with $q = 2$, using the reference measure $\mathbb{Q}_{\varepsilon}$ from Lemma 13.4. The average KL term is bounded by the worst KL term, so

$$\frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_{\theta_j} \| \mathbb{Q}_{\varepsilon}) \leq n\varepsilon^2 + \log \mathcal{N}(\varepsilon) \leq n\varepsilon^2 + c_2 \varepsilon^{-\alpha}.$$

Therefore

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^2(\hat{\theta}, \theta) \geq \frac{a^2 \varepsilon^2}{4} \left(1 - \frac{1 + n\varepsilon^2 + c_2 \varepsilon^{-\alpha}}{c_1 a^{-\alpha} \varepsilon^{-\alpha}} \right).$$

Choose a small enough that

$$c_1 a^{-\alpha} \geq 8c_2.$$

Then choose $A > 0$ small enough that

$$A^{2+\alpha} \leq \frac{c_1 a^{-\alpha}}{8}.$$

Set

$$\varepsilon_n = A n^{-1/(2+\alpha)}.$$

Then

$$n\varepsilon_n^2 = A^{2+\alpha} \varepsilon_n^{-\alpha} \leq \frac{c_1 a^{-\alpha}}{8} \varepsilon_n^{-\alpha}.$$

For all sufficiently large n , the remaining constant 1 is also bounded by

$$1 \leq \frac{c_1 a^{-\alpha}}{8} \varepsilon_n^{-\alpha}.$$

Hence

$$\frac{1 + n\varepsilon_n^2 + c_2 \varepsilon_n^{-\alpha}}{c_1 a^{-\alpha} \varepsilon_n^{-\alpha}} \leq \frac{1}{2}.$$

Substituting $\varepsilon = \varepsilon_n$ into the Fano bound gives

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} d^2(\hat{\theta}, \theta) \geq \frac{a^2}{8} \varepsilon_n^2 = \frac{a^2 A^2}{8} n^{-2/(2+\alpha)}.$$

This proves the theorem. □

Remark 13.7. Both entropy estimates are used. The packing lower bound creates many separated hypotheses. The covering upper bound keeps the KL divergence to the mixture under control. The rate comes from balancing

$$n\varepsilon^2 \quad \text{and} \quad \varepsilon^{-\alpha},$$

which gives $\varepsilon \asymp n^{-1/(2+\alpha)}$, and therefore squared risk $\varepsilon^2 \asymp n^{-2/(2+\alpha)}$.

13.5 Birgé's lemma as a finite-testing tool

The Yang–Barron method is well suited to entropy lower bounds. For finite constructions, especially when pairwise KL divergences are easy to compute, it is often more convenient to use Birgé's lemma. We state it without proof.

Lemma 13.8 (Birgé's lemma). *Let $\mathbb{P}_1, \dots, \mathbb{P}_N$ be probability measures, with $N \geq 2$, and let $A_1, \dots, A_N$ be disjoint events. Then*

$$\min_{1 \leq i \leq N} \mathbb{P}_i(A_i) \leq \max \left\{ \frac{2e}{2e+1}, \frac{\max_{i \neq j} \text{KL}(\mathbb{P}_i \| \mathbb{P}_j)}{\log N} \right\}.$$

Here is the testing consequence we will use. Given a selector $\hat{J}$, take

$$A_i = \{\hat{J} = i\}.$$

These events are disjoint. Therefore, if

$$\max_{i \neq j} \text{KL}(\mathbb{P}_i \| \mathbb{P}_j) \leq \frac{2e}{2e+1} \log N,$$

then, for every selector, at least one hypothesis has success probability at most $2e/(2e+1)$. Equivalently,

$$\inf_{\hat{J}} \max_{1 \leq i \leq N} \mathbb{P}_i(\hat{J} \neq i) \geq \frac{1}{2e+1}.$$

Remark 13.9 (Why use Birgé instead of Fano?). Birgé's lemma has two practical advantages in finite constructions. First, it avoids choosing an auxiliary reference measure $\mathbb{Q}$. Second, it uses the maximum pairwise KL divergence, which is often exactly what is easiest to compute for Gaussian shifts or two-point constructions. The tradeoff is that it is less flexible than the mixture version of Fano in entropy arguments, where the Yang–Barron choice of $\mathbb{Q}$ is the natural object.

Remark 13.10 (Where it is needed). The next natural use of Birgé's lemma is in sparse regression lower bounds. One chooses many sparse vectors that are well separated in Hamming distance, maps them through the fixed design, and checks that the induced Gaussian laws have small pairwise KL divergence. We do not develop the sparse/RIP argument here, since it needs a separate combinatorial construction and a discussion of restricted isometry.

13.6 Other lower-bound methods

We finish with two lower-bound tools that are useful in different situations. They are less geometric than Fano, but often more direct.

13.6.1 Bayes risk lower bounds

The minimax risk is always at least the Bayes risk for any prior. Indeed, for any prior π on Θ ,

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_{\theta} L(\hat{\theta}, \theta) \geq \inf_{\hat{\theta}} \int_{\Theta} \mathbb{E}_{\theta} L(\hat{\theta}, \theta) \pi(d\theta).$$

This is simply the fact that a supremum is at least an average. The usefulness of the bound is that the Bayes problem can sometimes be solved exactly.

Example 13.11 (Gaussian mean and the exact constant). Let

$$X = \theta + Z, \quad Z \sim N(0, \sigma^2),$$

where $\theta \in \mathbb{R}$ is unknown. Put the Gaussian prior

$$\theta \sim N(0, \tau^2),$$

independent of Z , and consider squared loss.

For squared loss, the Bayes estimator is the posterior mean $\mathbb{E}(\theta | X)$, and the Bayes risk is

$$\mathbb{E}[\text{Var}(\theta | X)].$$

Since (θ, X) is jointly Gaussian,

$$\text{Var}(\theta | X) = \text{Var}(\theta) - \frac{\text{Cov}(\theta, X)^2}{\text{Var}(X)}.$$

Here

$$\text{Var}(\theta) = \tau^2, \quad \text{Cov}(\theta, X) = \tau^2, \quad \text{Var}(X) = \tau^2 + \sigma^2.$$

Thus

$$\text{Var}(\theta | X) = \tau^2 - \frac{\tau^4}{\tau^2 + \sigma^2} = \frac{\tau^2 \sigma^2}{\tau^2 + \sigma^2}.$$

Consequently, for every $\tau > 0$,

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}_{\theta}(\hat{\theta}(X) - \theta)^2 \geq \frac{\tau^2 \sigma^2}{\tau^2 + \sigma^2}.$$

Letting $\tau \rightarrow \infty$, we get

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}_{\theta}(\hat{\theta}(X) - \theta)^2 \geq \sigma^2.$$

This matches the risk of the estimator $\hat{\theta} = X$. Therefore the exact minimax risk is σ^2 .

For n independent observations, the sufficient statistic $\bar{X}$ has noise variance σ^2/n , so the same computation gives the exact minimax risk σ^2/n in one dimension and $d\sigma^2/n$ in d dimensions.

Remark 13.12. Bayes-risk lower bounds are especially useful when the prior and the model are conjugate. In such cases the posterior variance can be computed exactly, and the constants are often much sharper than those produced by Fano-type arguments.

13.6.2 Two-point lower bounds

The simplest testing reduction uses only two distributions. This is often enough to recover the correct rate, especially for parametric problems.

Proposition 13.13 (Two-point testing principle). *Let $\theta_0, \theta_1 \in \Theta$, and write*

$$\Delta = d(\theta_0, \theta_1).$$

For every $q > 0$,

$$\inf_{\hat{\theta}} \max_{i=0,1} \mathbb{E}_{\theta_i} d^q(\hat{\theta}, \theta_i) \geq \frac{\Delta^q}{2^q} \inf_{\hat{J}} \max_{i=0,1} \mathbb{P}_{\theta_i}(\hat{J} \neq i),$$

where the infimum on the right is over all tests $\hat{J} \in \{0, 1\}$.

Proof. Given an estimator $\hat{\theta}$, define $\hat{J} \in \{0, 1\}$ to be a nearest-neighbor selector among θ_0 and θ_1 . If the true index is i and $\hat{J} \neq i$, then the triangle inequality gives

$$d(\hat{\theta}, \theta_i) \geq \frac{\Delta}{2}.$$

Therefore

$$\max_{i=0,1} \mathbb{E}_{\theta_i} d^q(\hat{\theta}, \theta_i) \geq \frac{\Delta^q}{2^q} \max_{i=0,1} \mathbb{P}_{\theta_i}(\hat{J} \neq i).$$

Now take the infimum over estimators. The induced nearest-neighbor tests form only a subclass of all tests, so lower bounding by the infimum over all tests is valid. $\square$

Example 13.14 (Mean estimation for random signs). Let $X_1, \dots, X_n \in \{-1, 1\}$ be independent, and suppose we want to estimate the mean

$$\theta = \mathbb{E}X_i \in [-1, 1].$$

Fix $\delta \in (0, 1/4)$, and define two distributions on $\{-1, 1\}$:

$$\mathbb{P}_+(X = 1) = \frac{1 + \delta}{2}, \quad \mathbb{P}_-(X = 1) = \frac{1 - \delta}{2}.$$

The corresponding means are

$$\theta_+ = \delta, \quad \theta_- = -\delta,$$

so their separation is 2δ .

For one observation,

$$\begin{aligned} \text{KL}(\mathbb{P}_+ \parallel \mathbb{P}_-) &= \frac{1 + \delta}{2} \log \left(\frac{1 + \delta}{1 - \delta} \right) + \frac{1 - \delta}{2} \log \left(\frac{1 - \delta}{1 + \delta} \right) \\ &= \delta \log \left(\frac{1 + \delta}{1 - \delta} \right). \end{aligned}$$

The same value is obtained for $\text{KL}(\mathbb{P}_- \parallel \mathbb{P}_+)$. For $0 < \delta < 1/4$,

$$\log \left(\frac{1 + \delta}{1 - \delta} \right) \leq 3\delta,$$

and hence

$$\text{KL}(\mathbb{P}_+^{\otimes n} \parallel \mathbb{P}_-^{\otimes n}) \leq 3n\delta^2, \quad \text{KL}(\mathbb{P}_-^{\otimes n} \parallel \mathbb{P}_+^{\otimes n}) \leq 3n\delta^2.$$

Choose $\delta = c_0 n^{-1/2}$, with $c_0 > 0$ small enough that

$$3n\delta^2 \leq \frac{2e}{2e+1} \log 2.$$

For all sufficiently large n , this choice also satisfies $\delta < 1/4$. Birgé's lemma, applied with $N = 2$, implies that every test has error at least a positive numerical constant for one of the two hypotheses. By Proposition 13.13, with squared loss,

$$\inf_{\hat{\theta}} \sup_{\theta \in [-1, 1]} \mathbb{E}_{\theta}(\hat{\theta} - \theta)^2 \geq c\delta^2 \geq \frac{c'}{n}$$

for universal constants $c, c' > 0$.

Thus the root mean-squared scale cannot be smaller than $n^{-1/2}$, which is exactly the usual parametric rate.

Remark 13.15. The two-point method is deliberately local. It does not try to understand the whole parameter space. It only finds two parameters that are separated in the loss but close enough statistically that no test can reliably distinguish them.

References

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